

Space–Time Algebra

A compact development of paravectors and their physical applications

Dr. Luke A. Wendt

Abstract

A paravector algebra is formed from a complex scalar and three-vector, with dot and cross products combined. Quaternions, rotations, Lorentz boosts, projections, spectral functions, spacetime geometry, relativistic dynamics, electromagnetism, gauge coupling, Klein–Gordon and Dirac equations, Schrödinger and Pauli limits, and spin amplitudes and probabilities are expressed in this algebra. These results are connected through shared definitions and short derivations: proper time, four-momentum, and the mass shell are obtained from spacetime determinants; Klein–Gordon is obtained from energy and momentum operators, and Dirac from first-order factorization. Spin one-half is determined by the rotation structure of paravector spinors; its magnetic coupling and measurement probabilities are obtained from the same products and projectors.

Contents

1	Algebra and paravectors	1
1.1	Notation and physical conventions	1
1.2	Sigmas and ordered multiplication	1
1.3	Hermitian and sigma conjugation	2
1.4	Complex paravectors and their components	3
1.5	Conjugates, determinant, and inverse	4
1.6	Real and imaginary extractions	5
2	Products and normalizations	5
2.1	Invertibility and normalization	5
2.2	Products and quadratic forms	6
2.3	Real products	7
2.4	Complex products	7
2.5	Norm and determinant	8
3	Projectors and spectral functions	8
3.1	Quadratic identity and eigenvalues	8
3.2	Projection operators	9
3.3	Spectral rule	10
3.4	Square roots and repeated eigenvalues	11
4	Exponentials and elementary functions	12
4.1	Exponential: projection and series derivations	12
4.2	Rotation and boost factors	13
4.3	Cayley transform	14
4.4	Composing the factors	15
4.5	Logarithm	16
5	Transformations and invariants	16
5.1	Similarity, determinant, and norm	16
5.2	Real paravectors and the Lorentz metric	17
5.3	Rotation action and Rodrigues' formula	19
5.4	Quaternions as paravectors	19
5.5	Boost action and its sign convention	21
6	Space–time kinematics	22
6.1	Intervals, light cones, and proper time	22
6.2	Four-velocity and worldlines	24
6.3	Fixed-axis motion and rapidity	24
6.4	Constant velocity	25
7	Inertial frames and particle dynamics	25
7.1	Coordinates, intervals, and light cones	25
7.2	Spacetime increments and velocity	26
7.3	Clock rates and measured lengths	26
7.4	Composing frames	27
7.5	Four-momentum and the mass shell	27

7.6	Free-particle action	28
8	Acceleration	29
8.1	Four-acceleration and proper acceleration	29
8.2	Force, work, and inertia	31
8.3	Fixed-axis acceleration	32
8.4	Constant coordinate acceleration	32
8.5	Constant proper acceleration	33
8.6	A trip with midpoint braking	34
9	Jerk	35
9.1	Coordinate derivatives, then proper-time derivatives	35
9.2	Projection into the instantaneous rest space	36
9.3	Fixed-axis jerk	37
9.4	Jerk at a thrust reversal	39
9.5	Constant jerk	39
10	Kinematics in a Frenet frame	42
10.1	Spatial frame and curvature	42
10.2	Coordinate velocity, acceleration, and jerk	42
10.3	Spacetime derivatives in the moving frame	43
11	Coordinates, frames, and differentiation	44
11.1	Paravector differential operators	44
11.2	Components and the product rule	45
11.3	Generalized coordinates and clock rates	45
11.4	Varying frame transformations	46
11.5	Physical derivatives in a carried frame	47
12	Electromagnetism	47
12.1	Maxwell's equations and wave propagation	48
12.2	Potentials and gauge freedom	49
12.3	The electromagnetic field action	50
12.4	Field invariants	50
12.5	Lorentz force from the action	51
12.6	Acceleration and the mass shell	52
12.7	Energy and momentum flux	54
13	Quantum amplitudes and differential operators	55
13.1	Spinor amplitudes and probability density	55
13.2	Weighted densities and local expectations	56
13.3	Energy and momentum as differential operators	57
13.4	Field strength from noncommuting derivatives	59
14	The Klein–Gordon equation	59
14.1	The mass shell as a wave equation	59
14.2	Positive and negative frequency	60
14.3	Square roots from Fourier series	60
14.4	Electromagnetic coupling	61

15 The Dirac equation	62
15.1 Block form and free factorization	62
15.2 Reconstructing free solutions from Klein–Gordon	63
15.3 Potential and paravector components	64
15.4 Interacting factorization and reconstruction	64
15.5 Gauge phase	66
16 Low-energy limits	66
16.1 Klein–Gordon to Schrödinger	66
16.2 Dirac to Pauli	67
16.3 Spin components in a uniform magnetic field	68
16.4 Evolution about an alternate spin axis	70
16.5 The next free order	71
17 Spin and probability	71
17.1 Spin one-half and rotations	71
17.2 Projectors and probabilities	72
17.3 Measurement along a second axis	73
17.4 Rotating the polarization	74
18 Lorentz covariance and electromagnetic transformations	74
18.1 Transformation laws for fields and derivatives	74
18.2 Maxwell covariance	75
18.3 Wave phases and spinor covariance	75
18.4 Boosted fields and energy flux	75
19 Position-dependent paravector bases	77
19.1 The basis and its interval	77
19.2 Algebra and differentiation	79
19.3 Expansion, free fall, and curvature	80
19.4 Newtonian motion and black-hole metrics	83
A Pauli-matrix representation	89
A.1 Sigmas and products	89
A.2 Conjugations and product order	89
A.3 Scalar part, determinant, and norm	90
A.4 Boost and rotation matrices	91
A.5 Matrix amplitudes and column spinors	91
A.6 Dirac columns	92
A.7 Four components and the standard Dirac basis	93
A.8 Quaternion matrix image	94
B Equation reference	95
C Conventions	99

1 Algebra and paravectors

1.1 Notation and physical conventions

Natural units $\hbar = c = 1$, rationalized electromagnetic units, and signature $(+, -, -, -)$ are used. SI conversions, including their unit assignments and explanations, appear in boxes labeled SI units. Within these boxes, the usual symbols carry the stated SI units; outside them, natural units apply. Vectors and matrices of objects are **bold cyan**; paravectors are **bold magenta**. Bold calligraphic Latin lettering is reserved for paravectors and paravector operators; Greek glyphs are retained and bold. No meaning is assigned by convention to uppercase or lowercase letters. The ordered sigma triple **σ** is bold; each individual sigma σ_k is non-bold.

Derivatives such as ∂_t and $\partial_{\mathbf{r}}$ retain bold vector subscripts. Scalar quantities and coefficient components remain unadorned. The scalar trip distance ℓ is distinguished from the position vector **$\mathbf{r}$** . Scalar and paravector quantities are distinguished by their stated domains. The paravector momentum **$\mathcal{P}$** is distinguished from its vector part **$\mathbf{P}$** by calligraphic lettering and color.

Space-time algebra is developed here using complex paravectors. Comparisons with terminology in the literature are collected in Appendix C.

$c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is identified with its coefficient pair $(c, \mathbf{Z}) \in \mathbb{C} \oplus \mathbb{C}^3$. Real paravectors lie in $\mathbb{R} \oplus \mathbb{R}^3$; mixed sums specify the two coefficient spaces separately. Values are specified by the direct sum; multiplication is defined by the sigma rules. The sigmas σ_k are algebraic objects with order-dependent multiplication rules; **σ** is their ordered triple, not a numerical vector in $\mathbb{R}^3$.

Domains persist within a section unless a specialization is stated. Primes and subscripts inherit the declared type unless explicitly redefined. For fields, membership states the value at each real spacetime point; function notation specifies the independent variables. The required regularity is assumed for all differentiations.

1.2 Sigmas and ordered multiplication

The sigmas σ_n , $n = 1, 2, 3$, are defined by the multiplication rules below. The general complex paravector is expressed in terms of these sigmas (1, pp. 4–5):

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ \mathbf{Z} &= (a + b\mathbf{i}) + (\mathbf{X} + \mathbf{Y}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{X}, \mathbf{Y} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations are defined by

$$\begin{aligned} \mathbf{X} \cdot \mathbf{Y} &= X_1 Y_1 + X_2 Y_2 + X_3 Y_3, \\ \mathbf{X} \times \mathbf{Y} &= \{X_2 Y_3 - X_3 Y_2, X_3 Y_1 - X_1 Y_3, X_1 Y_2 - X_2 Y_1\}, \\ \mathbf{X}^2 &:= \mathbf{X} \cdot \mathbf{X}. \end{aligned}$$

Thus the square of a vector always denotes its self dot product. For complex vectors, the same formulas are used for dot and cross products, with neither factor conjugated. The elementary Cartesian vectors used below are the real vectors

$$\mathbf{e}_1 = \{1, 0, 0\}, \quad \mathbf{e}_2 = \{0, 1, 0\}, \quad \mathbf{e}_3 = \{0, 0, 1\}.$$

Thus $\mathbf{e}_k \cdot \boldsymbol{\sigma} = \sigma_k$; the scalar basis element is 1. Unindexed e denotes only Euler's number.

A basis over $\mathbb{R}$ is provided by the eight elements

$$1, \quad \sigma_1, \sigma_2, \sigma_3, \quad i\sigma_1, i\sigma_2, i\sigma_3, \quad i.$$

By anticommutation, $i^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. Two sign changes are introduced when σ_n is moved through the other two sigmas, so

$$i\sigma_n = \sigma_n i.$$

Thus i commutes with every sigma and hence with every paravector.

For $\mathbf{x}, \mathbf{y} \in \mathbb{C} \oplus \mathbb{C}^3$, the commutator is

$$\begin{aligned} [\mathbf{x}, \mathbf{y}] &:= \mathbf{x}\mathbf{y} - \mathbf{y}\mathbf{x}, \\ [\mathbf{y}, \mathbf{x}] &= -[\mathbf{x}, \mathbf{y}]. \end{aligned}$$

Thus $[\mathbf{x}, \mathbf{y}] = 0$ exactly when $\mathbf{x}$ and $\mathbf{y}$ commute. For $n, m, k = 1, 2, 3$, the Kronecker delta is defined by

$$\delta_{nm} := \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m, \end{cases}$$

and the Levi-Civita symbol is defined by

$$\epsilon_{nmk} := \begin{cases} +1 & \text{if } (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{if two or more indices are equal.} \end{cases}$$

For $n, m = 1, 2, 3$, the ordered multiplication rule and commutator are (1, Eq. (3)) (1, Eq. (10))

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k, \\ [\sigma_n, \sigma_m] &= 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k. \end{aligned} \tag{1.1}$$

1.3 Hermitian and sigma conjugation

Hermitian conjugation is defined by (1.2), and *sigma conjugation* is defined by (1.3) (2, Eq. (3)):

$$(\cdot)^H := \text{the multiplication order of the sigmas is reversed}, \tag{1.2}$$

$$(\cdot)^* := \text{every } \sigma_n \text{ is replaced by } -\sigma_n \text{ with product order preserved.} \tag{1.3}$$

Thus, for two sigmas,

$$\begin{aligned} (\sigma_m \sigma_n)^H &= \sigma_n \sigma_m, \\ (\sigma_m \sigma_n)^* &= (-\sigma_m)(-\sigma_n). \end{aligned}$$

Both operations are applied term by term to sums, with real coefficients left unchanged. Since $i = \sigma_1 \sigma_2 \sigma_3$, under sigma conjugation each sigma is negated and multiplication order is preserved:

$$i^* = (-\sigma_1)(-\sigma_2)(-\sigma_3) = -\sigma_1 \sigma_2 \sigma_3 = -i.$$

Complex coefficients are therefore conjugated, and the sign of σ is reversed. The resulting signs on the basis elements are

element	1	σ_n	$i\sigma_n$	i
$(\cdot)^*$	+	-	+	-
$(\cdot)^H$	+	+	-	-
$(\cdot)^{*H}$	+	-	-	+

For $i\sigma_n$, the signs are obtained directly:

$$\begin{aligned} (i\sigma_n)^* &= i^* \sigma_n^* = i\sigma_n, \\ (i\sigma_n)^H &= \sigma_n^H i^H = -i\sigma_n. \end{aligned}$$

For $\mathbf{Z}, \mathbf{Z}' \in \mathbb{C} \oplus \mathbb{C}^3$, the definitions give the product laws (1, p. 4) (2, p. 2)

$$(\mathbf{Z}\mathbf{Z}')^H = \mathbf{Z}'^H \mathbf{Z}^H, \quad (1.4)$$

$$(\mathbf{Z}\mathbf{Z}')^* = \mathbf{Z}^* \mathbf{Z}'^*. \quad (1.5)$$

The operations commute with each other: $\mathbf{Z}^{*H} = \mathbf{Z}^{H*}$. Appendix A gives both operations in the Pauli-matrix representation, where H is the usual matrix Hermitian conjugation.

1.4 Complex paravectors and their components

$a, b \in \mathbb{R}$ and $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^3$ are assumed. The two real paravectors $\mathbf{X}, \mathbf{Y} \in \mathbb{R} \oplus \mathbb{R}^3$ are combined into the general complex paravector $\mathbf{Z} \in \mathbb{C} \oplus \mathbb{C}^3$:

$$\begin{aligned} \mathbf{X} &= a + \mathbf{X} \cdot \boldsymbol{\sigma}, \\ \mathbf{Y} &= b + \mathbf{Y} \cdot \boldsymbol{\sigma}, \\ \mathbf{Z} &= \mathbf{X} + i\mathbf{Y} \\ &= (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma} \\ &= c + \mathbf{Z} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here $c = a + bi \in \mathbb{C}$ and $\mathbf{Z} = \mathbf{X} + i\mathbf{Y} \in \mathbb{C}^3$ are its complex scalar and vector coefficients.

Thus $\text{sc}, \det, \text{tr}$ take values in $\mathbb{C}$, vec in $\mathbb{C}^3$, and re, im in $\mathbb{R} \oplus \mathbb{R}^3$. Conjugates, adjugates, and existing inverses take values in $\mathbb{C} \oplus \mathbb{C}^3$. The complex scalar and vector extractions are defined by

$$\begin{aligned} \text{sc}(\mathbf{Z}) &:= c = a + ib, \\ \text{vec}(\mathbf{Z}) &:= \mathbf{Z} = \mathbf{X} + i\mathbf{Y}. \end{aligned}$$

This notation is used throughout unless a real specialization is stated.

1.5 Conjugates, determinant, and inverse

On the coefficients, both conjugations are expressed by $c^* = c^H = a - bi$ and $Z^* = Z^H = X - iY$; on the paravector, the two conjugations differ in the vector sign (2, Eq. (45)):

$$Z^* = c^* - Z^* \cdot \sigma = (a - bi) + (-X + iY) \cdot \sigma, \quad (1.6)$$

$$Z^H = c^* + Z^* \cdot \sigma = (a - bi) + (X - iY) \cdot \sigma. \quad (1.7)$$

Their composition defines the adjugate (2, Eq. (39))

$$\text{adj } Z := Z^{*H} = c - Z \cdot \sigma. \quad (1.8)$$

The trace is twice the scalar part:

$$\text{tr}(Z) := Z + \text{adj } Z = 2 \text{sc}(Z) = 2c. \quad (1.9)$$

The determinant is the complex scalar (3, p. 104)

$$\det Z := \text{adj}(Z) Z = Z \text{adj } Z = c^2 - Z^2. \quad (1.10)$$

For $Z' \in \mathbb{C} \oplus \mathbb{C}^3$, adjugation reverses product order, giving determinant multiplicativity:

$$\begin{aligned} \text{adj}(ZZ') &= \text{adj}(Z') \text{adj}(Z), \\ \det(ZZ') &= \text{adj}(Z') (\text{adj}(Z) Z) Z' = \det(Z) \det(Z'). \end{aligned}$$

Since the determinants are scalars, the determinant of any product is unchanged by reordering its factors. For a scalar $z \in \mathbb{C}$,

$$\begin{aligned} \text{adj}(zZ) &= z \text{adj}(Z), \\ \det(zZ) &= z^2 \det(Z). \end{aligned} \quad (1.11)$$

When $\det(Z) \neq 0$, the inverse is (2, Eq. (47))

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)} = \frac{c - Z \cdot \sigma}{c^2 - Z^2}. \quad (1.12)$$

For $Z' = c' + Z' \cdot \sigma$, with $c' \in \mathbb{C}$, $Z' \in \mathbb{C}^3$, division is order-sensitive. An unambiguous quotient exists when Z is invertible and the factors commute:

$$\frac{Z'}{Z} := Z' Z^{-1} = Z^{-1} Z' \quad \text{if } Z \times Z' = 0.$$

Differentiating $Z^{-1} Z = 1$ gives the inverse rule

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}.$$

For $Z, Z', Z'' \in \mathbb{C} \oplus \mathbb{C}^3$, scalar part obeys the cyclic laws

$$\begin{aligned} \text{sc}(ZZ') &= \text{sc}(Z'Z), \\ \text{sc}(ZZ'Z'') &= \text{sc}(Z'Z''Z) = \text{sc}(Z''ZZ'). \end{aligned} \quad (1.13)$$

Multiplying these identities by 2 gives trace cyclicity. Their similarity invariants are derived in Section 5.

1.6 Real and imaginary extractions

The real and imaginary extractions are defined using Hermitian conjugation (1, Eq. (9)):

$$\text{re}(\mathcal{Z}) := \frac{1}{2}(\mathcal{Z} + \mathcal{Z}^H) = \mathcal{X}, \quad (1.14)$$

$$\text{im}(\mathcal{Z}) := \frac{\mathcal{Z} - \mathcal{Z}^H}{2i} = \mathcal{Y}. \quad (1.15)$$

Adjugation evaluates the scalar and vector extractions explicitly (1, Eq. (14)):

$$\begin{aligned} \text{sc}(\mathcal{Z}) &= \frac{1}{2}(\mathcal{Z} + \text{adj}(\mathcal{Z})) = c, \\ \text{vec}(\mathcal{Z}) \cdot \sigma &= \frac{1}{2}(\mathcal{Z} - \text{adj}(\mathcal{Z})) = \mathbf{Z} \cdot \sigma. \end{aligned}$$

The functions re and im also act on complex coefficients. Composing extractions recovers all four real parts:

$$\begin{aligned} a &= \text{re}(c) = \text{sc}(\text{re}(\mathcal{Z})), \\ b &= \text{im}(c) = \text{sc}(\text{im}(\mathcal{Z})), \\ \mathbf{X} &= \text{re}(\mathbf{Z}) = \text{vec}(\text{re}(\mathcal{Z})), \\ \mathbf{Y} &= \text{im}(\mathbf{Z}) = \text{vec}(\text{im}(\mathcal{Z})). \end{aligned}$$

Appendix A reproduces these intrinsic operations with Pauli matrices.

2 Products and normalizations

2.1 Invertibility and normalization

Notation: $\mathcal{Z} = c + \mathbf{Z} \cdot \sigma$, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. For $\mathbf{Z} \in \mathbb{C}^3$, the bilinear square and squared norm are distinguished:

$$\begin{aligned} \mathbf{Z}^2 &:= \mathbf{Z} \cdot \mathbf{Z}, \\ \|\mathbf{Z}\|^2 &:= \mathbf{Z} \cdot \mathbf{Z}^*. \end{aligned}$$

For $c \in \mathbb{C}$, $\|c\|^2 := cc^*$ is defined. The squared paravector norm is (2, Eq. (46))

$$\|\mathcal{Z}\|^2 := \|c\|^2 + \|\mathbf{Z}\|^2 = cc^* + \mathbf{Z} \cdot \mathbf{Z}^*. \quad (2.1)$$

All norms take values in $\mathbb{R}_{\geq 0}$ and denote nonnegative real square roots. Here $\mathbf{u} \in \mathbb{C}^3$ and $\mathcal{Z} \in \mathbb{C} \oplus \mathbb{C}^3$. Each name below specifies a condition; a letter alone implies none.

- **Vector of square one.** $\mathbf{u}$ satisfies

$$\mathbf{u}^2 = \mathbf{u} \cdot \mathbf{u} = 1. \quad (2.2)$$

For $\mathbf{Z}^2 \neq 0$, $\mathbf{u} = \mathbf{Z}/\sqrt{\mathbf{Z}^2}$ is defined with either square root. Its norm need not be one. A nonzero vector with $\mathbf{Z}^2 = 0$ cannot be normalized this way.

- **Vector of unit norm.** $\mathbf{u}$ satisfies

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}^* = 1.$$

For any $\mathbf{Z} \neq 0$, $\mathbf{u} = \mathbf{Z}/\|\mathbf{Z}\|$ is defined; then $\mathbf{u}^2 = \mathbf{Z}^2/\|\mathbf{Z}\|^2$ need not be one. For real vectors the two conditions coincide: a *real unit vector* has $\mathbf{u}^2 = \|\mathbf{u}\|^2 = 1$. For complex vectors, specify which holds.

- **Invertible paravector.** $\mathbf{Z}$ has an inverse on both sides:

$$\mathbf{Z}\mathbf{Z}^{-1} = \mathbf{Z}^{-1}\mathbf{Z} = 1. \quad (2.3)$$

Equivalently, $\det(\mathbf{Z}) \neq 0$. Neither unit norm nor determinant one is required.

- **Unitary paravector.** Its Hermitian conjugate is its inverse:

$$\mathbf{Z}^H \mathbf{Z} = 1. \quad (2.4)$$

Then $\mathbf{Z}^H = \mathbf{Z}^{-1}$ and $\mathbf{Z}\mathbf{Z}^H = 1$.

- **Paravector of determinant one.** $\mathbf{Z}$ satisfies

$$\det \mathbf{Z} = 1. \quad (2.5)$$

It is invertible with $\mathbf{Z}^{-1} = \text{adj}(\mathbf{Z})$; it need not be unitary. For real $\mathbf{Z}$, $\text{adj} \mathbf{Z} = \mathbf{Z}^*$, so this condition is also $\mathbf{Z}^* \mathbf{Z} = 1$. This equivalence need not hold for complex $\mathbf{Z}$.

- **Paravector of unit norm.** $\mathbf{Z}$ satisfies

$$\|\mathbf{Z}\|^2 = 1. \quad (2.6)$$

For $\mathbf{Z} \neq 0$, $\mathbf{Z}' = \mathbf{Z}/\|\mathbf{Z}\|$ has unit norm. It need not have an inverse: $(1 + \mathbf{u} \cdot \boldsymbol{\sigma})/\sqrt{2}$ has norm one and determinant zero for any real unit vector $\mathbf{u}$. Unitarity implies unit norm by (2.14).

- **Self-inverse paravector.** $\mathbf{Z}$ is its own inverse:

$$\mathbf{Z}^2 = 1. \quad (2.7)$$

Then $\mathbf{Z}^{-1} = \mathbf{Z}$; it is unitary exactly when $\mathbf{Z}^H = \mathbf{Z}$. Its parts obey $c^2 + \mathbf{Z}^2 = 1$ and $2c\mathbf{Z} = 0$: thus $\mathbf{Z} = \pm 1$ or $\mathbf{Z} = \mathbf{u} \cdot \boldsymbol{\sigma}$ with $\mathbf{u}^2 = 1$.

The physical roles of the paravector conditions are compared in Section 5.

2.2 Products and quadratic forms

For $\mathbf{X}, \mathbf{Y} \in \mathbb{C}^3$, multiplication combines the dot and cross products; both products are complex bilinear, with no conjugation. The sigma rule (1.1) gives the dot-cross identity (2, Eq. (38))

$$\begin{aligned} (\mathbf{X} \cdot \boldsymbol{\sigma})(\mathbf{Y} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 X_n Y_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 \sum_{m=1}^3 X_n Y_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 X_n Y_m \epsilon_{nmk} i \sigma_k \\ &= \mathbf{X} \cdot \mathbf{Y} + i(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (2.8)$$

For real $\mathbf{X}, \mathbf{Y}$, the product lies in $\mathbb{R} \oplus i\mathbb{R}^3$. Adding the product with $\mathbf{X}$ and $\mathbf{Y}$ interchanged gives

$$2(\mathbf{X} \cdot \mathbf{Y}) = (\mathbf{X} \cdot \boldsymbol{\sigma})(\mathbf{Y} \cdot \boldsymbol{\sigma}) + (\mathbf{Y} \cdot \boldsymbol{\sigma})(\mathbf{X} \cdot \boldsymbol{\sigma}).$$

Consequently, for every nonnegative integer n ,

$$\begin{aligned} (\mathbf{X} \cdot \boldsymbol{\sigma})^2 &= \mathbf{X}^2, \\ (\mathbf{X} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{X}^2)^n, \\ (\mathbf{X} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{X}^2)^n \mathbf{X} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Also,

$$(\mathbf{X} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{X} - \mathbf{i}\mathbf{X} \times \boldsymbol{\sigma}.$$

Subtracting the two orders gives

$$[\mathbf{X} \cdot \boldsymbol{\sigma}, \mathbf{Y} \cdot \boldsymbol{\sigma}] = 2\mathbf{i}(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}.$$

Hence

$$\begin{aligned} [\mathbf{X} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2\mathbf{i}\mathbf{X} \times \boldsymbol{\sigma}, \\ [\mathbf{X} \cdot \mathbf{i}\boldsymbol{\sigma}, \boldsymbol{\sigma}] &= 2\mathbf{X} \times \boldsymbol{\sigma}, \\ [\mathbf{X} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{Y} &= 2\mathbf{i}(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

2.3 Real products

For real $\mathbf{x} = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ and $\mathbf{y} = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $a, b \in \mathbb{R}$ and $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^3$, (2.8) gives

$$\begin{aligned} \mathbf{x}\mathbf{y} &= (ab + \mathbf{X} \cdot \mathbf{Y}) + (a\mathbf{Y} + b\mathbf{X} + \mathbf{i}\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \\ [\mathbf{x}, \mathbf{y}] &= 2\mathbf{i}(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \\ \mathbf{x}^2 &= (a^2 + \mathbf{X}^2) + 2a\mathbf{X} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Thus $\mathbf{x}\mathbf{y} \in \mathbb{R} \oplus \mathbb{C}^3$, $[\mathbf{x}, \mathbf{y}] \in \{0\} \oplus \mathbf{i}\mathbb{R}^3$, and $\mathbf{x}^2 \in \mathbb{R} \oplus \mathbb{R}^3$. Real paravectors are not closed under multiplication. The real quadratic identity is (3.1) with $c = a$. If $\mathbf{X} \times \mathbf{Y} = 0$, then $\mathbf{x}\mathbf{y} = \mathbf{y}\mathbf{x}$; when $\det(\mathbf{x}) \neq 0$, also $\mathbf{x}^{-1}\mathbf{y} = \mathbf{y}\mathbf{x}^{-1}$.

2.4 Complex products

The decompositions $\mathbf{z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $\mathbf{z}' = c' + \mathbf{Z}' \cdot \boldsymbol{\sigma}$ are used, where $c, c' \in \mathbb{C}$ and $\mathbf{Z}, \mathbf{Z}' \in \mathbb{C}^3$. Complex bilinearity extends (2.8) to the general product

$$\mathbf{z}\mathbf{z}' = (cc' + \mathbf{Z} \cdot \mathbf{Z}') + (c\mathbf{Z}' + c'\mathbf{Z} + \mathbf{i}\mathbf{Z} \times \mathbf{Z}') \cdot \boldsymbol{\sigma}. \quad (2.9)$$

The result $\mathbf{z}\mathbf{z}' \in \mathbb{C} \oplus \mathbb{C}^3$ has a complex scalar and complex vector part. Thus

$$\begin{aligned} \text{sc}(\mathbf{z}\mathbf{z}') &= cc' + \mathbf{Z} \cdot \mathbf{Z}', \\ \text{vec}(\mathbf{z}\mathbf{z}') &= c\mathbf{Z}' + c'\mathbf{Z} + \mathbf{i}\mathbf{Z} \times \mathbf{Z}'. \end{aligned}$$

Equivalently, $\mathbf{z} = \mathbf{x} + \mathbf{i}\mathbf{y}$ and $\mathbf{z}' = \mathbf{x}' + \mathbf{i}\mathbf{y}'$ are used with all four factors real. Their products need not be real, but distributivity gives

$$\mathbf{z}\mathbf{z}' = (\mathbf{x}\mathbf{x}' - \mathbf{y}\mathbf{y}') + \mathbf{i}(\mathbf{x}\mathbf{y}' + \mathbf{y}\mathbf{x}').$$

The decomposition $\mathbf{z} = (a + \mathbf{b}\mathbf{i}) + (\mathbf{X} + \mathbf{i}\mathbf{Y}) \cdot \boldsymbol{\sigma}$ is used with real $a, b, \mathbf{X}, \mathbf{Y}$, and primes are used for the second paravector's real parts. The real and imaginary scalar and vector parts of $\mathbf{z}\mathbf{z}'$ are

$$\begin{aligned} \text{real scalar: } & aa' - bb' + \mathbf{X} \cdot \mathbf{X}' - \mathbf{Y} \cdot \mathbf{Y}', \\ \text{imaginary scalar: } & ab' + ba' + \mathbf{X} \cdot \mathbf{Y}' + \mathbf{Y} \cdot \mathbf{X}', \\ \text{real vector: } & a\mathbf{X}' - b\mathbf{Y}' + a'\mathbf{X} - b'\mathbf{Y} - \mathbf{X} \times \mathbf{Y}' - \mathbf{Y} \times \mathbf{X}', \\ \text{imaginary vector: } & a\mathbf{Y}' + b\mathbf{X}' + a'\mathbf{Y} + b'\mathbf{X} + \mathbf{X} \times \mathbf{X}' - \mathbf{Y} \times \mathbf{Y}'. \end{aligned}$$

Specializing (2.9) gives the quadratic products:

$$\begin{aligned}\mathbf{Z}^2 &= (a^2 - b^2 + \mathbf{X}^2 - \mathbf{Y}^2) + 2i(ab + \mathbf{X} \cdot \mathbf{Y}) \\ &\quad + 2(a\mathbf{X} - b\mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} + b\mathbf{X}) \cdot \boldsymbol{\sigma},\end{aligned}$$

$$\mathbf{Z}\mathbf{Z}^* = (a^2 + b^2 - \mathbf{X}^2 - \mathbf{Y}^2) - 2(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} - b\mathbf{X}) \cdot \boldsymbol{\sigma}, \quad (2.10)$$

$$\mathbf{Z}^*\mathbf{Z} = (a^2 + b^2 - \mathbf{X}^2 - \mathbf{Y}^2) + 2(\mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{Y} - b\mathbf{X}) \cdot \boldsymbol{\sigma}, \quad (2.11)$$

$$\mathbf{Z}\mathbf{Z}^H = a^2 + b^2 + \mathbf{X}^2 + \mathbf{Y}^2 + 2(a\mathbf{X} + b\mathbf{Y} + \mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \quad (2.12)$$

$$\mathbf{Z}^H\mathbf{Z} = a^2 + b^2 + \mathbf{X}^2 + \mathbf{Y}^2 + 2(a\mathbf{X} + b\mathbf{Y} - \mathbf{X} \times \mathbf{Y}) \cdot \boldsymbol{\sigma}, \quad (2.13)$$

$$\mathbf{Z}\mathbf{Z}^{*H} = \det \mathbf{Z} = a^2 - b^2 - \mathbf{X}^2 + \mathbf{Y}^2 + 2i(ab - \mathbf{X} \cdot \mathbf{Y}).$$

Equations (2.10) and (2.11) have values in $\mathbb{R} \oplus \mathbb{C}^3$ and show $\mathbf{Z}\mathbf{Z}^* = \mathbf{Z}^*\mathbf{Z}$ exactly when $\mathbf{X} \times \mathbf{Y} = 0$. Equations (2.12) and (2.13) take values in $\mathbb{R} \oplus \mathbb{R}^3$ and give the same condition for $\mathbf{Z}\mathbf{Z}^H = \mathbf{Z}^H\mathbf{Z}$.

2.5 Norm and determinant

For the complex coefficients $c = a + bi$ and $\mathbf{Z} = \mathbf{X} + i\mathbf{Y}$, the squared norms defined above are

$$\begin{aligned}\|c\|^2 &= a^2 + b^2, \\ \|\mathbf{Z}\|^2 &= \mathbf{X}^2 + \mathbf{Y}^2.\end{aligned}$$

For the squared norm (2.1), the quadratic products and cyclicity give the Hermitian form (2, Eq. (46))

$$\begin{aligned}\|\mathbf{Z}\|^2 &= \text{sc}(\mathbf{Z}^H\mathbf{Z}) = \frac{1}{2} \text{tr}(\mathbf{Z}^H\mathbf{Z}) = \text{sc}(\mathbf{Z}\mathbf{Z}^H) \\ &= \text{sc}(\mathbf{Z}^*\mathbf{Z}^{*H}) = \text{sc}(\mathbf{Z}^{*H}\mathbf{Z}^*).\end{aligned} \quad (2.14)$$

For complex coefficients, c^2 and $\mathbf{Z}^2$ generally differ from these squared norms. Also $\det(\mathbf{Z})$ may be complex. For a complex scalar $z = re^{i\theta}$, with real $r \geq 0$ and θ , scalar scaling gives

$$\begin{aligned}\det(z\mathbf{Z}) &= r^2 e^{2i\theta} \det(\mathbf{Z}), \\ \|z\mathbf{Z}\|^2 &= \|z\|^2 \|\mathbf{Z}\|^2 = r^2 \|\mathbf{Z}\|^2.\end{aligned} \quad (2.15)$$

The determinant retains the scalar phase by (1.11); the norm loses it by (2.15).

3 Projectors and spectral functions

3.1 Quadratic identity and eigenvalues

Notation: $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. Since $\mathbf{Z} - c = \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $(\mathbf{Z} \cdot \boldsymbol{\sigma})^2 = \mathbf{Z}^2$,

$$\begin{aligned}(\mathbf{Z} - c)^2 &= \mathbf{Z}^2, \\ \mathbf{Z}^2 - 2c\mathbf{Z} + (c^2 - \mathbf{Z}^2) &= 0.\end{aligned}$$

Using $\text{tr}(\mathbf{Z}) = 2c$ and $\det \mathbf{Z} = c^2 - \mathbf{Z}^2$ gives the trace-determinant identity

$$\mathbf{Z}^2 - \text{tr}(\mathbf{Z})\mathbf{Z} + \det \mathbf{Z} = 0. \quad (3.1)$$

An eigenvalue is a scalar $\lambda \in \mathbb{C}$ for which $\mathbf{Z} - \lambda$ has no inverse. Thus $\det(\mathbf{Z} - \lambda) = \lambda^2 - 2c\lambda + \det \mathbf{Z} = 0$, the scalar quadratic corresponding to (3.1). Its roots are

$$\lambda_{\pm} = \frac{\operatorname{tr}(\mathbf{Z}) \pm \sqrt{\operatorname{tr}(\mathbf{Z})^2 - 4\det(\mathbf{Z})}}{2} = c \pm \sqrt{\mathbf{Z}^2}.$$

Because $\mathbf{Z}$ is generally complex, $\mathbf{Z}^2 = \mathbf{Z} \cdot \mathbf{Z}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(\mathbf{Z} - \lambda_-)(\mathbf{Z} - \lambda_+) = 0.$$

3.2 Projection operators

For a real unit vector $\mathbf{u}$, the paravector $\mathbf{u} \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{R}^3$ is Hermitian and self-inverse. Its two orthogonal projectors are defined by (3, p. 121)

$$\boldsymbol{\Pi}_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.2)$$

Here $\boldsymbol{\Pi}_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$ with scalar part 1/2. Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\begin{aligned} \boldsymbol{\Pi}_{\pm}^2 &= \boldsymbol{\Pi}_{\pm}, \\ \boldsymbol{\Pi}_+ \boldsymbol{\Pi}_- &= 0, \\ \boldsymbol{\Pi}_+ + \boldsymbol{\Pi}_- &= 1, \\ \boldsymbol{\Pi}_{\pm}^H &= \boldsymbol{\Pi}_{\pm}, \\ \boldsymbol{\Pi}_{\pm}^* &= \boldsymbol{\Pi}_{\mp}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \boldsymbol{\Pi}_{\pm} &= \pm \boldsymbol{\Pi}_{\pm}. \end{aligned}$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \boldsymbol{\Pi}_+ - \boldsymbol{\Pi}_-$. For a second real unit vector $\mathbf{u}'$,

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The projections $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$ are defined in $\mathbb{R}^3$. Then

$$\begin{aligned} \boldsymbol{\Pi}_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\boldsymbol{\Pi}_{\pm} &= \pm(\mathbf{u} \cdot \mathbf{u}')\boldsymbol{\Pi}_{\pm}, \\ \boldsymbol{\Pi}_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\boldsymbol{\Pi}_{\mp} &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned}$$

For $\boldsymbol{\Pi}'_{\pm} = (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})/2 \in \mathbb{R} \oplus \mathbb{R}^3$, all four products lie in $\mathbb{R} \oplus \mathbb{C}^3$:

$$\begin{aligned} \boldsymbol{\Pi}_{\pm}\boldsymbol{\Pi}'_{\pm} &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \pm \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \boldsymbol{\Pi}_{\pm}\boldsymbol{\Pi}'_{\mp} &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \mp \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

3.3 Spectral rule

The decomposition $\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is used, with $c \in \mathbb{C}$, $\mathbf{Z} \in \mathbb{C}^3$, and $\mathbf{Z}^2 \neq 0$. The complex scalars $z, \lambda_{\pm}$, complex vector $\mathbf{u}$, and projectors $\Pi_{\pm} \in \mathbb{R} \oplus \mathbb{C}^3$ are defined by

$$\begin{aligned} z &:= +\sqrt{\mathbf{Z}^2}, \\ \mathbf{u} &:= \mathbf{Z}/z, & \mathcal{Z} &= c + z\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \lambda_{\pm} &:= c \pm z, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The root is the principal scalar root. Thus $\mathbf{u}$ is a vector of square one; $\mathbf{u} \cdot \boldsymbol{\sigma}$ is self-inverse. By squaring,

$$\begin{aligned} \Pi_{\pm}^2 &= \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + 1) = \Pi_{\pm}, \\ \Pi_+ \Pi_- &= \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0, \\ \Pi_+ + \Pi_- &= 1, \\ \mathcal{Z} &= c(\Pi_+ + \Pi_-) + z(\Pi_+ - \Pi_-) = \lambda_+ \Pi_+ + \lambda_- \Pi_-. \end{aligned}$$

Here $z, \lambda_{\pm} \in \mathbb{C}$, $\mathbf{u} \in \mathbb{C}^3$, and $\mathcal{Z}, f(\mathcal{Z}) \in \mathbb{C} \oplus \mathbb{C}^3$ for the analytic functions below. These projectors are Hermitian when $\mathbf{u}$ is real; in general they need not be. The same formulas apply directly to any given $\mathcal{Z} = c + z\mathbf{u} \cdot \boldsymbol{\sigma}$, with $c, z \in \mathbb{C}$ and $\mathbf{u}^2 = 1$. If $z = 0$, $\mathcal{Z} = c$ is scalar and $f(\mathcal{Z}) = f(c)$. The complementary-projector identities give, for every nonnegative integer n ,

$$\mathcal{Z}^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n)\mathbf{u} \cdot \boldsymbol{\sigma}.$$

For a scalar power series $f(z) = \sum_{n=0}^{\infty} c_n z^n$ converging at both eigenvalues, with $c_n \in \mathbb{C}$, inserting these powers into $f(\mathcal{Z}) = \sum_n c_n \mathcal{Z}^n$ gives the spectral rule (4, p. 3)

$$f(\mathcal{Z}) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.3)$$

The same rule defines chosen analytic branches at the two eigenvalues. Only the complementary projector identities are used in the proof, so the result is valid for any decomposition with these identities.

Adjugation exchanges the eigenvalues, so

$$\begin{aligned} \text{adj } \mathcal{Z} &= \lambda_- \Pi_+ + \lambda_+ \Pi_-, \\ \det \mathcal{Z} &= \lambda_+ \lambda_-, \\ \mathcal{Z}^{-1} &= \frac{\text{adj}(\mathcal{Z})}{\det(\mathcal{Z})} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-. \end{aligned}$$

The inverse requires both eigenvalues to be nonzero. Independently choosing the signs of their principal square roots gives

$$\sqrt{\mathcal{Z}} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-.$$

There are four distinct roots when both eigenvalues are nonzero, and two when one vanishes. Changing z to $-z$ exchanges the eigenvalue-projector pairs and leaves the function unchanged when its branch choices are exchanged with them.

3.4 Square roots and repeated eigenvalues

3.4.1 Real paravectors and direct square roots

For real $\mathbf{x} = a + \mathbf{X} \cdot \boldsymbol{\sigma}$, with $a \in \mathbb{R}$, $\mathbf{X} \in \mathbb{R}^3$, and $\mathbf{X} \neq 0$, the spectral quantities above reduce to

$$z = \|\mathbf{X}\|, \quad \mathbf{u} = \mathbf{X}/\|\mathbf{X}\|, \quad \lambda_{\pm} = a \pm \|\mathbf{X}\|.$$

Here $\mathbf{x}, \boldsymbol{\Pi}_{\pm} \in \mathbb{R} \oplus \mathbb{R}^3$, $z, \lambda_{\pm} \in \mathbb{R}$, and $\mathbf{u} \in \mathbb{R}^3$. The projectors are now Hermitian; complex-valued functions or square roots may still give $f(\mathbf{x}) \in \mathbb{C} \oplus \mathbb{C}^3$. In particular, for a real unit vector $\mathbf{u}$,

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\boldsymbol{\Pi}_+ + f(-1)\boldsymbol{\Pi}_- \\ &= \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm \boldsymbol{\Pi}_+ \pm i\boldsymbol{\Pi}_-. \end{aligned}$$

The two root signs are independent. For a fixed analytic branch, $f(\mathbf{x}) \rightarrow f(a)$ as $\mathbf{X} \rightarrow 0$; independent root choices require the scalar case below.

A direct component derivation also determines all *real* roots. $\mathbf{y} = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$ is taken to be real, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and real roots $\mathbf{x} = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ are sought such that $\mathbf{x}^2 = \mathbf{y}$. By equating parts,

$$\begin{aligned} a^2 + \mathbf{X}^2 &= b, \\ 2a\mathbf{X} &= \mathbf{Y}. \end{aligned}$$

If $\mathbf{Y} \neq 0$, then $a \neq 0$ and $\mathbf{X} = \mathbf{Y}/(2a)$. By substitution,

$$\begin{aligned} a^4 - ba^2 + \frac{1}{4}\mathbf{Y}^2 &= 0, \\ a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{Y}^2} \right). \end{aligned} \tag{3.4}$$

For each nonnegative a^2 in (3.4), both scalar roots a are included, with $\mathbf{X} = \mathbf{Y}/(2a)$; then $\mathbf{X}^2 = b - a^2$. There are four real roots when $b > \|\mathbf{Y}\|$, two when $b = \|\mathbf{Y}\| > 0$, and none when $b < \|\mathbf{Y}\|$.

If $\mathbf{Y} = 0$, the equations instead give the scalar roots $\mathbf{x} = \pm\sqrt{b}$ and the pure-vector roots $\mathbf{x} = \mathbf{X} \cdot \boldsymbol{\sigma}$ with $\mathbf{X}^2 = b$ for $b > 0$. For $b = 0$ only $\mathbf{x} = 0$ is real; for $b < 0$ there is no real root.

3.4.2 Coalescing eigenvalues

For complex $\mathbf{z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 = 0$, $(\mathbf{Z} \cdot \boldsymbol{\sigma})^2 = 0$. A scalar function analytic at c has the terminating Taylor expansion (4, p. 4)

$$f(\mathbf{z}) = f(c) + f'(c)\mathbf{Z} \cdot \boldsymbol{\sigma}. \tag{3.5}$$

For $c \neq 0$, the chosen square root and logarithm are

$$\begin{aligned} \sqrt{\mathbf{z}} &= r + \frac{\mathbf{Z} \cdot \boldsymbol{\sigma}}{2r}, \\ \log \mathbf{z} &= \log c + \frac{\mathbf{Z} \cdot \boldsymbol{\sigma}}{c}. \end{aligned}$$

Here $r \in \mathbb{C}$ is a chosen scalar root of c ($r^2 = c$); a consistent logarithm branch is used. A nonzero $\mathbf{Z} \cdot \boldsymbol{\sigma}$ with $c = 0$ has no square root: such a root would have only zero eigenvalues, and its quadratic

identity would force its square to vanish. It has no logarithm because it is not invertible. If $\mathbf{Z} \cdot \boldsymbol{\sigma} = 0$, a fixed analytic branch gives $f(c)$; non-scalar square roots of a scalar must instead be found from the component equations $a^2 + \mathbf{X}^2 = c$, $2a\mathbf{X} = 0$, now allowing $a \in \mathbb{C}$ and $\mathbf{X} \in \mathbb{C}^3$. Independent branch choices need not remain finite as distinct eigenvalues coalesce.

4 Exponentials and elementary functions

Explicit functions and the factors used for rotations and boosts are obtained from the spectral rule. Algebraic functions are connected to geometric transformations through composition and logarithms.

4.1 Exponential: projection and series derivations

For complex $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 \neq 0$, the following notation is retained: $z = +\sqrt{\mathbf{Z}^2}$, the vector of square one $\mathbf{u} = \mathbf{Z}/z$, and $\lambda_{\pm} = c \pm z$. The spectral rule (3.3) gives

$$\begin{aligned} \exp(\mathbf{Z}) &= \frac{1}{2}e^{c+z}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{1}{2}e^{c-z}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^c \left(\frac{e^z + e^{-z}}{2} + \frac{e^z - e^{-z}}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Independently, c commutes with the vector part and $(z\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = z^{2n}$, $(z\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = z^{2n+1}\mathbf{u} \cdot \boldsymbol{\sigma}$. Separating the defining series therefore gives

$$\begin{aligned} \exp(\mathbf{Z}) &= e^c \sum_{n=0}^{\infty} \frac{(z\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= e^c \left(\sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

Both derivations give (2, Eq. (52))

$$\exp(\mathbf{Z}) = e^c (\cosh z + \sinh z \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (4.1)$$

This is independent of the choice z or $-z$: $\cosh$ is even, while both $\sinh z$ and $\mathbf{u}$ change sign. When $\mathbf{Z}^2 = 0$, (3.5) instead gives

$$\exp(c + \mathbf{Z} \cdot \boldsymbol{\sigma}) = e^c (1 + \mathbf{Z} \cdot \boldsymbol{\sigma}).$$

The defining series also gives

$$\begin{aligned} \exp(\mathbf{Z}^*) &= \exp(\mathbf{Z})^*, \\ \exp(\mathbf{Z}^H) &= \exp(\mathbf{Z})^H. \end{aligned}$$

For the real components $a, b, \mathbf{X}, \mathbf{Y}$ of $\mathbf{Z} = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}$, the following form is used:

$$z = +\sqrt{\mathbf{X}^2 - \mathbf{Y}^2 + 2i\mathbf{X} \cdot \mathbf{Y}}.$$

Substituting $\mathbf{u} = (\mathbf{X} + i\mathbf{Y})/z$ into (4.1) gives the component form

$$\exp(\mathbf{Z}) = e^a (\cos b + i \sin b) \left(\cosh z + \frac{\sinh z}{z} (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma} \right).$$

Its real and imaginary specializations are

$$\exp(a + \mathbf{X} \cdot \boldsymbol{\sigma}) = e^a \left(\cosh \|\mathbf{X}\| + \frac{\sinh \|\mathbf{X}\|}{\|\mathbf{X}\|} \mathbf{X} \cdot \boldsymbol{\sigma} \right), \quad (4.2)$$

$$\exp(i(b + \mathbf{Y} \cdot \boldsymbol{\sigma})) = (\cos b + i \sin b) \left(\cos \|\mathbf{Y}\| + i \frac{\sin \|\mathbf{Y}\|}{\|\mathbf{Y}\|} \mathbf{Y} \cdot \boldsymbol{\sigma} \right). \quad (4.3)$$

Setting $a = 0$ in (4.2) or $b = 0$ in (4.3) gives the pure-vector exponentials. Every quotient uses its analytic limit at a zero argument.

4.2 Rotation and boost factors

For a general real exponent $a + \mathbf{X} \cdot \boldsymbol{\sigma}$, where $a \in \mathbb{R}$ and $\mathbf{X} \in \mathbb{R}^3$, (4.2) gives

$$\det(\exp(a + \mathbf{X} \cdot \boldsymbol{\sigma})) = e^{2a}. \quad (4.4)$$

Thus a real exponential has determinant one exactly when its scalar exponent is zero. Its value still has a nonzero scalar part, supplied by the hyperbolic cosine.

A real parameter $\theta \in \mathbb{R}$ is used, and $\mathbf{u} \in \mathbb{R}^3$ is taken to be a **real unit vector**, $\mathbf{u}^2 = 1$. The two special exponentials just derived define the complex rotation factor $\mathcal{R} \in \mathbb{R} \oplus i\mathbb{R}^3$ and the real boost factor $\mathcal{L} \in \mathbb{R} \oplus \mathbb{R}^3$ (1, Eq. (6)) (5, p. 23):

$$\mathcal{R}(\theta, \mathbf{u}) := \exp\left(-i\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} - i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.5)$$

$$\mathcal{L}(\theta, \mathbf{u}) := \exp\left(-\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh \frac{\theta}{2} - \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (4.6)$$

In (4.5) and (4.6), the parameter is a rotation angle for $\mathcal{R}$ and a rapidity for $\mathcal{L}$; both are real dimensionless numbers. Here these names specify algebraic objects. Their actions on a general paravector are derived in Section 5; their physical applications follow in Sections 7 and 18.

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, reversing the parameter gives the inverses

$$\begin{aligned} \mathcal{R}^{-1} &= \mathcal{R}(-\theta, \mathbf{u}) = \cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \mathcal{L}^{-1} &= \mathcal{L}(-\theta, \mathbf{u}) = \cosh \frac{\theta}{2} + \sinh \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (4.7)$$

Sigma conjugation changes both i and $\boldsymbol{\sigma}$ in sign; Hermitian conjugation changes i but fixes a real vector. Hence

$$\begin{aligned} \mathcal{R}^* &= \mathcal{R}, \\ \mathcal{R}^H &= \mathcal{R}^{-1}, \end{aligned} \quad (4.8)$$

$$\begin{aligned} \mathcal{L}^* &= \mathcal{L}^{-1}, \\ \mathcal{L}^H &= \mathcal{L}. \end{aligned} \quad (4.9)$$

Equation (4.8) makes $\mathcal{R}$ unitary; (4.9) makes $\mathcal{L}$ Hermitian, so $\mathcal{L}^H \mathcal{L} = \mathcal{L}^2$. Their determinants and squared norms follow from the circular and hyperbolic identities:

$$\det \mathcal{R} = \cos^2(\theta/2) + \sin^2(\theta/2) = 1, \quad (4.10)$$

$$\det \mathcal{L} = \cosh^2(\theta/2) - \sinh^2(\theta/2) = 1, \quad (4.11)$$

$$\|\mathcal{R}\|^2 = 1, \quad (4.12)$$

$$\|\mathcal{L}\|^2 = \cosh \theta. \quad (4.13)$$

Equations (4.10) and (4.11) make both **paravectors of determinant one**, so their adjugates equal their inverses. Equations (4.12) and (4.13) distinguish their squared norms; only $\mathcal{R}$ is unitary for nonzero θ . The half-angle gives the sign and period:

$$\begin{aligned}\mathcal{R}(\theta + 2\pi, \mathbf{u}) &= -\mathcal{R}(\theta, \mathbf{u}), \\ \mathcal{R}(\theta + 4\pi, \mathbf{u}) &= +\mathcal{R}(\theta, \mathbf{u}).\end{aligned}\tag{4.14}$$

Gamma form. Notation: $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6). Its real velocity parameter and Lorentz factor are defined by

$$\begin{aligned}\beta &:= \tanh \theta, \\ \gamma &:= \cosh \theta = \frac{1}{\sqrt{1 - \beta^2}}.\end{aligned}$$

Here $-1 < \beta < 1$, $\gamma \geq 1$, and $\gamma\beta = \sinh \theta$. By the half-angle identities,

$$\begin{aligned}\cosh \frac{\theta}{2} &= \sqrt{\frac{\gamma + 1}{2}}, \\ \sinh \frac{\theta}{2} &= \frac{\gamma\beta}{\sqrt{2(\gamma + 1)}}.\end{aligned}$$

Hence the boost factor itself is

$$\mathcal{L} = \sqrt{\frac{\gamma + 1}{2}} \left(1 - \frac{\gamma\beta}{\gamma + 1} \mathbf{u} \cdot \boldsymbol{\sigma} \right).\tag{4.15}$$

Squaring doubles the exponent, so the full-rapidity paravector is

$$\mathcal{L}^2 = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}).\tag{4.16}$$

Equation (4.16) has prefactor γ ; $\mathcal{L}$ has the half-angle prefactor in (4.15). Equivalently, for any real vector $\mathbf{X} \neq 0$, $\mathbf{v} = \tanh\|\mathbf{X}\| \mathbf{X}/\|\mathbf{X}\|$ and $\gamma = \cosh\|\mathbf{X}\|$ are defined. Then

$$\exp(\mathbf{X} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).$$

The zero-vector case follows by continuity. This last real paravector has **determinant one** and represents four-velocity in Section 6.

4.3 Cayley transform

$\mathcal{Y} = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$ is taken to be real, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and $0 < \|\mathbf{Y}\| < b$. The **real unit vector** $\mathbf{u} = \mathbf{Y}/\|\mathbf{Y}\|$ is used, with $\mathbf{\Pi}_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Its sigma conjugate $\mathcal{Y}^* = b - \mathbf{Y} \cdot \boldsymbol{\sigma} = \text{adj}(\mathcal{Y})$ commutes with $\mathcal{Y}$, so

$$\frac{\mathcal{Y}}{\mathcal{Y}^*} = \frac{\mathcal{Y}^2}{\det(\mathcal{Y})} = \frac{b^2 + \mathbf{Y}^2 + 2b\mathbf{Y} \cdot \boldsymbol{\sigma}}{b^2 - \mathbf{Y}^2}.$$

The real parameter $\theta = 2 \tanh^{-1}(\|\mathbf{Y}\|/b)$ is defined. By the double-angle identities,

$$\begin{aligned}\cosh \theta &= \frac{b^2 + \mathbf{Y}^2}{b^2 - \mathbf{Y}^2}, \\ \sinh \theta &= \frac{2b\|\mathbf{Y}\|}{b^2 - \mathbf{Y}^2}.\end{aligned}$$

Thus the quotient is the full-rapidity exponential

$$\begin{aligned}\frac{b + \mathbf{Y} \cdot \boldsymbol{\sigma}}{b - \mathbf{Y} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta \boldsymbol{\Pi}_+} + e^{-\theta \boldsymbol{\Pi}_-}.\end{aligned}\tag{4.17}$$

Equation (4.17) is $\mathcal{L}(\theta, \mathbf{u})^{-2}$ from (4.16). Its real exponent is written using $\mathbf{X} = \theta \mathbf{u}$, while the Cayley vector is determined by

$$\mathbf{Y} = b \tanh\left(\frac{\|\mathbf{X}\|}{2}\right) \frac{\mathbf{X}}{\|\mathbf{X}\|}.$$

Thus $\|\mathbf{Y}\|/b = \tanh(\theta/2)$, the half-rapidity parameter. The corresponding real Lorentz factor and velocity vector are

$$\begin{aligned}\gamma &= \frac{b^2 + \mathbf{Y}^2}{b^2 - \mathbf{Y}^2}, \\ \mathbf{v} &= \frac{2b\mathbf{Y}}{b^2 + \mathbf{Y}^2}.\end{aligned}$$

They give the equivalent forms

$$\exp(\mathbf{X} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{Y} \cdot \boldsymbol{\sigma}}{b - \mathbf{Y} \cdot \boldsymbol{\sigma}} = \frac{\mathcal{Y}}{\mathcal{Y}^*}.$$

Choosing $b = 1$ normalizes the scalar part of $\mathcal{Y}$ to one; $\mathbf{Y} = 0$ follows by continuity.

4.4 Composing the factors

Notation: $\mathcal{L}(\theta, \mathbf{u}) = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $\mathcal{R}(\theta, \mathbf{u}) = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6) and (4.5). For a fixed real unit direction $\mathbf{u}$ and real parameters θ, θ' , the parameters can be added because the exponents commute:

$$\mathcal{L}(\theta, \mathbf{u})\mathcal{L}(\theta', \mathbf{u}) = \mathcal{L}(\theta + \theta', \mathbf{u}),\tag{4.18}$$

$$\begin{aligned}\mathcal{R}(\theta, \mathbf{u})\mathcal{R}(\theta', \mathbf{u}) &= \mathcal{R}(\theta + \theta', \mathbf{u}), \\ \mathcal{L}(\theta, \mathbf{u})\mathcal{R}(\theta', \mathbf{u}) &= \mathcal{R}(\theta', \mathbf{u})\mathcal{L}(\theta, \mathbf{u}).\end{aligned}$$

For a common axis, the following complex parameter is defined: $z = -(\theta + i\theta')/2 \in \mathbb{C}$. Its exponential is

$$\exp(z \mathbf{u} \cdot \boldsymbol{\sigma}) = \mathcal{L}(\theta, \mathbf{u})\mathcal{R}(\theta', \mathbf{u}) = \cosh z + \sinh z \mathbf{u} \cdot \boldsymbol{\sigma}.$$

By expansion of the complex scalar coefficients,

$$\begin{aligned}\cosh z &= +\cosh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \sinh \frac{\theta}{2} \sin \frac{\theta'}{2}, \\ \sinh z &= -\sinh \frac{\theta}{2} \cos \frac{\theta'}{2} - i \cosh \frac{\theta}{2} \sin \frac{\theta'}{2}.\end{aligned}$$

This factorization is valid when the real and imaginary vector parts of the exponent are parallel. For different real unit directions $\mathbf{u}, \mathbf{v}$, the vector product instead gives

$$\left[-\frac{1}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, -\frac{1}{2}\mathbf{v} \cdot \boldsymbol{\sigma}\right] = \frac{i}{2}(\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.\tag{4.19}$$

This commutator has the imaginary-vector form used in the exponent of a rotation; boosts along nonparallel directions cannot be interchanged.

4.5 Logarithm

For complex $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ with $\mathbf{Z}^2 \neq 0$, the following notation is retained: $z = +\sqrt{\mathbf{Z}^2}$, the vector of square one $\mathbf{u} = \mathbf{Z}/z$, $\lambda_{\pm} = c \pm z$, and $\boldsymbol{\Pi}_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$. Both eigenvalues are assumed to be nonzero. For a nonzero complex scalar λ and any integer n , the scalar branches are

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi).$$

One branch is chosen for each eigenvalue. By the spectral rule,

$$\begin{aligned} \log \mathbf{Z} &= \log(\lambda_+) \boldsymbol{\Pi}_+ + \log(\lambda_-) \boldsymbol{\Pi}_- \\ &= \frac{1}{2}(\log \lambda_+ + \log \lambda_-) + \frac{1}{2}(\log \lambda_+ - \log \lambda_-) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (4.20)$$

Exchanging z and $-z$ exchanges both eigenvalue–projector pairs and preserves the same chosen logarithm. Independent scalar branches give the different logarithms.

For a real unit vector $\mathbf{u}$, (4.20) uses eigenvalues 1 and -1 , giving, for integers n, m ,

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \boldsymbol{\Pi}_+ + (2m+1)\pi i \boldsymbol{\Pi}_-.$$

For real $\mathbf{Y} = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$, $\mathbf{Y} \in \mathbb{R}^3$, and $b > \|\mathbf{Y}\|$, both eigenvalues are positive. Combining their real logarithms yields

$$\log \mathbf{Y} = \frac{1}{2} \log(b^2 - \mathbf{Y}^2) + \tanh^{-1} \left(\frac{\|\mathbf{Y}\|}{b} \right) \frac{\mathbf{Y}}{\|\mathbf{Y}\|} \cdot \boldsymbol{\sigma}.$$

The vector term vanishes at $\mathbf{Y} = 0$; the general coalescing case was treated above.

5 Transformations and invariants

Here $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is a complex paravector: $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$, hence $\mathbf{Z} \in \mathbb{C} \oplus \mathbb{C}^3$. Transformation factors $\mathcal{T}, \mathcal{T}' \in \mathbb{C} \oplus \mathbb{C}^3$ are constant. A distinction is made between the scalar part $\text{sc}(\mathbf{Z}) = c = \frac{1}{2} \text{tr} \mathbf{Z}$ from (1.9), the complex determinant $c^2 - \mathbf{Z}^2$ from (1.10), and the squared norm $\text{sc}(\mathbf{Z}^H \mathbf{Z})$ from (2.14). Scalar part and trace carry exactly the same information; sc is used below.

5.1 Similarity, determinant, and norm

For a complex invertible paravector $\mathcal{T}$, similarity means $\mathbf{Z}' = \mathcal{T} \mathbf{Z} \mathcal{T}^{-1}$. By scalar-part cyclicity,

$$\text{sc}(\mathbf{Z}') = \text{sc}(\mathcal{T} \mathbf{Z} \mathcal{T}^{-1}) = \text{sc}(\mathbf{Z} \mathcal{T}^{-1} \mathcal{T}) = \text{sc}(\mathbf{Z}).$$

Thus trace is also preserved. By multiplicativity,

$$\det \mathbf{Z}' = \det(\mathcal{T}) \det(\mathbf{Z}) \det \mathcal{T}^{-1} = \det(\mathbf{Z}). \quad (5.1)$$

The opposite similarity $\mathbf{Z}' = \mathcal{T}^{-1} \mathbf{Z} \mathcal{T}$ has the same invariants. For independent complex factors $\mathcal{T}, \mathcal{T}'$, the law $\mathbf{Z}' = \mathcal{T}' \mathbf{Z} \mathcal{T}$ instead gives

$$\det \mathbf{Z}' = \det(\mathcal{T}') \det(\mathcal{T}) \det(\mathbf{Z}).$$

This preserves the determinant for every $\mathbf{Z}$ exactly when $\det(\mathcal{T}') \det \mathcal{T} = 1$. This alone does not preserve scalar part, norm, or the real paravector subspace.

If the factor $\mathcal{T}$ is unitary, then $\mathcal{T}^H = \mathcal{T}^{-1}$ by (2.4). For $\mathcal{Z}' = \mathcal{T}\mathcal{Z}\mathcal{T}^H$, cancellation and scalar-part cyclicity give

$$\begin{aligned}\|\mathcal{Z}'\|^2 &= \text{sc}\left(\mathcal{T}\mathcal{Z}^H\mathcal{T}^H\mathcal{T}\mathcal{Z}\mathcal{T}^H\right) \\ &= \text{sc}\left(\mathcal{Z}^H\mathcal{Z}\mathcal{T}^H\mathcal{T}\right) = \|\mathcal{Z}\|^2.\end{aligned}$$

The norm is also preserved by $\mathcal{Z}' = \mathcal{T}\mathcal{Z}$ and $\mathcal{Z}' = \mathcal{Z}\mathcal{T}$. The congruence additionally preserves scalar part and determinant; a one-sided unitary product need not.

5.1.1 Determinant one is not Hermitian unitarity

Invertibility, determinant one, and Hermitian unitarity are distinct conditions. Invertibility means that a two-sided inverse exists, as in (2.3); determinant one means $\det \mathcal{Z} = 1$; and Hermitian unitarity means $\mathcal{Z}^H\mathcal{Z} = 1$, as in (2.4). The phrase *unit determinant* means only that the determinant equals one. It does not mean unit norm or Hermitian unitarity.

Determinant one implies invertibility, but does not imply Hermitian unitarity. For comparison, a real paravector $\mathcal{X} = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ with $a \in \mathbb{R}$, $\mathbf{X} \in \mathbb{R}^3$, and $a^2 - \mathbf{X}^2 = 1$ is considered. Then

$$\mathcal{X}^*\mathcal{X} = \det \mathcal{X} = 1, \quad \mathcal{X}^{-1} = \mathcal{X}^* = a - \mathbf{X} \cdot \boldsymbol{\sigma}.$$

Thus determinant one implies invertibility. Hermitian conjugation fixes $\mathcal{X}$, whereas sigma conjugation reverses its vector part, so

$$\begin{aligned}\mathcal{X}^H\mathcal{X} &= \mathcal{X}^2 = a^2 + \mathbf{X}^2 + 2a\mathbf{X} \cdot \boldsymbol{\sigma}, \\ \|\mathcal{X}\|^2 &= a^2 + \mathbf{X}^2 = 1 + 2\mathbf{X}^2.\end{aligned}$$

Both equal 1 only when $\mathbf{X} = 0$, in which case $\mathcal{X} = \pm 1$. Choosing any nonzero real $\mathbf{X}$ and $a = \sqrt{1 + \mathbf{X}^2}$ therefore gives an invertible paravector of determinant one that is not unitary. These properties follow from the defining equations, not from the letter used for the paravector.

5.2 Real paravectors and the Lorentz metric

$\mathcal{X} = a + \mathbf{X} \cdot \boldsymbol{\sigma}$ is taken to be real, with $a \in \mathbb{R}$ and $\mathbf{X} \in \mathbb{R}^3$. The determinant and squared norm give different quadratic forms:

$$\begin{aligned}\det \mathcal{X} &= a^2 - \mathbf{X}^2, \\ \|\mathcal{X}\|^2 &= a^2 + \mathbf{X}^2.\end{aligned}$$

The first is the Lorentz quadratic form of signature $(+, -, -, -)$; the second is the positive sum of squared coefficients. A real-linear isometry of the first form must preserve the real subspace and its determinant. For a constant complex invertible paravector $\mathcal{T}$ with $\det \mathcal{T} = 1$, the following Hermitian congruence is used (1, Eq. (17))

$$\boxed{\mathcal{X}' = \mathcal{T}\mathcal{X}\mathcal{T}^H.} \tag{5.2}$$

Hermitian conjugation and determinant multiplication give

$$\begin{aligned}\mathcal{X}'^H &= \mathcal{T}\mathcal{X}^H\mathcal{T}^H = \mathcal{X}', \\ \det \mathcal{X}' &= \det(\mathcal{T}) \det(\mathcal{X}) \det \mathcal{T}^H = \det(\mathcal{X}).\end{aligned}$$

For a second real paravector $\mathbf{y} = b + \mathbf{Y} \cdot \boldsymbol{\sigma}$, with $b \in \mathbb{R}$ and $\mathbf{Y} \in \mathbb{R}^3$, the associated metric pairing is (1, Eq. (15))

$$\text{sc}(\mathbf{x}^* \mathbf{y}) = ab - \mathbf{X} \cdot \mathbf{Y} = \frac{1}{2}(\det(\mathbf{x} + \mathbf{y}) - \det(\mathbf{x}) - \det(\mathbf{y})). \quad (5.3)$$

Linearity and determinant invariance therefore give $\text{sc}(\mathbf{x}'^* \mathbf{y}') = \text{sc}(\mathbf{x}^* \mathbf{y})$: the full metric is preserved. These maps preserve the positive cone as well. For $a \geq \|\mathbf{X}\|$, the eigenvalues $a \pm \|\mathbf{X}\|$ are nonnegative, so $\mathbf{x}$ has a real square root and

$$\text{sc}(\mathbf{x}') = \|\sqrt{\mathbf{x}} \mathcal{T}^H\|^2 \geq 0.$$

Together with determinant invariance, this keeps $\mathbf{x}'$ in the cone $a \geq \|\mathbf{X}\|$. These are the proper Lorentz transformations preserving that cone; $\mathcal{T}$ and $-\mathcal{T}$ give the same map.

Why the scalar exponent vanishes. More generally, the congruence multiplies $\det(\mathbf{x})$ by $\|\det(\mathcal{T})\|^2$. For real $\mathcal{T} = \exp(a + \mathbf{X} \cdot \boldsymbol{\sigma})$, (4.4) therefore gives

$$\det(\mathcal{T} \mathbf{x} \mathcal{T}^H) = e^{4a} \det(\mathbf{x}).$$

A nonzero real scalar exponent dilates the metric; it is excluded by determinant invariance. An imaginary scalar exponent contributes a phase e^{ib} , $b \in \mathbb{R}$, which cancels between $\mathcal{T}$ and $\mathcal{T}^H$. Requiring $\det \mathcal{T} = 1$ removes this continuous phase freedom. Thus a pure boost needs only a real vector exponent.

Why the squared norm is different. For $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real θ and real unit $\mathbf{u}$, applying (5.2) to $\mathbf{x} = 1$ gives

$$\begin{aligned} \mathbf{x}' &= \mathcal{L}^2 = \cosh \theta - \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \det \mathbf{x}' &= 1, \\ \text{sc}(\mathbf{x}') &= \cosh \theta, \\ \|\mathbf{x}'\|^2 &= \cosh(2\theta). \end{aligned}$$

A nonzero boost preserves the determinant while changing the scalar part and squared norm. In fact, a determinant-one congruence preserves the squared norm for every real $\mathbf{x}$ exactly when $\mathcal{T}$ is unitary. Necessity follows by taking $\mathbf{x} = 1$: the real, positive $\mathcal{T} \mathcal{T}^H$ can have both determinant and norm one only if it is 1, which makes $\mathcal{T}$ unitary. Sufficiency was proved above. Normalizing $\mathcal{L}$ to $\|\mathcal{L}\| = 1$ would change its determinant and lose the Lorentz metric.

Every factor separates into a boost and a rotation. For any complex $\mathcal{T}$, the Hermitian product rule gives

$$(\mathcal{T} \mathcal{T}^H)^H = (\mathcal{T}^H)^H \mathcal{T}^H = \mathcal{T} \mathcal{T}^H.$$

By (1.7), a paravector equals its Hermitian conjugate exactly when its scalar and vector coefficients are real. Thus $\mathcal{T} \mathcal{T}^H \in \mathbb{R} \oplus \mathbb{R}^3$, without any determinant constraint. For a determinant-one complex $\mathcal{T}$, i.e., $\det(\mathcal{T}) = 1$, this real paravector also has determinant one and positive scalar part $\|\mathcal{T}\|^2$. Both eigenvalues are therefore positive. Using the positive real square root,

$$\mathcal{L} = \sqrt{\mathcal{T} \mathcal{T}^H}, \quad \mathcal{R} = \mathcal{L}^{-1} \mathcal{T}, \quad \mathcal{T} = \mathcal{L} \mathcal{R}.$$

The factors satisfy

$$\det \mathcal{L} = \det \mathcal{R} = 1, \quad \mathcal{L}^H = \mathcal{L}, \quad \mathcal{R}^H = \mathcal{R}^{-1},$$

with $\mathcal{L} \in \mathbb{R} \oplus \mathbb{R}^3$ and $\mathcal{R} \in \mathbb{R} \oplus i\mathbb{R}^3$. The real logarithm of $\mathcal{L}$ has zero scalar part by (4.4), so $\mathcal{L}$ is a real vector exponential. For $\mathcal{R}$, the identity $\mathcal{R}^H = \mathcal{R}^{-1} = \text{adj}(\mathcal{R})$ makes its scalar coefficient real and its vector coefficient imaginary; determinant one then gives $\mathcal{R} = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}$ for some real θ and real unit $\mathbf{u}$. Consequently,

$$\mathcal{X}' = \mathcal{T} \mathcal{X} \mathcal{T}^H = \mathcal{L} \mathcal{R} \mathcal{X} \mathcal{R}^H \mathcal{L}, \quad \det \mathcal{X}' = \det \mathcal{X}.$$

The rotation is followed by the boost. Their component actions follow next.

5.3 Rotation action and Rodrigues' formula

Notation: the unitary factor $\mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.5), with real angle θ and real unit axis $\mathbf{u}$. For the complex paravector $\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, the following Hermitian congruence is used (1, Eq. (5))

$$\boxed{\mathcal{Z}' = \mathcal{R} \mathcal{Z} \mathcal{R}^H.} \quad (5.4)$$

Since $\mathcal{R}^H = \mathcal{R}^{-1}$ by (4.8), this keeps $c' = c$. The vector product (2.8) gives

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{Z} \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2(\mathbf{u} \cdot \mathbf{Z})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{Z} \cdot \boldsymbol{\sigma}, \\ [\mathbf{Z} \cdot \boldsymbol{\sigma}, \mathbf{u} \cdot \boldsymbol{\sigma}] &= -2i(\mathbf{u} \times \mathbf{Z}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Inserting the sine and cosine forms of $\mathcal{R}$ and $\mathcal{R}^H$ therefore yields

$$\begin{aligned} \mathbf{Z}' \cdot \boldsymbol{\sigma} &= \cos^2 \frac{\theta}{2} \mathbf{Z} \cdot \boldsymbol{\sigma} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} (\mathbf{u} \times \mathbf{Z}) \cdot \boldsymbol{\sigma} \\ &\quad + \sin^2 \frac{\theta}{2} (2(\mathbf{u} \cdot \mathbf{Z})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{Z} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Rodrigues' formula is obtained by collecting terms (5, p. 14):

$$\boxed{\mathcal{Z}' = \mathbf{Z} \cos \theta + (\mathbf{u} \times \mathbf{Z}) \sin \theta + (\mathbf{u} \cdot \mathbf{Z})(1 - \cos \theta)\mathbf{u}.} \quad (5.5)$$

This is the active right-handed rotation through $+\theta$ about $\mathbf{u}$. Rotating the coordinate axes through $+\theta$ instead uses the inverse map. It preserves sc, determinant, norm, and the real paravector subspace. For complex $\mathbf{Z}$, it rotates its real and imaginary parts by the same real rotation. The sign change in (4.14) affects both factors in (5.4), so $\mathcal{Z}'$ has period 2π .

5.4 Quaternions as paravectors

The quaternion multiplication rules are satisfied by $(-i\sigma_k)$:

$$\begin{aligned} (-i\sigma_1)^2 &= (-i\sigma_2)^2 = (-i\sigma_3)^2 = (-i\sigma_1)(-i\sigma_2)(-i\sigma_3) = -1, \\ (-i\sigma_1)(-i\sigma_2) &= (-i\sigma_3), \\ (-i\sigma_2)(-i\sigma_3) &= (-i\sigma_1), \\ (-i\sigma_3)(-i\sigma_1) &= (-i\sigma_2). \end{aligned}$$

The sign is reversed when two distinct factors are exchanged. Here i remains the commuting complex scalar.

For $a \in \mathbb{R}$ and $\mathbf{Q} \in \mathbb{R}^3$, the quaternion representation is (3, p. 92)

$$\mathbf{Q} := a + \mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = a - i\mathbf{Q} \cdot \boldsymbol{\sigma}. \quad (5.6)$$

Here $\mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = Q_1(-i\sigma_1) + Q_2(-i\sigma_2) + Q_3(-i\sigma_3)$. In the decomposition $\mathbf{Z} = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}$, $b = 0$, $\mathbf{X} = 0$, and $\mathbf{Y} = -\mathbf{Q}$ are selected. Thus $\mathbf{Q}, \mathbf{Q}' \in \mathbb{R} \oplus i\mathbb{R}^3$: a real scalar coefficient and purely imaginary vector coefficients. This four-real-component form is preserved under addition and multiplication. Indeed, for $\mathbf{Q}' = a' + \mathbf{Q}' \cdot (-i\boldsymbol{\sigma})$, the paravector product becomes (6, p. 9)

$$\mathbf{Q}\mathbf{Q}' = (aa' - \mathbf{Q} \cdot \mathbf{Q}') + (a\mathbf{Q}' + a'\mathbf{Q} + \mathbf{Q} \times \mathbf{Q}') \cdot (-i\boldsymbol{\sigma}). \quad (5.7)$$

Since $(-i\sigma_k)^H = -(-i\sigma_k)$ and $(-i\sigma_k)^* = (-i\sigma_k)$, quaternion conjugation is the existing Hermitian conjugation:

$$\begin{aligned} \mathbf{Q}^H &= a - \mathbf{Q} \cdot (-i\boldsymbol{\sigma}) = a + i\mathbf{Q} \cdot \boldsymbol{\sigma}, \\ \mathbf{Q}^* &= \mathbf{Q}, \quad \text{adj } \mathbf{Q} = \mathbf{Q}^H, \\ \mathbf{Q}\mathbf{Q}^H &= \mathbf{Q}^H\mathbf{Q} = a^2 + \mathbf{Q}^2 = \det \mathbf{Q} = \|\mathbf{Q}\|^2. \end{aligned} \quad (5.8)$$

For nonzero $\mathbf{Q}$, (5.8) gives $\mathbf{Q}^{-1} = \mathbf{Q}^H/(a^2 + \mathbf{Q}^2)$. Nonzero quaternions are normalized with

$$\mathbf{Q} \leftarrow \frac{\mathbf{Q}}{\sqrt{\mathbf{Q}\mathbf{Q}^H}},$$

which, with $\|\mathbf{Q}\|^2 = \mathbf{Q}^2$ for real $\mathbf{Q}$, gives

$$\mathbf{Q}\mathbf{Q}^H = a^2 + \|\mathbf{Q}\|^2 = 1.$$

Thus a unit quaternion has $\mathbf{Q}^{-1} = \mathbf{Q}^H$.

For rotation, $\mathbf{r} \in \mathbb{R}^3$ is represented by the pure quaternion $\mathbf{r} \cdot (-i\boldsymbol{\sigma}) = -i\mathbf{r} \cdot \boldsymbol{\sigma}$. For a unit quaternion $\mathbf{Q} = a + \mathbf{Q} \cdot (-i\boldsymbol{\sigma})$, the quaternion sandwich is used:

$$\mathbf{r}' \cdot (-i\boldsymbol{\sigma}) = \mathbf{Q}(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))\mathbf{Q}^H, \quad \mathbf{r}' \cdot \boldsymbol{\sigma} = \mathbf{Q}(\mathbf{r} \cdot \boldsymbol{\sigma})\mathbf{Q}^H. \quad (5.9)$$

The two forms agree because $-i$ commutes with every paravector. Applying (5.7) twice gives

$$\begin{aligned} \mathbf{Q}(\mathbf{r} \cdot (-i\boldsymbol{\sigma})) &= -\mathbf{Q} \cdot \mathbf{r} + (a\mathbf{r} + \mathbf{Q} \times \mathbf{r}) \cdot (-i\boldsymbol{\sigma}), \\ \mathbf{Q}(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))\mathbf{Q}^H &= ((a^2 - \mathbf{Q}^2)\mathbf{r} + 2(\mathbf{Q} \cdot \mathbf{r})\mathbf{Q} + 2a\mathbf{Q} \times \mathbf{r}) \cdot (-i\boldsymbol{\sigma}). \end{aligned}$$

The scalar part cancels. For a unit quaternion with $\mathbf{Q} \neq 0$, its real unit axis $\mathbf{u}$ and rotation angle θ satisfy

$$\mathbf{u} = \frac{\mathbf{Q}}{\|\mathbf{Q}\|}, \quad a = \cos \frac{\theta}{2}, \quad \mathbf{Q} = \mathbf{u} \sin \frac{\theta}{2}.$$

Thus the quaternion is the rotation factor

$$\mathbf{Q} = \cos \frac{\theta}{2} - i\mathbf{u} \cdot \boldsymbol{\sigma} \sin \frac{\theta}{2} = \mathcal{R}.$$

Then $a^2 - \mathbf{Q}^2 = \cos \theta$, $2a\mathbf{Q} = \mathbf{u} \sin \theta$, and $2(\mathbf{Q} \cdot \mathbf{r})\mathbf{Q} = (1 - \cos \theta)(\mathbf{u} \cdot \mathbf{r})\mathbf{u}$. Consequently (5.9) recovers

$$\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \theta)\mathbf{u}.$$

This is precisely Rodrigues' formula (5.5) for $\mathbf{Z} = \mathbf{r}$. Thus $\mathcal{R} = \mathbf{Q}$, and the quaternion sandwich is exactly $\mathcal{R}(\mathbf{r} \cdot \boldsymbol{\sigma})\mathcal{R}^H$ from (5.4). The reversed sandwich $\mathbf{Q}^H(\mathbf{r} \cdot \boldsymbol{\sigma})\mathbf{Q}$ gives the inverse rotation.

5.5 Boost action and its sign convention

Notation: $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ from (4.6); $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. The same congruence is used with $\mathcal{T} = \mathcal{L}$ and $\mathcal{L}^H = \mathcal{L}$ from (4.9) (5, Eq. (24)):

$$\boxed{\mathcal{Z}' = \mathcal{L} \mathcal{Z} \mathcal{L}} \quad (5.10)$$

The paravector $\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ may be complex; the congruence also preserves its real subspace. $\mathbf{Z} \in \mathbb{C}^3$ is resolved into parallel and perpendicular parts in $\mathbb{C}^3$:

$$\begin{aligned} \mathbf{Z}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{Z}) \mathbf{u}, \\ \mathbf{Z}_{\perp} &:= \mathbf{Z} - \mathbf{Z}_{\parallel}. \end{aligned}$$

The parallel term commutes with $\mathbf{u} \cdot \boldsymbol{\sigma}$; the perpendicular term anticommutes with it. Consequently,

$$\begin{aligned} \mathcal{L}(c + \mathbf{Z}_{\parallel} \cdot \boldsymbol{\sigma}) \mathcal{L} &= (c + \mathbf{Z}_{\parallel} \cdot \boldsymbol{\sigma}) \mathcal{L}^2, \\ \mathcal{L}(\mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}) \mathcal{L} &= \mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Using $\mathcal{L}^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma})$ from (4.16), multiplication gives (7, Eq. (5.1))

$$\boxed{c' = \gamma(c - \beta \mathbf{u} \cdot \mathbf{Z})}, \quad (5.11)$$

$$\boxed{\mathbf{Z}' = \mathbf{Z}_{\perp} + \gamma(\mathbf{Z}_{\parallel} - \beta c \mathbf{u})}. \quad (5.12)$$

The equivalent longitudinal combinations expose the invariant directly:

$$c' \pm \mathbf{u} \cdot \mathbf{Z}' = e^{\mp \theta} (c \pm \mathbf{u} \cdot \mathbf{Z}). \quad (5.13)$$

Their product is $c^2 - (\mathbf{u} \cdot \mathbf{Z})^2$; the transverse part is unchanged. Thus (5.13) preserves $c^2 - \mathbf{Z}^2$ and, for real $\mathcal{Z}$, the null cone $c^2 - \mathbf{Z}^2 = 0$.

The inverse map is $\mathcal{Z} = \mathcal{L}^{-1} \mathcal{Z}' \mathcal{L}^{-1}$, obtained by reversing θ . Generally $\mathcal{L} \mathcal{Z} \neq \mathcal{Z} \mathcal{L}$: the scalar and parallel parts commute with $\mathcal{L}$, while a nonzero perpendicular part does not for a nonzero boost. For any transverse vector $\mathbf{Z}_{\perp}$ with $\mathbf{u} \cdot \mathbf{Z}_{\perp} = 0$,

$$\mathcal{L}(\mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}) = (\mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}) \mathcal{L}^{-1}, \quad \mathcal{L}(\mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}) \mathcal{L} = \mathbf{Z}_{\perp} \cdot \boldsymbol{\sigma}.$$

Thus the two-sided action leaves every transverse component unchanged. For real inputs, the right factor cancels the imaginary transverse term introduced by the left factor. A single left factor, even at full rapidity, cannot represent the boost of every real paravector.

Similarity action of the boost factor. For $\mathcal{Z}' = \mathcal{L} \mathcal{Z} \mathcal{L}^{-1}$, the same commutation rules give $c' = c$ and

$$\mathbf{Z}' = \mathbf{Z}_{\parallel} + \gamma(\mathbf{Z}_{\perp} - i\beta \mathbf{u} \times \mathbf{Z}). \quad (5.14)$$

This preserves scalar part and determinant, but fixes $\mathcal{Z} = 1$ and generally makes a real vector complex. It does not mix scalar and vector components as the congruence does in (5.11)–(5.12). In particular, similarity preserves the scalar-free subspace; a nonzero boost congruence gives a nonzero scalar part to $\mathcal{Z} = z \mathbf{u} \cdot \boldsymbol{\sigma}$ for any nonzero $z \in \mathbb{C}$.

6 Space–time kinematics

In an inertial frame, an event is $\mathbf{x} = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$. A massive particle is described by a future-directed time-like path $\mathbf{r}(t)$, with coordinate velocity $\mathbf{v} = \dot{\mathbf{r}}$ and $v^2 < 1$. Coordinate time is denoted by t ; the carried clock’s proper time is represented by s in the natural units used here. Dots denote total t derivatives of scalars, vectors, and paravectors, even when a path is parametrized by s .

6.1 Intervals, light cones, and proper time

The displacement and its Minkowski interval are (5, Eq. (7))

$$\begin{aligned} d\mathbf{x} &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\ \det(d\mathbf{x}) &= d\mathbf{x} d\mathbf{x}^* = dt^2 - d\mathbf{r}^2. \end{aligned} \quad (6.1)$$

The interval is positive, zero, or negative for time-like, null, or space-like displacements. Along a massive particle’s path it equals ds^2 ; a null path has no elapsed proper time. By (5.2), a constant complex invertible paravector $\mathcal{T}$ with $\det \mathcal{T} = 1$ gives

$$d\mathbf{x}' = \mathcal{T} d\mathbf{x} \mathcal{T}^H, \quad \det(d\mathbf{x}') = \det(d\mathbf{x}).$$

For a rotation, $\mathbf{x}' = \mathcal{R} \mathbf{x} \mathcal{R}^H$ leaves $t' = t$ and acts on $\mathbf{r}$ by Rodrigues’ formula (5.5). For real displacements, the product also gives

$$\begin{aligned} [d\mathbf{x}, d\mathbf{x}'] &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}, \\ (d\mathbf{x})^{-1} &= \frac{d\mathbf{x}^*}{\det(d\mathbf{x})}, \quad \text{if } \det(d\mathbf{x}) \neq 0. \end{aligned}$$

A null interval has no inverse.

Light cones. For a finite real event separation $\Delta\mathbf{x} = \Delta t + \Delta\mathbf{r} \cdot \boldsymbol{\sigma}$, the eigenvalues are $\Delta t \pm \|\Delta\mathbf{r}\|$. Both are nonnegative precisely in the future causal cone:

$$\Delta t \geq \|\Delta\mathbf{r}\|. \quad (6.2)$$

The future null boundary has $\det(\Delta\mathbf{x}) = 0$ and $\Delta t \geq 0$; for a real unit vector $\mathbf{u}$, its nonzero future rays are

$$\Delta\mathbf{x} = \Delta t(1 + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Delta t > 0, \quad \mathbf{u}^2 = 1.$$

The Hermitian congruence (5.2) preserves nonnegativity of the eigenvalues and the determinant, hence the cone and its boundary. For a future time-like or null path,

$$\|\Delta\mathbf{r}\| = \left\| \int \mathbf{v} dt \right\| \leq \int \|\mathbf{v}\| dt \leq \Delta t.$$

Its endpoints therefore satisfy (6.2); a space-like separation cannot carry a causal signal (9, p. 118).

Coordinate time and proper time. The coordinate t is read from synchronized clocks of the chosen inertial frame. Along a future-directed time-like path, $dt > 0$ and the positive spacetime arc-length increment $ds = \sqrt{\det(d\mathbf{x})}$ are chosen. Proper time is commonly denoted by τ .

SI units: Proper time Reverting to SI units, with s measured in metres and τ in seconds, $ds = c \, d\tau.$

With $c = 1$, the same elapsed quantity is represented by s and τ ; only s is used for proper time below. This interpretation is fixed by the positive interval normalization, not by an arbitrary choice of path parameter. Null paths have $ds = 0$ and no proper-time parametrization; space-like paths have no real proper-time increment (9, pp. 133–134).

Dividing by dt gives

$$\begin{aligned} \dot{s} &= \frac{ds}{dt} = \frac{\sqrt{dt^2 - d\mathbf{r}^2}}{dt} \\ &= \sqrt{\frac{dt^2 - d\mathbf{r}^2}{dt^2}} \\ &= \sqrt{1 - \left(\frac{d\mathbf{r}}{dt}\right)^2} \\ &= \sqrt{1 - \dot{\mathbf{r}}^2} = \sqrt{1 - \mathbf{v}^2}. \end{aligned}$$

The Lorentz factor and the derivative conversion are therefore (9, Eq. (7.18))

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{dt}{ds}, \quad (6.3)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}. \quad (6.4)$$

The coordinate-time and proper-time rates coincide only at rest in the chosen frame.

Elapsed time and clock comparison. Between coordinate times t_0 and t the clock records (9, Eq. (7.19))

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} \, dt'. \quad (6.5)$$

Fixed, time-like separated events are denoted by $\mathbf{x}_0 = \mathbf{x}(s_0)$ and $\mathbf{x}_1 = \mathbf{x}(s_1)$, with $\mathbf{x}_1$ to the future of $\mathbf{x}_0$. The decomposition $\mathbf{x}_k = t_k + \mathbf{r}_k \cdot \boldsymbol{\sigma}$ is used for $k = 0, 1$. In their separation's rest frame, $\mathbf{r}_1 = \mathbf{r}_0$ and $t_1 - t_0 = \sqrt{\det(\mathbf{x}_1 - \mathbf{x}_0)}$. Since $\sqrt{1 - \mathbf{v}^2} \leq 1$, (6.5) gives (7, p. 122)

$$s_1 - s_0 \leq \sqrt{\det(\mathbf{x}_1 - \mathbf{x}_0)}. \quad (6.6)$$

Equality requires the straight inertial path. The determinant of a finite displacement therefore does not generally equal the squared elapsed proper time of an accelerated clock.

For a return trip lasting real coordinate time $T > 0$, velocities $+\beta\mathbf{u}$ and $-\beta\mathbf{u}$ are maintained for $T/2$ each, with real unit vector $\mathbf{u}$ and $0 < \beta < 1$. The ideal instantaneous reversal is the limit of a short smooth turn. By the traveling clock, the following interval is recorded:

$$s(T) - s(0) = T\sqrt{1 - \beta^2},$$

while T is recorded by the clock at rest. This is the twin-clock comparison (9, p. 121).

6.2 Four-velocity and worldlines

The real four-velocity is the proper-time tangent (5, Eq. (8)):

$$\mathbf{u} := \frac{d\mathbf{x}}{ds} = \frac{dt}{ds} (1 + \dot{\mathbf{r}} \cdot \boldsymbol{\sigma}) = \gamma (1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.7)$$

Its normalization and conjugates are

$$\mathbf{u}^* \mathbf{u} = \det \mathbf{u} = \frac{\det(d\mathbf{x})}{ds^2} = 1, \quad \mathbf{u}^H = \mathbf{u}, \quad \mathbf{u}^{-1} = \mathbf{u}^* = \gamma (1 - \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.8)$$

Determinant one is distinct from Hermitian unitarity:

$$\mathbf{u}^H \mathbf{u} = \mathbf{u}^2 = 2\gamma^2 - 1 + 2\gamma^2 \mathbf{v} \cdot \boldsymbol{\sigma}, \quad \|\mathbf{u}\|^2 = 2\gamma^2 - 1.$$

Both equal 1 only at rest. Thus four-velocity has determinant one in every inertial frame but is Hermitian unitary only in its rest frame. For a real unit direction $\mathbf{u}$, $\mathbf{v} = \beta \mathbf{u}$ and $\theta = \tanh^{-1} \beta$ are used. Equation (4.16) gives $\mathbf{u} = \mathcal{L}^{-2}$ for $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$: the direct boost $\mathcal{L} \mathbf{u} \mathcal{L} = 1$ takes this tangent to rest.

Integrating four-velocity. Integrating $\mathbf{u} = d\mathbf{x}/ds$ between $s_0 < s_1$ gives

$$\mathbf{x}(s_1) - \mathbf{x}(s_0) = \int_{s_0}^{s_1} \mathbf{u}(s) ds. \quad (6.9)$$

Its real components are

$$\begin{aligned} t(s_1) - t(s_0) &= \int_{s_0}^{s_1} \gamma(s) ds, \\ \mathbf{r}(s_1) - \mathbf{r}(s_0) &= \int_{s_0}^{s_1} \gamma(s) \mathbf{v}(s) ds. \end{aligned}$$

The path is determined by the velocity history; an endpoint boost alone is insufficient.

6.3 Fixed-axis motion and rapidity

For a fixed real unit vector $\mathbf{u}$, the signed velocity along the axis is $\beta = \tanh \theta$, with $\mathbf{v} = \beta \mathbf{u}$ and $\gamma = \cosh \theta$. With the projectors $\mathbf{\Pi}_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ from (3.2), the four-velocity and longitudinal rest direction have equivalent hyperbolic, exponential, and projected forms:

$$\begin{aligned} \mathbf{u} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \mathbf{\Pi}_+ + e^{-\theta} \mathbf{\Pi}_-, \\ \mathcal{N} &= \sinh \theta + \cosh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= (\mathbf{u} \cdot \boldsymbol{\sigma}) \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \mathbf{\Pi}_+ - e^{-\theta} \mathbf{\Pi}_-. \end{aligned}$$

Here $\det \mathbf{u} = 1$, $\det \mathcal{N} = -1$, and $\text{sc}(\mathbf{u}^* \mathcal{N}) = 0$. The pair is an orthonormal time and spatial basis in the plane of motion. Although rapidity is given as $\theta(s)$, its coordinate-time derivatives are those of the composition $\theta(s(t))$.

6.4 Constant velocity

Subscript 0 denotes the initial event $t = s = 0$. For $\mathbf{v}(t) = \mathbf{v}_0$, $v_0^2 < 1$, and $\mathbf{r}(0) = \mathbf{r}_0$,

$$\gamma_0 = \frac{1}{\sqrt{1 - v_0^2}}, \quad \mathcal{X}(t) = t + (\mathbf{r}_0 + \mathbf{v}_0 t) \cdot \boldsymbol{\sigma}.$$

With $s(0) = 0$, the clock integral gives $s = t/\gamma_0$. Hence

$$\mathcal{X}(s) = \mathbf{r}_0 \cdot \boldsymbol{\sigma} + s\mathcal{U}_0, \quad \mathcal{U}_0 = \gamma_0 (1 + \mathbf{v}_0 \cdot \boldsymbol{\sigma}). \quad (6.10)$$

This straight time-like worldline has $\dot{\mathbf{v}} = 0$ and $\dot{\mathcal{U}} = d\mathcal{U}/ds = 0$. For any two events on this inertial path, $\Delta\mathcal{X} = \Delta s\mathcal{U}_0$ and $\det\mathcal{U}_0 = 1$ give

$$\det(\Delta\mathcal{X}) = (\Delta s)^2 = (\Delta t)^2(1 - v_0^2).$$

In the particle's rest frame, $\mathbf{v}_0 = 0$, so $\mathcal{U}_0 = 1$ and $s = t$.

7 Inertial frames and particle dynamics

Constant Lorentz transformations are applied to coordinates, clocks, and particle quantities. The proper time s defined in Section 6.1 is invariant; t, t' are frame-dependent coordinate times. Dots continue to denote total t derivatives.

7.1 Coordinates, intervals, and light cones

Notation: $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , real unit vector $\mathbf{u}$, $\beta = \tanh \theta$, and $\gamma = \cosh \theta$. Here γ is the factor for frame velocity $\beta \mathbf{u}$. The gamma form (4.15) and its square (4.16) are the same exponential at half and full rapidity. For real $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, coincident inertial origins are chosen at $t = t' = 0$. Equation (5.2) with $\mathcal{T} = \mathcal{L}$ gives $\mathcal{X}' = \mathcal{L}\mathcal{X}\mathcal{L}$: the primed frame moves at $+\beta \mathbf{u}$. Equations (5.11)–(5.12) become (7, Eq. (5.1))

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (7.1)$$

$$\mathbf{r}' = \mathbf{r}_\perp + \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \quad (7.2)$$

where $\mathbf{r}_\parallel = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ and $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$. In particular, $\mathbf{r} = \beta t \mathbf{u}$ maps to $\mathbf{r}' = 0$, fixing the physical meaning of the sign. Reversing θ as in (4.7) gives $\mathcal{X} = \mathcal{L}^{-1}\mathcal{X}'\mathcal{L}^{-1}$.

Specializing (5.13) to the real event $\mathcal{X}$ gives (9, p. 154)

$$t' \pm \mathbf{u} \cdot \mathbf{r}' = e^{\mp\theta}(t \pm \mathbf{u} \cdot \mathbf{r}). \quad (7.3)$$

Multiplying the two signs in (7.3) and using $\mathbf{r}'_\perp = \mathbf{r}_\perp$ gives

$$t'^2 - \mathbf{r}'^2 = (t' + \mathbf{u} \cdot \mathbf{r}')(t' - \mathbf{u} \cdot \mathbf{r}') - \mathbf{r}'_\perp^2 = t^2 - \mathbf{r}^2.$$

Thus the exponential stretches one light-cone coordinate and compresses the other reciprocally. The same identity holds for event differences; in particular,

$$dt'^2 - d\mathbf{r}'^2 = \det(d\mathcal{X}') = \det(d\mathcal{X}) = dt^2 - d\mathbf{r}^2. \quad (7.4)$$

Equation (7.4) preserves the interval (6.1) and light cones in every direction. Future causal increments stay future because $dt' = \gamma(dt - \beta \mathbf{u} \cdot d\mathbf{r}) > 0$ when $dt > 0$ and $dt \geq \|d\mathbf{r}\|$ (9, p. 118).

Along a future-directed time-like path, $ds^2 = dt'^2 - d\mathbf{r}'^2$. With $\mathbf{v}' = d\mathbf{r}'/dt'$ and particle Lorentz factor $\gamma' = dt'/ds = (1 - \mathbf{v}'^2)^{-1/2}$, the squared four-velocity norm is

$$\begin{aligned} \left\| \frac{d\mathbf{x}'}{ds} \right\|^2 &= \left(\frac{dt'}{ds} \right)^2 + \left(\frac{d\mathbf{r}'}{ds} \right)^2 \\ &= 2 \left(\frac{dt'}{ds} \right)^2 - 1 \\ &= 2\gamma'^2 - 1 = \frac{1 + \mathbf{v}'^2}{1 - \mathbf{v}'^2}. \end{aligned}$$

Likewise, in the unprimed frame,

$$\left\| \frac{d\mathbf{x}}{ds} \right\|^2 = 2 \left(\frac{dt}{ds} \right)^2 - 1 = \frac{1 + \mathbf{v}^2}{1 - \mathbf{v}^2}.$$

Each norm equals 1 only at rest in its corresponding frame, whereas $\det(d\mathbf{x}'/ds) = 1$ in every inertial frame. Proper time and causality are determined by the determinant; these Lorentz transformations would not be obtained by preserving the squared norm.

7.2 Spacetime increments and velocity

Notation: the real four-velocity $\mathbf{u} = d\mathbf{x}/ds$, an invertible paravector with $\mathbf{u}\mathbf{u}^{-1} = 1$, along a real world line $\mathbf{x}(s)$. For constant $\mathcal{L}$, differentiation gives

$$d\mathbf{x}' = \mathcal{L} d\mathbf{x} \mathcal{L}. \quad (7.5)$$

Because ds is invariant, $\mathbf{u}' = \mathcal{L}\mathbf{u}\mathcal{L}$. At a point where $\mathbf{u} = \mathcal{L}^{-2} = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma})$, this gives $\mathbf{u}' = 1$ and $d\mathbf{x}' = ds$: the boost takes the particle's instantaneous tangent to rest. For varying velocity, a different $\mathcal{L}$ is needed at each point; Section 11.4 supplies its derivative terms.

The particle velocity $\mathbf{v} = \dot{\mathbf{r}}$ is resolved along $\mathbf{u}$. Dividing the spatial increment by $dt' = \gamma(1 - \beta \mathbf{u} \cdot \mathbf{v}) dt$ gives (9, Eq. (7.22))

$$\boxed{\mathbf{v}' = \frac{\mathbf{v}_\perp/\gamma + \mathbf{v}_\parallel - \beta \mathbf{u}}{1 - \beta \mathbf{u} \cdot \mathbf{v}}}. \quad (7.6)$$

By interval invariance,

$$1 - \mathbf{v}'^2 = \frac{1 - \mathbf{v}^2}{\gamma^2(1 - \beta \mathbf{u} \cdot \mathbf{v})^2}. \quad (7.7)$$

Thus (7.6) preserves subluminal motion and light speed. Since $\mathcal{L}$ is constant, the same transformation applies to the integrated worldline in (6.9).

7.3 Clock rates and measured lengths

For a clock at rest in the primed frame, (7.1) uses $d\mathbf{r} = \beta \mathbf{u} dt$, hence (9, p. 119)

$$dt' = \gamma(1 - \beta^2) dt = dt/\gamma. \quad (7.8)$$

Here $dt' = ds$; finite elapsed time follows from (6.5).

For a rod at rest in the unprimed frame, the fixed endpoint separation is denoted by $\Delta \mathbf{r} \in \mathbb{R}^3$. A simultaneous measurement in the primed frame requires $\Delta t' = 0$, so $\Delta t = \beta \mathbf{u} \cdot \Delta \mathbf{r}$. Substituting in (7.2) gives (9, p. 123)

$$\Delta \mathbf{r}'_{\parallel} = \Delta \mathbf{r}_{\parallel} / \gamma. \quad (7.9)$$

The transverse separation stays unchanged; thus a rod of any orientation has

$$\|\Delta \mathbf{r}'\|^2 = \|\Delta \mathbf{r}_{\perp}\|^2 + \|\Delta \mathbf{r}_{\parallel}\|^2 / \gamma^2.$$

Longitudinal contraction (7.9) therefore compares endpoint events at equal primed times. Equal unprimed times instead give $\Delta t' = -\gamma \beta \mathbf{u} \cdot \Delta \mathbf{r}$; simultaneity is frame-dependent. The reciprocal statement follows by exchanging primes and reversing β .

7.4 Composing frames

For constant determinant-one factors $\mathcal{T}, \mathcal{T}'$, applying the frame change $\mathcal{T}$ and then $\mathcal{T}'$ gives

$$\mathcal{Z}'' = \mathcal{T}'(\mathcal{T}\mathcal{Z}\mathcal{T}^H)\mathcal{T}'^H = (\mathcal{T}'\mathcal{T})\mathcal{Z}(\mathcal{T}'\mathcal{T})^H.$$

Thus the combined factor is $\mathcal{T}'\mathcal{T}$: the rightmost factor acts first.

Collinear boosts add rapidities by (4.18). For a second real rapidity θ' , the parameter $\beta' = \tanh \theta'$ is defined, and $\gamma' = \cosh \theta'$. Their combined parameters are

$$\begin{aligned} \beta'' &= \tanh(\theta + \theta') = \frac{\beta + \beta'}{1 + \beta\beta'}, \\ \gamma'' &= \gamma\gamma'(1 + \beta\beta'). \end{aligned}$$

These are the collinear velocity and clock-factor composition laws. Noncollinear boosts instead have the rotation commutator (4.19), producing the Thomas–Wigner rotation.

7.5 Four-momentum and the mass shell

For constant invariant mass $m > 0$, the real four-momentum $\mathcal{P} \in \mathbb{R} \oplus \mathbb{R}^3$ is defined by (9, Eq. (7.24))

$$\mathcal{P} := m\mathcal{U} = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (7.10)$$

Energy and momentum are

$$\begin{aligned} E &= \text{sc}(\mathcal{P}) = m \frac{dt}{ds} = \frac{m}{\sqrt{1 - \mathbf{v}^2}}, \\ \mathbf{P} &= \text{vec}(\mathcal{P}) = E\mathbf{v}. \end{aligned}$$

Since $\det \mathcal{U} = 1$, scalar scaling gives the mass shell (9, Eq. (7.30)):

$$E^2 - \mathbf{P}^2 = \det \mathcal{P} = m^2, \quad E > 0. \quad (7.11)$$

SI units: Momentum

With dimensionless four-velocity $\mathbf{u}$, mass m in kilograms, energy E in joules, and momentum $\mathbf{P}$ in kg m s^{-1} ,

$$\mathcal{P} = mc\mathbf{u} = \frac{E}{c} + \mathbf{P} \cdot \boldsymbol{\sigma},$$

$$\det \mathcal{P} = m^2 c^2, \quad E^2 = m^2 c^4 + c^2 \mathbf{P}^2.$$

The scalar and vector coefficients of $\mathcal{P}$ now both have momentum units.

Returning to natural units, the inverse and speed follow directly:

$$\mathcal{P}^{-1} = \frac{E - \mathbf{P} \cdot \boldsymbol{\sigma}}{m^2}, \quad v^2 = \frac{\mathbf{P}^2}{E^2} = 1 - \frac{m^2}{E^2} < 1.$$

For fixed $m > 0$, approaching light speed requires unbounded energy. A massless particle has no proper-time four-velocity. Its momentum is defined directly by $\mathcal{P} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, with $E > 0$ and $E^2 = \mathbf{P}^2$.

For the real momentum $\mathcal{P} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, the same congruence gives (21, Eq. (17.12))

$$E' = \gamma(E - \beta \mathbf{u} \cdot \mathbf{P}), \quad (7.12)$$

$$\mathbf{P}' = \mathbf{P}_\perp + \gamma(\mathbf{P}_\parallel - \beta E \mathbf{u}). \quad (7.13)$$

Here the projections are along $\mathbf{u}$. The factor $\gamma = \cosh \theta$ belongs to the frame boost. Since $\mathbf{P} = E\mathbf{v}$, energy and coordinate-time increments acquire the same factor along a massive particle's path:

$$\frac{E'}{E} = \frac{dt'}{dt} = \gamma(1 - \beta \mathbf{u} \cdot \mathbf{v}). \quad (7.14)$$

This is also the time factor used to obtain the velocity transformation. Determinant invariance gives $E'^2 - \mathbf{P}'^2 = E^2 - \mathbf{P}^2 = m^2$ for a particle of rest mass $m > 0$. The energy increase is obtained directly from $\mathcal{P}' = m$ in the rest frame by inversion of the boost:

$$\mathcal{P} = \mathcal{L}^{-1} m \mathcal{L}^{-1} = m\gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}).$$

Thus $E = m\gamma$ and $\mathbf{P} = m\gamma\beta\mathbf{u}$, while m remains unchanged (9, Eqs. (7.25), (7.28)).

7.6 Free-particle action

Notation: real $m > 0$, $\mathbf{v} = \dot{\mathbf{r}}$, particle factor $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, and $ds = dt/\gamma$. Here $L : \mathbb{R} \times \mathbb{R}^3 \times \{\mathbf{v} \in \mathbb{R}^3 : \mathbf{v}^2 < 1\} \rightarrow \mathbb{R}$ is the particle Lagrangian; the boost factor is $\mathcal{L} \in \mathbb{R} \oplus \mathbb{R}^3$. $\partial_{\mathbf{v}}$ denotes its vector derivative with respect to the three independent velocity components, while $\partial_{\mathbf{r}}$ denotes the spatial gradient. The free action is the physical choice (7, Eqs. (5.38)–(5.39))

$$\int L dt = -m \int ds. \quad (7.15)$$

Using $ds = dt/\gamma$ gives the Lagrangian

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m/\gamma.$$

For endpoint-fixed real path variations $\delta \mathbf{r}(t)$, varying the Lagrangian and integrating by parts gives

$$\delta \int L dt = \int \left(\partial_{\mathbf{r}} L - \frac{d(\partial_{\mathbf{v}} L)}{dt} \right) \cdot \delta \mathbf{r} dt.$$

The Euler–Lagrange equation is obtained from stationarity for every variation. With Hamiltonian $H := \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \in \mathbb{R}$, the chain rule gives

$$\frac{d(\partial_{\mathbf{v}} L)}{dt} = \partial_{\mathbf{r}} L, \quad (7.16)$$

$$\dot{H} = \mathbf{v} \cdot \left(\frac{d(\partial_{\mathbf{v}} L)}{dt} - \partial_{\mathbf{r}} L \right) - \partial_t L = -\partial_t L. \quad (7.17)$$

By (7.16) and (7.17), spatial and temporal translation invariance conserve canonical momentum and energy. For the free action,

$$\begin{aligned} \partial_{\mathbf{v}} L &= m\gamma \mathbf{v} = \mathbf{P}, \\ H &= m\gamma \mathbf{v}^2 + m/\gamma = m\gamma = E. \end{aligned}$$

The notation $v = \|\mathbf{v}\| \ll 1$ is used. Here $O(v^n)$ denotes terms bounded by a constant times v^n as $v \rightarrow 0$. The binomial series gives

$$\begin{aligned} \sqrt{1 - v^2} &= 1 - \tfrac{1}{2}v^2 + O(v^4), \\ \gamma &= 1 + \tfrac{1}{2}v^2 + O(v^4), \\ L &= -m + \tfrac{1}{2}mv^2 + O(mv^4), \\ H &= m + \tfrac{1}{2}mv^2 + O(mv^4). \end{aligned}$$

By the mass shell, the exact kinetic energy above rest is

$$K := H - m = m(\gamma - 1) = \frac{\mathbf{P}^2}{E + m} \in \mathbb{R}_{\geq 0}.$$

This and both expansions recover $mv^2/2$ at low speed after removal of the constant rest term.

8 Acceleration

Coordinate acceleration is $\dot{\mathbf{v}} = d\mathbf{v}/dt$; four-acceleration is $d\mathbf{u}/ds$. Proper acceleration is measured by an accelerometer.

8.1 Four-acceleration and proper acceleration

Coordinate-time derivatives are first obtained:

$$\begin{aligned} \dot{\gamma} &= \gamma^3 (\mathbf{v} \cdot \dot{\mathbf{v}}), \\ \mathbf{u} &= \dot{\gamma} + \frac{d(\gamma \mathbf{v})}{dt} \cdot \boldsymbol{\sigma}, \\ \frac{d(\gamma \mathbf{v})}{dt} &= \gamma \dot{\mathbf{v}} + \dot{\gamma} \mathbf{v}. \end{aligned}$$

For $\mathbf{v} \neq 0$, the coordinate acceleration is resolved into tangential (parallel to the velocity) and perpendicular components:

$$\dot{\mathbf{v}}_{\parallel} := \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v}}{v^2}, \quad \dot{\mathbf{v}}_{\perp} := \dot{\mathbf{v}} - \dot{\mathbf{v}}_{\parallel}, \quad \dot{\mathbf{v}} = \dot{\mathbf{v}}_{\perp} + \dot{\mathbf{v}}_{\parallel}.$$

The projections of $\dot{\mathbf{v}}$ are indicated by these subscripts; at $\mathbf{v} = 0$, the unsplit formulas are used. Since $(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v} = v^2 \dot{\mathbf{v}}_{\parallel}$ and $1 + \gamma^2 v^2 = \gamma^2$, the spatial derivative becomes

$$\begin{aligned} \frac{d(\gamma \mathbf{v})}{dt} &= \gamma \dot{\mathbf{v}}_{\perp} + (\gamma + \gamma^3 v^2) \dot{\mathbf{v}}_{\parallel} \\ &= \gamma \dot{\mathbf{v}}_{\perp} + \gamma^3 \dot{\mathbf{v}}_{\parallel}. \end{aligned}$$

Thus the full coordinate-time derivative of four-velocity is

$$\dot{\mathbf{u}} = \gamma^3 (\mathbf{v} \cdot \dot{\mathbf{v}}) + (\gamma \dot{\mathbf{v}}_{\perp} + \gamma^3 \dot{\mathbf{v}}_{\parallel}) \cdot \boldsymbol{\sigma}. \quad (8.1)$$

Orthogonality and determinant. Differentiating $\mathbf{u}^* \mathbf{u} = 1$ gives

$$\dot{\mathbf{u}}^* \mathbf{u} + \mathbf{u}^* \dot{\mathbf{u}} = 0.$$

By taking scalar parts,

$$\text{sc}(\mathbf{u}^* \dot{\mathbf{u}}) = 0.$$

Thus $\mathbf{u}$ and $\dot{\mathbf{u}}$ are Minkowski-orthogonal. Although $\dot{\mathbf{u}}$ can have a scalar part, the product $\mathbf{u}^* \dot{\mathbf{u}}$ is scalar-free and belongs to $\{0\} \oplus \mathbb{C}^3$. Multiplication on the right by $\mathbf{u}^*$ also gives

$$\text{adj} \dot{\mathbf{u}} = -\mathbf{u}^* \dot{\mathbf{u}} \mathbf{u}^*.$$

By expansion of the commutator and the product,

$$\begin{aligned} [\mathbf{u}, \dot{\mathbf{u}}] &= 2i\gamma^2 (\mathbf{v} \times \dot{\mathbf{v}}) \cdot \boldsymbol{\sigma}, \\ \mathbf{u}^* \dot{\mathbf{u}} &= \gamma^2 (\dot{\mathbf{v}} - i\mathbf{v} \times \dot{\mathbf{v}}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The cross product depends only on acceleration perpendicular to $\mathbf{v}$. Since $\det \mathbf{u}^* = 1$, multiplying by $\mathbf{u}^*$ preserves the determinant. The determinant of a scalar-free paravector is the negative of its square. Using $\dot{\mathbf{v}} \cdot (\mathbf{v} \times \dot{\mathbf{v}}) = 0$ therefore gives

$$\begin{aligned} \det \dot{\mathbf{u}} &= -(\mathbf{u}^* \dot{\mathbf{u}})^2 \\ &= -\gamma^4 (\dot{\mathbf{v}}^2 - (\mathbf{v} \times \dot{\mathbf{v}})^2) \\ &= -\gamma^2 (\dot{\mathbf{v}}^2 + \gamma^2 (\mathbf{v} \cdot \dot{\mathbf{v}})^2). \end{aligned}$$

The last equality uses $(\mathbf{v} \times \dot{\mathbf{v}})^2 = v^2 \dot{\mathbf{v}}^2 - (\mathbf{v} \cdot \dot{\mathbf{v}})^2$ and $1 - v^2 = \gamma^{-2}$. Using the parallel and perpendicular components defined above, $\dot{\mathbf{v}}^2 = \dot{\mathbf{v}}_{\perp}^2 + \dot{\mathbf{v}}_{\parallel}^2$ and $(\mathbf{v} \cdot \dot{\mathbf{v}})^2 = v^2 \dot{\mathbf{v}}_{\parallel}^2$ give

$$\det \dot{\mathbf{u}} = -\gamma^2 \dot{\mathbf{v}}_{\perp}^2 - \gamma^4 \dot{\mathbf{v}}_{\parallel}^2.$$

Conversion to proper time. Multiplying $\dot{\mathbf{u}}$ by γ gives (9, p. 141)

$$\frac{d\mathbf{u}}{ds} = \gamma \dot{\mathbf{u}} = \gamma^4 (\mathbf{v} \cdot \dot{\mathbf{v}}) + (\gamma^2 \dot{\mathbf{v}} + \gamma^4 (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (8.2)$$

The determinant is quadratic, so

$$\begin{aligned} \det \left(\frac{d\mathbf{u}}{ds} \right) &= \gamma^2 \det (\dot{\mathbf{u}}) \\ &= -\gamma^4 \dot{\mathbf{v}}_{\perp}^2 - \gamma^6 \dot{\mathbf{v}}_{\parallel}^2. \end{aligned}$$

The same scaling gives $\text{sc}(\mathbf{u}^* d\mathbf{u}/ds) = 0$, so four-velocity and four-acceleration are Minkowski-orthogonal. Nonzero four-acceleration is space-like. Its negative determinant is the squared magnitude of the acceleration measured in the instantaneous rest frame, called the proper acceleration. At rest, $\gamma = 1$ and $d\mathbf{u}/ds = \dot{\mathbf{v}} \cdot \boldsymbol{\sigma}$, so the determinant is $-\dot{\mathbf{v}}^2$ and the proper acceleration has magnitude $\|\dot{\mathbf{v}}\|$.

SI units: Proper acceleration

Here t, τ are measured in seconds, $s = c\tau$ and $\mathbf{r}$ in metres, and $\mathbf{v} = d\mathbf{r}/dt$ in metres per second. Dots denote derivatives with respect to this SI time. With $\mathbf{x} = ct + \mathbf{r} \cdot \boldsymbol{\sigma}$,

$$\begin{aligned} \gamma &= \frac{1}{\sqrt{1 - \mathbf{v}^2/c^2}}, \quad \mathbf{u} = \frac{d\mathbf{x}}{ds} = \gamma \left(1 + \frac{\mathbf{v}}{c} \cdot \boldsymbol{\sigma} \right), \\ c^2 \sqrt{-\det \left(\frac{d\mathbf{u}}{ds} \right)} &= \sqrt{\gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2}. \end{aligned}$$

Four-velocity remains dimensionless; both sides of the last equation give the accelerometer magnitude in m s^{-2} .

8.2 Force, work, and inertia

Here $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ is the particle Lorentz factor. For constant invariant mass $m > 0$, four-momentum is $\mathbf{P} = m\mathbf{u}$, so $\dot{\mathbf{P}} = m\dot{\mathbf{u}}$. Its scalar and spatial parts are $E = m\gamma$ and $\mathbf{P} = m\gamma\mathbf{v}$. The acceleration projections are parallel and perpendicular to $\mathbf{v}$, as in Section 8.1. Multiplying the spatial part of (8.1) by m therefore gives

$$\dot{\mathbf{P}} = m\gamma \dot{\mathbf{v}}_{\perp} + m\gamma^3 \dot{\mathbf{v}}_{\parallel}. \quad (8.3)$$

Taking its dot product with $\mathbf{v}$ gives the work rate

$$\dot{E} = m\gamma^3 \mathbf{v} \cdot \dot{\mathbf{v}} = \mathbf{v} \cdot \dot{\mathbf{P}}. \quad (8.4)$$

At rest the force is $m\dot{\mathbf{v}}$.

SI units: Force and energy

In the following SI formulas, t is in seconds, $\mathbf{v}$ in metres per second, and $\mathbf{P}$ in newtons:

$$\frac{d\mathbf{P}}{dt} = m\gamma^3\dot{\mathbf{v}}_{\parallel} + m\gamma\dot{\mathbf{v}}_{\perp},$$

$$E = \gamma mc^2.$$

In the natural units used elsewhere, invariant m is retained: the longitudinal and transverse inertia in (8.3) differ. For the Lorentz force, the corresponding acceleration components are derived in (12.22).

8.3 Fixed-axis acceleration

For the orthonormal pair $\mathbf{u}, \mathbf{N}$ in Section 6.3, the first derivatives exchange the two directions:

$$\dot{\mathbf{u}} = \dot{\theta}\mathbf{N}, \quad \dot{\mathbf{N}} = -\dot{\theta}\mathbf{u}, \quad \det \dot{\mathbf{u}} = -\dot{\theta}^2.$$

Differentiating $\beta = \tanh \theta$ gives

$$\dot{\beta} = \gamma^{-2}\dot{\theta}.$$

Since $\dot{s} = 1/\gamma$, the chain rule gives

$$\dot{\theta} = \frac{d\theta}{ds}\dot{s}, \quad \frac{d\theta}{ds} = \gamma\dot{\theta} = \gamma^3\dot{\beta}.$$

These are ordinary derivatives of the same rapidity along the path. The signed proper acceleration is $d\theta/ds$. By scaling the coordinate derivatives,

$$\frac{d\mathbf{u}}{ds} = \frac{d\theta}{ds}\mathbf{N} = \gamma\dot{\theta}\mathbf{N}, \quad \frac{d\mathbf{N}}{ds} = -\gamma\dot{\theta}\mathbf{u}, \quad \det \left(\frac{d\mathbf{u}}{ds} \right) = -\gamma^2\dot{\theta}^2.$$

The proper-acceleration magnitude is therefore $\gamma|\dot{\theta}|$.

8.4 Constant coordinate acceleration

In the following examples, subscript 0 denotes the initial event $t = s = 0$. Constant coordinate acceleration means that $\dot{\mathbf{v}} = \dot{\mathbf{v}}_0$ is a fixed real vector in a chosen inertial frame. With initial position $\mathbf{r}(0) = \mathbf{r}_0$ and velocity $\mathbf{v}(0) = \mathbf{v}_0$, integration gives

$$\mathbf{v}(t) = \mathbf{v}_0 + \dot{\mathbf{v}}_0 t, \tag{8.5}$$

$$\mathbf{x}(t) = t + \left(\mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\dot{\mathbf{v}}_0 t^2 \right) \cdot \boldsymbol{\sigma}. \tag{8.6}$$

These familiar coordinate-time formulas also describe a relativistic massive particle, provided $\|\mathbf{v}_0 + \dot{\mathbf{v}}_0 t\| < 1$ throughout the interval considered. A nonzero constant coordinate acceleration can therefore persist only for a finite coordinate-time interval.

For motion from rest at the origin, the notation $\dot{\mathbf{v}}_0 = \dot{v}_0 \mathbf{u}$ is used, where $\dot{v}_0 > 0$ is the scalar acceleration along the fixed real unit vector $\mathbf{u}$. Then

$$\mathbf{v}(t) = \dot{v}_0 t \mathbf{u}, \quad \mathbf{x}(t) = t + \frac{1}{2}\dot{v}_0 t^2 \mathbf{u} \cdot \boldsymbol{\sigma}, \quad \gamma(t) = \frac{1}{\sqrt{1 - \dot{v}_0^2 t^2}}, \quad 0 \leq t < \frac{1}{\dot{v}_0}.$$

The particle's clock runs at $\dot{s} = 1/\gamma(t)$. Taking $s(0) = 0$ in (6.5) gives

$$s(t) = \int_0^t \sqrt{1 - \dot{v}_0^2 t'^2} dt' = \frac{\arcsin(\dot{v}_0 t) + \dot{v}_0 t \sqrt{1 - \dot{v}_0^2 t^2}}{2\dot{v}_0}. \quad (8.7)$$

Four-velocity and four-acceleration follow by differentiating the world line:

$$\begin{aligned} \mathcal{U}(t) &= \gamma(t)(1 + \dot{v}_0 t \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \frac{d\mathcal{U}}{ds} &= \gamma \dot{\mathcal{U}} = \gamma^4 \dot{v}_0^2 t + \gamma^4 \dot{v}_0 \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Consequently,

$$\det \left(\frac{d\mathcal{U}}{ds} \right) = \gamma^8 \dot{v}_0^2 (\dot{v}_0^2 t^2 - 1) = -\gamma^6 \dot{v}_0^2.$$

The proper-acceleration magnitude is therefore

$$\sqrt{-\det \left(\frac{d\mathcal{U}}{ds} \right)} = \gamma^3 \dot{v}_0 = \frac{\dot{v}_0}{(1 - \dot{v}_0^2 t^2)^{3/2}}.$$

Although the coordinate acceleration stays fixed, the acceleration measured in the instantaneous rest frame grows without bound as $t \rightarrow 1/\dot{v}_0$. The next example instead holds the proper acceleration fixed.

8.5 Constant proper acceleration

A fixed real unit axis $\mathbf{u}$ is chosen, and the initial conditions $\mathcal{X}(0) = 0$, $\mathcal{U}(0) = 1$ are imposed, with initial coordinate acceleration $\dot{\mathbf{v}}_0 = \dot{v}_0 \mathbf{u}$, where $\dot{v}_0 > 0$. At rest $\gamma_0 = 1$ and $\dot{\theta}_0 = \dot{v}_0$. Holding the signed proper acceleration along this axis constant gives $\theta(s) = \dot{v}_0 s$, and hence

$$\mathcal{U}(s) = \mathcal{L}(\dot{v}_0 s, \mathbf{u})^{-2} = \cosh(\dot{v}_0 s) + \sinh(\dot{v}_0 s) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

The stated acceleration is verified by differentiation:

$$\det \left(\frac{d\mathcal{U}}{ds} \right) = -\dot{v}_0^2 (\cosh^2(\dot{v}_0 s) - \sinh^2(\dot{v}_0 s)) = -\dot{v}_0^2.$$

Integrating (6.9) yields (9, Eq. (7.33), pp. 142–143)

$$\mathcal{X}(s) = \frac{\sinh(\dot{v}_0 s)}{\dot{v}_0} + \frac{\cosh(\dot{v}_0 s) - 1}{\dot{v}_0} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (8.8)$$

In components,

$$\begin{aligned} \mathbf{v}(t(s)) &= \tanh(\dot{v}_0 s) \mathbf{u}, \\ (1 + \dot{v}_0 \mathbf{u} \cdot \mathbf{r})^2 - (\dot{v}_0 t)^2 &= 1. \end{aligned}$$

Thus (8.8) is the usual hyperbolic motion, with speed below light speed at every finite s . Eliminating s using $\dot{v}_0 t = \sinh(\dot{v}_0 s)$ makes the coordinate acceleration explicit:

$$\mathbf{v}(t) = \frac{\dot{v}_0 t}{\sqrt{1 + \dot{v}_0^2 t^2}} \mathbf{u}, \quad \dot{\mathbf{v}}(t) = \frac{\dot{v}_0}{(1 + \dot{v}_0^2 t^2)^{3/2}} \mathbf{u}.$$

Here $\gamma(t) = \sqrt{1 + \dot{v}_0^2 t^2}$, so $\gamma^3 \|\dot{\mathbf{v}}(t)\| = \dot{v}_0$ is constant while the coordinate acceleration decreases as the particle gains speed.

8.6 A trip with midpoint braking

Departure and destination are at rest in one inertial frame, separated by $\ell > 0$ along a fixed real unit vector $\mathbf{u}$. Flat spacetime and initial conditions $t = s = 0$, $\mathbf{r} = 0$, $\mathbf{v} = 0$ are assumed. Two equal burns at constant proper-acceleration magnitude $\dot{v}_0 > 0$ are joined by an instantaneous thrust reversal; arrival is at rest. This is a prescribed trajectory; fuel consumption and the duration of the reversal are not modeled.

Rapidity and derivatives. The reversal times are denoted by s_m and t_m ; arrival occurs at $2s_m$ and $2t_m$. With $d\theta/ds = +\dot{v}_0$ during acceleration and $-\dot{v}_0$ during braking, integration gives

$$\theta(s) = \begin{cases} \dot{v}_0 s, & 0 \leq s \leq s_m, \\ \dot{v}_0(2s_m - s), & s_m \leq s \leq 2s_m. \end{cases} \quad (8.9)$$

Throughout the trip,

$$v = \tanh \theta, \quad \mathbf{v} = v\mathbf{u}, \quad \gamma = \cosh \theta, \quad \dot{s} = \gamma^{-1}.$$

On the open burn intervals, the upper sign denotes acceleration and the lower sign braking. The derivatives are obtained directly:

$$\begin{aligned} \mathcal{U} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \frac{d\mathcal{U}}{ds} &= \pm \dot{v}_0 (\sinh \theta + \cosh \theta \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \dot{\mathcal{U}} &= \frac{1}{\gamma} \frac{d\mathcal{U}}{ds} = \pm \dot{v}_0 (v + \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \dot{v} &= \pm \frac{\dot{v}_0}{\gamma^3}, \quad \dot{\mathbf{v}} = \dot{v}\mathbf{u}, \quad \dot{\gamma} = \pm \dot{v}_0 v. \end{aligned} \quad (8.10)$$

The acceleration is entirely tangential; the perpendicular component vanishes. Although $\|\dot{\mathbf{v}}\|$ varies, the proper-acceleration magnitude is fixed:

$$\det \mathcal{U} = 1, \quad \sqrt{-\det \left(\frac{d\mathcal{U}}{ds} \right)} = \gamma^3 \|\dot{\mathbf{v}}\| = \dot{v}_0.$$

Position, clocks, and arrival. By integrating $dt/ds = \cosh \theta$ and $d\mathbf{r}/ds = \sinh \theta \mathbf{u}$, the continuous worldline is obtained:

$$\begin{aligned} t(s) &= \begin{cases} \sinh \theta(s)/\dot{v}_0, & 0 \leq s \leq s_m, \\ 2t_m - \sinh \theta(s)/\dot{v}_0, & s_m \leq s \leq 2s_m, \end{cases} \\ \mathbf{r}(s) &= \begin{cases} (\cosh \theta(s) - 1)\mathbf{u}/\dot{v}_0, & 0 \leq s \leq s_m, \\ \mathbf{u}(\ell - (\cosh \theta(s) - 1)/\dot{v}_0), & s_m \leq s \leq 2s_m. \end{cases} \end{aligned} \quad (8.11)$$

Continuity at the reversal fixes the midpoint and burn durations:

$$\begin{aligned} \mathbf{r}(s_m) &= \frac{1}{2}\mathbf{u}\ell, \quad \gamma_m := \cosh(\dot{v}_0 s_m) = 1 + \frac{\dot{v}_0 \ell}{2}, \\ s_m &= \frac{\text{arcosh}(1 + \dot{v}_0 \ell/2)}{\dot{v}_0}, \quad t_m = \sqrt{\frac{\ell}{\dot{v}_0} + \frac{\ell^2}{4}}, \\ v_{\max} &= \sqrt{1 - \gamma_m^{-2}}. \end{aligned} \quad (8.12)$$

Thus $\mathbf{r}(2s_m) = \mathbf{u}\ell$ and $\mathbf{u}(0) = \mathbf{u}(2s_m) = 1$. For evaluation in coordinate time, the same expressions are used with

$$\theta(t) = \begin{cases} \operatorname{arsinh}(\dot{v}_0 t), & 0 \leq t \leq t_m, \\ \operatorname{arsinh}(\dot{v}_0(2t_m - t)), & t_m \leq t \leq 2t_m. \end{cases}$$

In particular, $\|\dot{\mathbf{v}}\|$ decreases to $\dot{v}_0/\gamma_m^3$ before the reversal and increases back to $\dot{v}_0$ during braking.

The thrust reversal. Position, both clocks, velocity, and four-velocity are continuous at the midpoint. The one-sided values of $\dot{\mathbf{v}}$, $\dot{\mathbf{u}}$, and $d\mathbf{u}/ds$ are opposite; these derivatives are undefined at the ideal instantaneous reversal. The thrust is reversed, not the direction of travel. The jerk during each burn and through a finite reversal is discussed in Section 9.4.

SI units: Trip parameters

In the following SI formulas, $\dot{v}_0$ is the initial acceleration in metres per second squared, also equal to the constant proper-acceleration magnitude. The distance ℓ is in metres, t_{trip} is the elapsed departure-frame time in seconds, and τ_{trip} is the elapsed onboard proper time in seconds. Restoring units in (8.12) gives

$$\begin{aligned} t_{\text{trip}} &= \sqrt{\frac{\ell^2}{c^2} + \frac{4\ell}{\dot{v}_0}}, \\ \tau_{\text{trip}} &= \frac{2c}{\dot{v}_0} \operatorname{arcosh}\left(1 + \frac{\dot{v}_0 \ell}{2c^2}\right), \\ v_{\text{max}} &= c \sqrt{1 - \left(1 + \frac{\dot{v}_0 \ell}{2c^2}\right)^{-2}}. \end{aligned}$$

For a specified onboard duration and peak speed, the required proper acceleration is fixed by the midpoint rapidity:

$$\dot{v}_0 = \frac{2c}{\tau_{\text{trip}}} \operatorname{artanh}\left(\frac{v_{\text{max}}}{c}\right).$$

9 Jerk

Three quantities are distinguished: the coordinate jerk $\ddot{\mathbf{v}}$, the proper-time derivative of four-acceleration, and its projection into the instantaneous rest space. Proper jerk is the vector rate of change of measured acceleration per unit proper time, referred to nonrotating carried axes. Dots still mean t derivatives.

9.1 Coordinate derivatives, then proper-time derivatives

The expressions for $\dot{\gamma}$ and $\dot{\mathbf{u}}$ in Section 8.1 are differentiated once more with respect to t . Using

$$\frac{d(\mathbf{v} \cdot \dot{\mathbf{v}})}{dt} = \dot{\mathbf{v}}^2 + \mathbf{v} \cdot \ddot{\mathbf{v}},$$

the following is obtained by a second differentiation:

$$\begin{aligned}\ddot{\gamma} &= \gamma^3 (\dot{\mathbf{v}}^2 + \mathbf{v} \cdot \ddot{\mathbf{v}}) + \frac{3\dot{\gamma}^2}{\gamma}, \\ \ddot{\mathbf{u}} &= \ddot{\gamma} + (\gamma \ddot{\mathbf{v}} + 2\dot{\gamma} \dot{\mathbf{v}} + \ddot{\gamma} \mathbf{v}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The conversion $d/ds = \gamma d/dt$ is now applied. The first derivative is $d\mathbf{u}/ds = \gamma \dot{\mathbf{u}}$, and the second is

$$\frac{d^2 \mathbf{u}}{ds^2} = \gamma \frac{d(\gamma \dot{\mathbf{u}})}{dt} = \gamma^2 \ddot{\mathbf{u}} + \gamma \dot{\gamma} \dot{\mathbf{u}}. \quad (9.1)$$

The same conversion applies to every twice-differentiable scalar, vector, or paravector along the worldline. In particular,

$$\begin{aligned}\frac{d^2 \gamma}{ds^2} &= \gamma^2 \ddot{\gamma} + \gamma \dot{\gamma}^2, \\ \frac{d^2 \mathbf{u}}{ds^2} &= \frac{d^2 \gamma}{ds^2} + \left(\gamma^3 \ddot{\mathbf{v}} + 3\gamma^2 \dot{\gamma} \dot{\mathbf{v}} + \frac{d^2 \gamma}{ds^2} \mathbf{v} \right) \cdot \boldsymbol{\sigma}.\end{aligned} \quad (9.2)$$

This derivative of four-acceleration is the *unprojected four-jerk*:

$$\frac{d}{ds} \left(\frac{d\mathbf{u}}{ds} \right) = \frac{d^2 \mathbf{u}}{ds^2} = \frac{d^3 \mathbf{x}}{ds^3}. \quad (9.3)$$

By comparison, $d\dot{\mathbf{v}}/ds = \gamma \ddot{\mathbf{v}}$ still differentiates the coordinate acceleration of the chosen inertial frame.

9.2 Projection into the instantaneous rest space

The coordinate-time normalization identities are used:

$$\text{sc}(\mathbf{u}^* \dot{\mathbf{u}}) = 0, \quad \text{sc}(\mathbf{u}^* \ddot{\mathbf{u}}) = -\det(\dot{\mathbf{u}}).$$

The second follows by differentiating the first. Using

$$\det\left(\frac{d\mathbf{u}}{ds}\right) = \gamma^2 \det \dot{\mathbf{u}},$$

substitution into (9.1) gives

$$\text{sc}\left(\mathbf{u}^* \frac{d^2 \mathbf{u}}{ds^2}\right) = -\det\left(\frac{d\mathbf{u}}{ds}\right).$$

Thus the derivative of four-acceleration need not be orthogonal to four-velocity. Since $\det \mathbf{u} = 1$, the right-hand side is its coefficient of $\mathbf{u}$ in the unprojected four-jerk. Removing that component defines the projected relativistic jerk (22, Eqs. (2.8)–(2.13)). With the signature $(+, -, -, -)$, its orthogonality is

$$\text{sc}\left(\mathbf{u}^* \left(\frac{d^2 \mathbf{u}}{ds^2} + \det\left(\frac{d\mathbf{u}}{ds}\right) \mathbf{u} \right)\right) = 0. \quad (9.4)$$

At an instant of rest, $\mathbf{u} = 1$ and

$$\begin{aligned}\frac{d^2 \mathbf{u}}{ds^2} &= \dot{\mathbf{v}}^2 + \ddot{\mathbf{v}} \cdot \boldsymbol{\sigma} \\ &= -\det\left(\frac{d\mathbf{u}}{ds}\right) \mathbf{u} + \ddot{\mathbf{v}} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Here the coordinate derivatives are evaluated in the inertial frame in which the particle is instantaneously at rest. The projected expression is therefore the measured proper jerk there; it is space-like whenever nonzero.

Invariant magnitude. Expanding the determinant using the metric pairing and (9.4) gives

$$\det \left(\frac{d^2 \mathbf{u}}{ds^2} + \det \left(\frac{d\mathbf{u}}{ds} \right) \mathbf{u} \right) = \det \left(\frac{d^2 \mathbf{u}}{ds^2} \right) - \det \left(\frac{d\mathbf{u}}{ds} \right)^2 \leq 0. \quad (9.5)$$

The negative of this determinant is the squared proper-jerk magnitude. The unprojected four-jerk can instead be time-like, null, or space-like; its determinant includes the positive contribution from the component along $\mathbf{u}$.

When proper jerk vanishes. In (9.2), subtracting the scalar coefficient times $\mathbf{v}$ from the vector coefficient leaves $\gamma^3 (\ddot{\mathbf{v}} + 3\dot{\gamma}\dot{\mathbf{v}}/\gamma)$. Its vanishing says that the unprojected four-jerk is parallel to $\mathbf{u}$, so its projection vanishes. Hence (22, Eq. (2.14))

$$\frac{d^2 \mathbf{u}}{ds^2} + \det \left(\frac{d\mathbf{u}}{ds} \right) \mathbf{u} = 0 \quad \Longleftrightarrow \quad \ddot{\mathbf{v}} = -3 \frac{\dot{\gamma}}{\gamma} \dot{\mathbf{v}}. \quad (9.6)$$

Zero proper jerk therefore need not mean zero coordinate jerk. An equivalent first integral is

$$\frac{d(\gamma^3 \dot{\mathbf{v}})}{dt} = 0, \quad \gamma^3 \dot{\mathbf{v}} = \gamma_0^3 \dot{\mathbf{v}}_0.$$

Here subscript 0 denotes any reference event. Any nonzero coordinate acceleration therefore retains a fixed direction, although the velocity need not be parallel to it; the spatial trajectory is confined to a plane. Also, differentiating the acceleration determinant gives

$$\frac{d}{ds} \det \left(\frac{d\mathbf{u}}{ds} \right) = 2 \operatorname{sc} \left(\left(\frac{d\mathbf{u}}{ds} \right)^* \frac{d^2 \mathbf{u}}{ds^2} \right).$$

The pairing is unchanged if the second factor is replaced by its projected expression, because $d\mathbf{u}/ds$ is orthogonal to $\mathbf{u}$. Consequently, zero projected jerk implies constant proper-acceleration magnitude. The converse does not hold in general: the direction can still change, as in uniform circular motion.

9.3 Fixed-axis jerk

Differentiating the paired first derivatives in Section 8.3 gives

$$\begin{aligned} \ddot{\mathbf{u}} &= \dot{\theta}^2 \mathbf{u} + \ddot{\theta} \mathbf{N}, \\ \ddot{\mathbf{N}} &= \dot{\theta}^2 \mathbf{N} + \ddot{\theta} \mathbf{u}. \end{aligned}$$

The $\dot{\theta}^2$ terms come from differentiating the moving basis; the $\ddot{\theta}$ terms come from differentiating its rate of change. By orthonormality,

$$\det \ddot{\mathbf{u}} = -\det \ddot{\mathbf{N}} = \dot{\theta}^4 - \ddot{\theta}^2.$$

Differentiating $\dot{\beta} = \gamma^{-2} \dot{\theta}$ also gives

$$\ddot{\beta} = \gamma^{-2} \ddot{\theta} - 2\gamma^{-3} \dot{\gamma} \dot{\theta}.$$

Conversion to proper time. By (9.1), the signed proper jerk is

$$\frac{d^2\theta}{ds^2} = \gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta}.$$

Applying (9.1) to each member of the pair gives

$$\begin{aligned}\frac{d^2\mathcal{U}}{ds^2} &= \gamma^2\dot{\theta}^2\mathcal{U} + (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})\mathcal{N}, \\ \frac{d^2\mathcal{N}}{ds^2} &= \gamma^2\dot{\theta}^2\mathcal{N} + (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})\mathcal{U}.\end{aligned}\tag{9.7}$$

Removing the component along $\mathcal{U}$ leaves

$$\frac{d^2\mathcal{U}}{ds^2} - \gamma^2\dot{\theta}^2\mathcal{U} = (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})\mathcal{N} = \frac{d^2\theta}{ds^2}\mathcal{N}.$$

The proper-jerk magnitude is $|\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta}|$. The corresponding determinant identities are

$$\begin{aligned}\det\left(\frac{d^2\mathcal{U}}{ds^2} - \gamma^2\dot{\theta}^2\mathcal{U}\right) &= -(\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})^2, \\ \det\left(\frac{d^2\mathcal{U}}{ds^2}\right) &= -\det\left(\frac{d^2\mathcal{N}}{ds^2}\right) \\ &= \gamma^4\dot{\theta}^4 - (\gamma^2\ddot{\theta} + \gamma\dot{\gamma}\dot{\theta})^2.\end{aligned}$$

The sign reversal again comes from exchanging the unit time and spatial directions. Using $\dot{\theta} = \gamma^2\dot{\beta}$ and $\dot{\gamma} = \gamma^3\dot{\beta}$ also gives

$$\begin{aligned}\frac{d^2\theta}{ds^2} &= \gamma^4\ddot{\beta} + 3\gamma^3\dot{\gamma}\dot{\beta} = \gamma^4\ddot{\beta} + 3\gamma^6\dot{\beta}^2, \\ \ddot{\beta} &= \gamma^{-4}\frac{d^2\theta}{ds^2} - 3\frac{\dot{\gamma}}{\gamma}\dot{\beta}.\end{aligned}\tag{9.8}$$

For fixed-axis motion, zero proper jerk is exactly $\gamma\ddot{\theta} + \dot{\gamma}\dot{\theta} = 0$, or equivalently $d^2\theta/ds^2 = 0$. Thus $\gamma\dot{\theta}$ is constant and rapidity is affine in proper time. The unprojected four-jerk remains nonzero whenever that constant proper acceleration is nonzero.

Null combinations and constant acceleration. The symmetry of the paired derivatives becomes especially simple on adding or subtracting $\mathcal{U}$ and $\mathcal{N}$:

$$\mathcal{U} \pm \mathcal{N} = e^{\pm\theta} (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \det(\mathcal{U} \pm \mathcal{N}) = 0.$$

These two null directions are fixed; only their scalar factors change. Their coordinate-time derivatives are

$$\begin{aligned}\dot{\mathcal{U}} \pm \dot{\mathcal{N}} &= \pm\dot{\theta}(\mathcal{U} \pm \mathcal{N}), \\ \ddot{\mathcal{U}} \pm \ddot{\mathcal{N}} &= (\dot{\theta}^2 \pm \ddot{\theta})(\mathcal{U} \pm \mathcal{N}).\end{aligned}$$

The proper-time derivatives follow by the same conversion:

$$\begin{aligned}\frac{d(\mathcal{U} \pm \mathcal{N})}{ds} &= \pm \gamma \dot{\theta} (\mathcal{U} \pm \mathcal{N}), \\ \frac{d^2(\mathcal{U} \pm \mathcal{N})}{ds^2} &= \left(\gamma^2 \dot{\theta}^2 \pm \left(\gamma^2 \ddot{\theta} + \gamma \dot{\theta} \right) \right) (\mathcal{U} \pm \mathcal{N}).\end{aligned}$$

These formulas allow varying acceleration. If the signed proper acceleration $d\theta/ds = \gamma \dot{\theta}$ is constant, then

$$\frac{d^2\theta}{ds^2} = \gamma^2 \ddot{\theta} + \gamma \dot{\theta} = 0.$$

The second-derivative formula therefore reduces to

$$\frac{d^2(\mathcal{U} \pm \mathcal{N})}{ds^2} = \gamma^2 \dot{\theta}^2 (\mathcal{U} \pm \mathcal{N}) = \left(\pm \gamma \dot{\theta} \right)^2 (\mathcal{U} \pm \mathcal{N}).$$

Only under this constant-acceleration assumption is each null combination multiplied by the same constant factor at every proper-time differentiation. For integers $n \geq 0$, the $n = 2$ case above is included in

$$\frac{d^n(\mathcal{U} \pm \mathcal{N})}{ds^n} = \left(\pm \gamma \dot{\theta} \right)^n (\mathcal{U} \pm \mathcal{N}).$$

Adding and subtracting recovers the derivatives of $\mathcal{U}$ and $\mathcal{N}$: even orders preserve each direction, while odd orders exchange them. These ordinary derivatives remain nonzero for nonzero constant proper acceleration, even though the projected jerk is zero. The constant-proper-acceleration worldline in Section 8.5 realizes this pattern; Section 9.5 treats nonzero constant proper jerk.

9.4 Jerk at a thrust reversal

For the trip in Section 8.6, away from the reversal,

$$\frac{d^2\mathcal{U}}{ds^2} = \dot{v}_0^2 \mathcal{U}, \quad \ddot{v} = -\frac{3\dot{v}_0^2 v}{\gamma^4} \mathcal{U}.$$

The projected proper jerk is zero during each constant-acceleration burn, although coordinate jerk is generally nonzero. At the ideal instantaneous reversal, acceleration is discontinuous and jerk is undefined. A finite reversal must be described by a transition with nonzero proper jerk.

9.5 Constant jerk

A distinction is made between constant coordinate jerk in a chosen inertial frame, constant signed proper jerk along a fixed axis, and constant components of the unprojected four-jerk. The last condition is preserved by constant Lorentz transformations. Subscript 0 denotes the initial event $t = s = 0$.

9.5.1 Constant coordinate jerk

For $\ddot{v} = \ddot{v}_0$ in a chosen inertial frame, integration gives

$$\begin{aligned}\dot{v}(t) &= \dot{v}_0 + \ddot{v}_0 t, \\ v(t) &= v_0 + \dot{v}_0 t + \frac{1}{2} \ddot{v}_0 t^2, \\ \mathcal{X}(t) &= t + \left(r_0 + v_0 t + \frac{1}{2} \dot{v}_0 t^2 + \frac{1}{6} \ddot{v}_0 t^3 \right) \cdot \sigma.\end{aligned}\tag{9.9}$$

These formulas apply on the interval where $\mathbf{v}(t)^2 < 1$. Equations (6.3), (6.7), and (6.5) give γ , $\mathbf{u}$, and $s(t)$; (9.2) gives the generally varying four-jerk. For example, from rest with zero initial acceleration, the following is chosen: $\ddot{\mathbf{v}}_0 = \ddot{v}_0 \mathbf{u}$, where $\ddot{v}_0 > 0$ is the scalar jerk along the fixed real unit axis $\mathbf{u}$. Then

$$\mathbf{v}(t) = \frac{1}{2} \ddot{v}_0 t^2 \mathbf{u}, \quad \mathbf{r}(t) = \mathbf{r}_0 + \frac{1}{6} \ddot{v}_0 t^3 \mathbf{u}, \quad 0 \leq t < \sqrt{\frac{2}{\ddot{v}_0}}.$$

Here $\dot{\beta} = \ddot{v}_0 t$ and $\beta = \ddot{v}_0 t^2/2$, so (9.8) gives $d^2\theta/ds^2 = \gamma^4 \ddot{v}_0 + 3\gamma^6 \beta (\ddot{v}_0 t)^2$. Thus constant coordinate jerk does not give constant proper jerk.

9.5.2 Constant proper jerk

For fixed-axis motion, the signed quantity $d^2\theta/ds^2$ is held constant. Its value and the initial signed proper acceleration are expressed in coordinate-time data using $\gamma_0 = \cosh \theta_0$ and $\dot{\gamma}_0 = \sinh \theta_0 \dot{\theta}_0$. By integration with respect to proper time,

$$\begin{aligned} \frac{d\theta}{ds} &= \gamma_0 \dot{\theta}_0 + \left(\gamma_0^2 \ddot{\theta}_0 + \gamma_0 \dot{\gamma}_0 \dot{\theta}_0 \right) s, \\ \theta(s) &= \theta_0 + \gamma_0 \dot{\theta}_0 s + \frac{1}{2} \left(\gamma_0^2 \ddot{\theta}_0 + \gamma_0 \dot{\gamma}_0 \dot{\theta}_0 \right) s^2. \end{aligned} \quad (9.10)$$

If the particle starts at rest, $\theta_0 = 0$, $\gamma_0 = 1$, and $\dot{\gamma}_0 = 0$, so this reduces to $\theta(s) = \dot{\theta}_0 s + \frac{1}{2} \ddot{\theta}_0 s^2$. The corresponding world line, with $t(0) = 0$, is

$$\begin{aligned} t(s) &= \int_0^s \cosh \theta(s') ds', \\ \mathbf{r}(s) &= \mathbf{r}_0 + \mathbf{u} \int_0^s \sinh \theta(s') ds', \quad \mathcal{X}(s) = t(s) + \mathbf{r}(s) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (9.11)$$

The trajectory is specified exactly by these integrals; a related derivation is given in (22, Eqs. (5.8)–(5.9)). Because $dt/ds = \cosh \theta > 0$, they also define a unique coordinate-time trajectory. Its velocity, acceleration, and jerk are

$$\begin{aligned} \mathbf{v}(t(s)) &= \tanh \theta(s) \mathbf{u}, \\ \dot{\mathbf{v}}(t(s)) &= \frac{1}{\cosh^3 \theta(s)} \frac{d\theta}{ds} \mathbf{u}, \\ \ddot{\mathbf{v}}(t(s)) &= \frac{1}{\cosh^4 \theta(s)} \left(\frac{d^2\theta}{ds^2} - 3 \left(\frac{d\theta}{ds} \right)^2 \tanh \theta(s) \right) \mathbf{u}. \end{aligned}$$

The projected jerk is $(d^2\theta/ds^2) \mathcal{N}$, as derived in Section 9.3. Its magnitude is constant, while its inertial-frame components change with $\mathcal{N}$. The unprojected derivative also contains the changing component $(d\theta/ds)^2 \mathbf{u}$. Setting $\theta_0 = 0$, $\ddot{\theta}_0 = 0$, and $\dot{\theta}_0 = \dot{v}_0 > 0$ recovers the constant-proper-acceleration example in Section 8.5.

9.5.3 Constant unprojected four-jerk

If the components of $d^2\mathbf{u}/ds^2$ are constant in an inertial frame, its value is determined by the initial coordinate-time derivatives:

$$\frac{d^2\mathbf{u}}{ds^2} = \gamma_0^2 \ddot{\mathbf{u}}_0 + \gamma_0 \dot{\gamma}_0 \dot{\mathbf{u}}_0.$$

Here γ_0 and $\dot{\gamma}_0$ are the scalar parts of $\mathbf{u}_0$ and $\dot{\mathbf{u}}_0$, respectively. By direct integration,

$$\begin{aligned}\mathbf{u}(s) &= \mathbf{u}_0 + s\gamma_0 \dot{\mathbf{u}}_0 + \frac{s^2}{2} \left(\gamma_0^2 \ddot{\mathbf{u}}_0 + \gamma_0 \dot{\gamma}_0 \dot{\mathbf{u}}_0 \right), \\ \mathbf{x}(s) &= \mathbf{x}_0 + s\mathbf{u}_0 + \frac{s^2}{2} \gamma_0 \dot{\mathbf{u}}_0 + \frac{s^3}{6} \left(\gamma_0^2 \ddot{\mathbf{u}}_0 + \gamma_0 \dot{\gamma}_0 \dot{\mathbf{u}}_0 \right).\end{aligned}$$

With $\mathbf{u}_0$ future-directed, the following coefficient conditions are necessary and sufficient for $\det \mathbf{u}(s) = 1$. Equating powers of s using the metric pairing in (5.3) gives

$$\begin{aligned}\det \mathbf{u}_0 &= 1, & \text{sc}(\mathbf{u}_0^* \dot{\mathbf{u}}_0) &= 0, & \text{sc}(\mathbf{u}_0^* \ddot{\mathbf{u}}_0) &= -\det(\dot{\mathbf{u}}_0), \\ \gamma_0 \text{sc}(\dot{\mathbf{u}}_0^* \ddot{\mathbf{u}}_0) + \dot{\gamma}_0 \det(\dot{\mathbf{u}}_0) &= 0, & \det(\gamma_0 \ddot{\mathbf{u}}_0 + \dot{\gamma}_0 \dot{\mathbf{u}}_0) &= 0.\end{aligned}$$

In particular, a nonzero constant $d^2\mathbf{u}/ds^2$ must be null. If it is zero, the normalization conditions force $\dot{\mathbf{u}}_0 = 0$, so the motion is inertial. Thus a constant four-acceleration paravector must vanish; constant proper acceleration does not mean constant inertial-frame components. The same conditions imply constant $\det(d\mathbf{u}/ds)$. By (9.5), the proper-jerk magnitude is then the square of the proper-acceleration magnitude for this entire class.

For a concrete example (22, Eq. (3.40)), an initial rest state at the origin is chosen, with nonzero initial coordinate acceleration $\dot{\mathbf{v}}_0$ and jerk $\ddot{\mathbf{v}}_0$ constrained by

$$\dot{\mathbf{v}}_0 \cdot \ddot{\mathbf{v}}_0 = 0, \quad \ddot{\mathbf{v}}_0^2 = (\dot{\mathbf{v}}_0^2)^2.$$

Here $\gamma_0 = 1$ and $\dot{\gamma}_0 = 0$, so the initial paravector data are

$$\mathbf{x}_0 = 0, \quad \mathbf{u}_0 = 1, \quad \dot{\mathbf{u}}_0 = \dot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}, \quad \ddot{\mathbf{u}}_0 = \dot{\mathbf{v}}_0^2 + \ddot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}.$$

The resulting world line is

$$\mathbf{x}(s) = s + \frac{1}{6} \dot{\mathbf{v}}_0^2 s^3 + \left(\frac{1}{2} \dot{\mathbf{v}}_0 s^2 + \frac{1}{6} \ddot{\mathbf{v}}_0 s^3 \right) \cdot \boldsymbol{\sigma}. \quad (9.12)$$

By differentiation, the following is verified:

$$\frac{d^2\mathbf{u}}{ds^2} = \dot{\mathbf{v}}_0^2 + \ddot{\mathbf{v}}_0 \cdot \boldsymbol{\sigma}, \quad \det \mathbf{u} = 1.$$

Its invariants are

$$\det \left(\frac{d\mathbf{u}}{ds} \right) = -\dot{\mathbf{v}}_0^2, \quad \det \left(\frac{d^2\mathbf{u}}{ds^2} \right) = 0.$$

Thus the proper-acceleration magnitude $\|\dot{\mathbf{v}}_0\|$ and, by (9.5), the proper-jerk magnitude $\dot{\mathbf{v}}_0^2 = \|\ddot{\mathbf{v}}_0\|$ are both constant. Also $dt/ds = 1 + \dot{\mathbf{v}}_0^2 s^2/2 > 0$, so s is indeed future-directed proper time. This curved spatial path is a different motion from the fixed-axis constant-proper-jerk example.

10 Kinematics in a Frenet frame

Velocity, acceleration, and jerk are resolved along the tangent and two normals of the spatial path. Turning and twisting are described by curvature and torsion.

10.1 Spatial frame and curvature

In the chosen inertial frame, the speed is $v := \|\mathbf{v}\| \in (0, 1)$ and spatial arc length is ℓ , with $\dot{\ell} = v$. Dots denote t derivatives. On a curved portion of the spatial path, the real unit tangent, principal normal, and binormal are defined by

$$\mathbf{u} := \frac{\mathbf{v}}{v}, \quad k := \left\| \frac{d\mathbf{u}}{d\ell} \right\| = \frac{\|\dot{\mathbf{u}}\|}{v} > 0, \quad \mathbf{u}' := \frac{\dot{\mathbf{u}}}{vk}, \quad \mathbf{u}'' := \mathbf{u} \times \mathbf{u}'.$$

Here $\mathbf{u}, \mathbf{u}', \mathbf{u}'' \in \mathbb{R}^3$ form a right-handed orthonormal frame. The primes are frame labels; derivatives are indicated by dots or fractions. The instantaneous plane of curvature is spanned by $\mathbf{u}, \mathbf{u}'$. Spatial curvature k is the turning angle per unit distance along the path; the radius of its local osculating circle is $1/k$. The signed torsion $\tau \in \mathbb{R}$ is defined by

$$\tau := -\frac{d\mathbf{u}''}{d\ell} \cdot \mathbf{u}' = -\frac{\dot{\mathbf{u}}'' \cdot \mathbf{u}'}{v}.$$

Here τ denotes spatial torsion; proper time remains s . The turning of the curvature plane per unit distance is described by τ ; for a planar curved path, $\tau = 0$.

By differentiating the orthonormality relations and using $d/dt = v d/d\ell$, the Frenet equations are obtained (23, Secs. 2.4–2.6):

$$\dot{\mathbf{u}} = vk\mathbf{u}', \quad \dot{\mathbf{u}}' = -vk\mathbf{u} + v\tau\mathbf{u}'', \quad \dot{\mathbf{u}}'' = -v\tau\mathbf{u}'. \quad (10.1)$$

At $v = 0$ the tangent is undefined; at $k = 0$ the principal normal and binormal are not fixed by the path. The unsplit kinematic formulas remain valid there. On a straight segment, fixed transverse axes may be chosen and the turning terms set to zero.

10.2 Coordinate velocity, acceleration, and jerk

Differentiating $\mathbf{v} = v\mathbf{u}$ and applying (10.1) gives

$$\begin{aligned} \mathbf{v} &= v\mathbf{u}, \\ \dot{\mathbf{v}} &= \underbrace{\dot{\mathbf{v}}}_{\dot{\mathbf{v}}_{\parallel}: \text{tangential}} + \underbrace{v^2 k \mathbf{u}'}_{\dot{\mathbf{v}}_{\perp}: \text{perpendicular}}, \\ \ddot{\mathbf{v}} &= (\ddot{v} - v^3 k^2)\mathbf{u} + (3v\dot{v}k + v^2 \dot{k})\mathbf{u}' + v^3 k \tau \mathbf{u}''. \end{aligned} \quad (10.2)$$

Speed is changed by tangential acceleration; direction is changed by perpendicular acceleration. Acceleration is confined to the tangent–normal plane; a binormal component is introduced in jerk through torsion. Curvature is therefore the perpendicular-acceleration magnitude divided by speed squared. Both geometric quantities are recovered from coordinate derivatives as

$$\begin{aligned} k &= \frac{\|\dot{\mathbf{v}}_{\perp}\|}{v^2} = \frac{\|\mathbf{v} \times \dot{\mathbf{v}}\|}{v^3}, \\ \tau &= \frac{(\mathbf{v} \times \dot{\mathbf{v}}) \cdot \ddot{\mathbf{v}}}{\|\mathbf{v} \times \dot{\mathbf{v}}\|^2}. \end{aligned} \quad (10.3)$$

10.3 Spacetime derivatives in the moving frame

The earlier longitudinal rest direction $\mathcal{N}$ is extended to the varying tangent $\mathbf{u}(t)$:

$$\mathbf{u} = \gamma(1 + v\mathbf{u} \cdot \boldsymbol{\sigma}), \quad \mathcal{N} := \gamma(v + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \mathbf{u}, \mathcal{N} \in \mathbb{R} \oplus \mathbb{R}^3.$$

The four directions $\mathbf{u}, \mathcal{N}, \mathbf{u}' \cdot \boldsymbol{\sigma}, \mathbf{u}'' \cdot \boldsymbol{\sigma}$ are Minkowski-orthogonal, with determinants $1, -1, -1, -1$. Thus a spatial Frenet frame is used to resolve spacetime quantities; $\mathbf{u}$ is the spacetime tangent, while $\mathbf{u}$ is the spatial tangent. The three rest directions generally rotate relative to nonrotating carried axes.

With $\dot{\gamma} = \gamma^3 v \dot{v}$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \gamma(\mathcal{N} - v\mathbf{u})$, the complete frame evolution is

$$\begin{aligned} \dot{\mathbf{u}} &= \gamma^2 \dot{v} \mathcal{N} + \gamma v^2 k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \dot{\mathcal{N}} &= \gamma^2 \dot{v} \mathbf{u} + \gamma v k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \dot{\mathbf{u}}' \cdot \boldsymbol{\sigma} &= \gamma v^2 k \mathbf{u} - \gamma v k \mathcal{N} + v \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}, \\ \dot{\mathbf{u}}'' \cdot \boldsymbol{\sigma} &= -v \tau \mathbf{u}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

Coefficients are paired with the same sign between time and spatial directions, and with opposite signs between spatial directions. Thus Minkowski orthonormality is preserved. Differentiating the first row gives

$$\begin{aligned} \ddot{\mathbf{u}} &= (\gamma^4 \dot{v}^2 + \gamma^2 v^4 k^2) \mathbf{u} \\ &+ (\gamma^2 (\ddot{v} - v^3 k^2) + 2\gamma \dot{\gamma} \dot{v}) \mathcal{N} \\ &+ (3\gamma v \dot{v} k + 2\dot{\gamma} v^2 k + \gamma v^2 \dot{k}) \mathbf{u}' \cdot \boldsymbol{\sigma} + \gamma v^3 k \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}. \end{aligned}$$

The proper-time derivatives follow by $d/ds = \gamma d/dt$:

$$\begin{aligned} \frac{d\mathbf{u}}{ds} &= \gamma \dot{\mathbf{u}} = \gamma^3 \dot{v} \mathcal{N} + \gamma^2 v^2 k \mathbf{u}' \cdot \boldsymbol{\sigma}, \\ \frac{d^2\mathbf{u}}{ds^2} &= \gamma^2 \ddot{\mathbf{u}} + \gamma \dot{\gamma} \dot{\mathbf{u}}. \end{aligned} \tag{10.4}$$

The first is four-acceleration and the second is unprojected four-jerk. In the instantaneous rest frame, the tangential and perpendicular acceleration components are scaled by γ^3 and γ^2 , respectively, relative to their coordinate values. By orthogonality, the projection of four-jerk as in (9.4) is

$$\begin{aligned} \frac{d^2\mathbf{u}}{ds^2} + \det\left(\frac{d\mathbf{u}}{ds}\right) \mathbf{u} &= (\gamma^4 (\ddot{v} - v^3 k^2) + 3\gamma^6 v \dot{v}^2) \mathcal{N} \\ &+ (3\gamma^5 v \dot{v} k + \gamma^3 v^2 \dot{k}) \mathbf{u}' \cdot \boldsymbol{\sigma} + \gamma^3 v^3 k \tau \mathbf{u}'' \cdot \boldsymbol{\sigma}. \end{aligned} \tag{10.5}$$

The squared proper-jerk magnitude is the sum of the squares of these three rest-space coefficients. In particular, differentiating only the acceleration coefficients would omit the rotation of the frame.

Spatial curvature versus worldline curvature. The invariant curvature of the spacetime worldline is the magnitude of its proper-time tangent's derivative:

$$\sqrt{-\det\left(\frac{d\mathbf{u}}{ds}\right)} = \sqrt{\gamma^6 \dot{v}^2 + \gamma^4 v^4 k^2}. \tag{10.6}$$

In the natural units used here, this is the proper-acceleration magnitude. Both changing speed and changing spatial direction contribute. Spatial curvature k depends on the inertial frame; worldline curvature is Lorentz invariant. A spatially straight accelerated path has $k = 0$ but nonzero worldline curvature. For the trip in Section 8.6, these are $k = 0$ and $\dot{v}_0$ on each burn, respectively. In both cases the curvature is that of a path in flat spacetime; no curvature of spacetime itself is implied.

For uniform circular motion, v and k are constant and $\tau = 0$. The coordinate derivatives reduce to

$$\dot{\mathbf{v}} = v^2 k \mathbf{u}', \quad \ddot{\mathbf{v}} = -v^3 k^2 \mathbf{u}.$$

The proper-acceleration magnitude is the constant $\gamma^2 v^2 k$, while (10.5) reduces to

$$\frac{d^2 \mathcal{U}}{ds^2} + \det \left(\frac{d\mathcal{U}}{ds} \right) \mathcal{U} = -\gamma^4 v^3 k^2 \mathcal{N} \neq 0.$$

Thus constant acceleration magnitude is compatible with nonzero proper jerk: the acceleration direction is changing.

11 Coordinates, frames, and differentiation

Fields have domain $\mathbb{R}^4$ (or an open coordinate region). The symbols ∂_t , $\partial_{\mathbf{r}}$, ∂ , ∂^* , and $\square$ are differential operators. On a scalar field $f : \mathbb{R}^4 \rightarrow \mathbb{C}$, $\partial_t f, \square f \in \mathbb{C}$, $\partial_{\mathbf{r}} f \in \mathbb{C}^3$, and $\partial f, \partial^* f \in \mathbb{C} \oplus \mathbb{C}^3$ pointwise. For real f , these outputs lie in the corresponding real spaces. On a paravector field, ∂ and ∂^* use algebra multiplication, so even a real input may have a complex output.

11.1 Paravector differential operators

Cartesian inertial coordinates are denoted by $t \in \mathbb{R}$ and $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$. The scalar and vector derivatives are defined by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}.$$

The partial paravector and its sigma conjugate are defined by (8, Eq. (176))

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (11.1)$$

$$\partial^* = \text{adj } \partial = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.$$

The sigmas are constant. Derivatives act from the left on everything to their right unless parentheses specify their scope. For fields with commuting second derivatives, anticommutation cancels all mixed spatial terms:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The mixed time-space terms likewise cancel. Both orders therefore give the scalar d'Alembertian (8, Eq. (178))

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} =: \square. \quad (11.2)$$

These are operator identities: $\partial \partial^* \mathcal{Z}$ means $\partial(\partial^* \mathcal{Z})$.

11.2 Components and the product rule

Real scalar functions $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$ and real vector functions $\mathbf{X}, \mathbf{Y} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the complex paravector field

$$\mathbf{Z}(t, \mathbf{r}) = (a + bi) + (\mathbf{X} + i\mathbf{Y}) \cdot \boldsymbol{\sigma}.$$

Ordered vector multiplication gives the real and imaginary scalar and vector parts of $\partial \mathbf{Z}$:

$$\begin{aligned} \partial \mathbf{Z} &= (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{X}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{Y}) \\ &\quad + (\partial_{\mathbf{r}} a + \partial_t \mathbf{X} - \partial_{\mathbf{r}} \times \mathbf{Y}) \cdot \boldsymbol{\sigma} \\ &\quad + i(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{X} + \partial_t \mathbf{Y}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Multiplication by i exchanges real and imaginary parts, with the former imaginary parts acquiring a minus sign.

Here $\mathbf{Z} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$. For a second field $\mathbf{Z}' : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, the prime labels the field rather than a derivative. The scalar product rule is applied before any factors are reordered:

$$\begin{aligned} \partial(\mathbf{Z}\mathbf{Z}') &= (\partial_t \mathbf{Z})\mathbf{Z}' + \mathbf{Z}(\partial_t \mathbf{Z}') \\ &\quad + \sum_{n=1}^3 \sigma_n \left(\frac{\partial \mathbf{Z}}{\partial r_n} \mathbf{Z}' + \mathbf{Z} \frac{\partial \mathbf{Z}'}{\partial r_n} \right) \\ &= (\partial \mathbf{Z})\mathbf{Z}' + \mathbf{Z}(\partial \mathbf{Z}') + \sum_{n=1}^3 [\sigma_n, \mathbf{Z}] \frac{\partial \mathbf{Z}'}{\partial r_n}. \end{aligned}$$

For a complex scalar first factor, the commutators vanish and the usual product rule is recovered.

Partial derivatives are taken at fixed remaining coordinates. Along a path $\mathbf{r}(t)$ with $\mathbf{v} = \dot{\mathbf{r}}$, the total derivative is instead

$$\dot{\mathbf{Z}} = \partial_t \mathbf{Z} + \sum_{n=1}^3 \dot{r}_n \frac{\partial \mathbf{Z}}{\partial r_n} = (\partial_t + \mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{Z}.$$

All fields and partial derivatives are evaluated at $(t, \mathbf{r}(t))$.

11.3 Generalized coordinates and clock rates

Here $(t, \mathbf{r})$ are real chart coordinates. Smooth functions $a : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\mathbf{X} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, restricted to an open region with nonsingular Jacobian, specify inertial coordinates:

$$\mathbf{X} = a(t, \mathbf{r}) + \mathbf{X}(t, \mathbf{r}) \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3.$$

Here a is scalar-valued inertial time and $\mathbf{X}$ is vector-valued inertial position. Chart time t and chart position $\mathbf{r}$ need not be clock readings and Cartesian distances.

Interval. By differentiation,

$$\begin{aligned} d\mathbf{X} &= (\partial_t a + \partial_t \mathbf{X} \cdot \boldsymbol{\sigma}) dt + \sum_{k=1}^3 (\partial_{r_k} a + \partial_{r_k} \mathbf{X} \cdot \boldsymbol{\sigma}) dr_k, \\ \det(d\mathbf{X}) &= da^2 - d\mathbf{X}^2. \end{aligned}$$

This is the flat Minkowski interval in changed coordinates; varying coefficients alone do not imply spacetime curvature.

Motion and clock rates. Along a future-directed time-like path, s is proper time, $ds^2 = \det(d\mathbf{x}) > 0$, and $da > 0$. On a segment parametrized by increasing chart time t , overdots denote total t derivatives. The inertial velocity and particle Lorentz factor are

$$\mathbf{v} := \frac{d\mathbf{X}}{da} = \frac{\dot{\mathbf{X}}}{\dot{a}} \in \mathbb{R}^3, \quad \mathbf{v}^2 < 1, \quad \gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{da}{ds}.$$

Consequently,

$$\dot{s} = \sqrt{\dot{a}^2 - \dot{\mathbf{X}}^2} = \frac{\dot{a}}{\gamma}, \quad \frac{d}{ds} = \frac{\gamma}{\dot{a}} \frac{d}{dt}, \quad \mathbf{u} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (11.3)$$

Here $\dot{a} > 0$. The earlier rule $d/ds = \gamma d/dt$ is recovered when $t = a$. The two acceleration derivatives are related by

$$\dot{\mathbf{v}} = \frac{\ddot{\mathbf{X}}\dot{a} - \dot{\mathbf{X}}\ddot{a}}{\dot{a}^2}, \quad \frac{d\mathbf{v}}{da} = \frac{\dot{\mathbf{v}}}{\dot{a}}.$$

Equation (8.2) is applied with inertial time a and acceleration $d\mathbf{v}/da$. For inertial motion, $d\mathbf{u}/ds = 0$ implies $d\mathbf{v}/da = 0$, hence

$$\mathbf{X} = \mathbf{X}_0 + \mathbf{v}_0 a, \quad \mathbf{X}_0, \mathbf{v}_0 \in \mathbb{R}^3, \quad \mathbf{v}_0^2 < 1.$$

The reference inertial time is $a = 0$; the same path may be nonlinear in chart coordinates.

11.4 Varying frame transformations

A coordinate chart is held fixed while the local frame is varied along a curve. Here s is any real curve parameter; it is specialized to proper time in Section 11.5. A smooth complex paravector $\mathcal{T}(s)$ with $\det \mathcal{T} = 1$ is chosen. Here $\mathcal{T}(s), \mathcal{Z}(s) \in \mathbb{C} \oplus \mathbb{C}^3$. The paravector $\Omega(s) := (d\mathcal{T}/ds)\mathcal{T}^{-1} \in \{0\} \oplus \mathbb{C}^3$ is defined. Its scalar part vanishes because $d(\det \mathcal{T})/ds = 2\text{sc}(\Omega) = 0$. By the inverse rule,

$$\frac{d\mathcal{T}^{-1}}{ds} = -\mathcal{T}^{-1}\Omega.$$

For a complex paravector $\mathcal{Z}(s)$ transformed by similarity, $\mathcal{Z}' = \mathcal{T}\mathcal{Z}\mathcal{T}^{-1}$, product differentiation yields

$$\frac{d\mathcal{Z}'}{ds} = \mathcal{T} \frac{d\mathcal{Z}}{ds} \mathcal{T}^{-1} + [\Omega, \mathcal{Z}']. \quad (11.4)$$

Differentiating again and replacing $\mathcal{T}(d\mathcal{Z}/ds)\mathcal{T}^{-1}$ by its first-derivative expression gives

$$\begin{aligned} \frac{d^2\mathcal{Z}'}{ds^2} &= \mathcal{T} \frac{d^2\mathcal{Z}}{ds^2} \mathcal{T}^{-1} + 2 \left[\Omega, \frac{d\mathcal{Z}'}{ds} \right] \\ &\quad + \left[\frac{d\Omega}{ds}, \mathcal{Z}' \right] - [\Omega, [\Omega, \mathcal{Z}']]. \end{aligned}$$

For a real paravector transformed instead by congruence, $\mathcal{Z}' = \mathcal{T}\mathcal{Z}\mathcal{T}^H$, the same rule gives

$$\frac{d\mathcal{Z}'}{ds} = \mathcal{T} \frac{d\mathcal{Z}}{ds} \mathcal{T}^H + \Omega\mathcal{Z}' + \mathcal{Z}'\Omega^H. \quad (11.5)$$

Equations (11.4) and (11.5) include the frame terms omitted by a constant-factor transformation. For a varying rotation $\mathcal{T} = \mathcal{R} = \exp(-i\theta(s)\mathbf{u}(s) \cdot \boldsymbol{\sigma}/2)$, with real $\theta(s)$ and real unit vector $\mathbf{u}(s)$, unitarity gives $\Omega^H = -\Omega$ and $\Omega \in \{0\} \oplus i\mathbb{R}^3$, so both formulas agree.

Position-dependent sigma bases and metric constructions are developed in Section 19.

11.5 Physical derivatives in a carried frame

Here s is proper time along a future-directed time-like worldline, $\mathbf{u} = d\mathbf{x}/ds \in \mathbb{R} \oplus \mathbb{R}^3$, and $\det \mathbf{u} = 1$. From (11.5), the physical four-acceleration expressed in the local frame is

$$\mathcal{T} \frac{d\mathbf{u}}{ds} \mathcal{T}^H = \frac{d\mathbf{u}'}{ds} - \boldsymbol{\Omega} \mathbf{u}' - \mathbf{u}' \boldsymbol{\Omega}^H. \quad (11.6)$$

The derivative of the transformed components includes motion of the frame; its contribution is removed by the two $\boldsymbol{\Omega}$ terms. Similarly, $d\mathbf{x}'/ds$ includes frame terms and generally differs from the transformed tangent $\mathbf{u}'$.

If the frame is kept instantaneously at rest with the particle, then $\mathbf{u}' = \mathcal{T} \mathbf{u} \mathcal{T}^H = 1$ along the curve, so

$$\mathcal{T} \frac{d\mathbf{u}}{ds} \mathcal{T}^H = -(\boldsymbol{\Omega} + \boldsymbol{\Omega}^H) = -2 \operatorname{re}(\boldsymbol{\Omega}).$$

A constant rest-frame four-velocity therefore does not imply zero physical acceleration. Its components are encoded in the Hermitian part of $\boldsymbol{\Omega}$.

Differentiating this expression, removing the frame terms, and applying the projection in (9.4) gives

$$\mathcal{T} \left(\frac{d^2 \mathbf{u}}{ds^2} + \det \left(\frac{d\mathbf{u}}{ds} \right) \mathbf{u} \right) \mathcal{T}^H = -\frac{d(\boldsymbol{\Omega} + \boldsymbol{\Omega}^H)}{ds} + [\boldsymbol{\Omega}, \boldsymbol{\Omega}^H]. \quad (11.7)$$

Rotation of the carried axes is accounted for by the commutator. For axes kept aligned with ideal torque-free gyroscopes, $\boldsymbol{\Omega} = \boldsymbol{\Omega}^H$, so the proper jerk is $-2 d\boldsymbol{\Omega}/ds$: the proper-time rate of the measured acceleration. The fixed-axis results of Section 9.3 are recovered with $\mathcal{T} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and fixed $\mathbf{u}$.

12 Electromagnetism

All coefficient fields depend on real inertial coordinates $(t, \mathbf{r})$. The electric and magnetic fields $\mathbf{E}, \mathbf{B} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the complex field vector $\mathbf{F} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$ and its scalar-free paravector $\mathcal{F} \in \{0\} \oplus \mathbb{C}^3$ (5, Eq. (23)):

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}, \quad \mathcal{F} := \mathbf{F} \cdot \boldsymbol{\sigma}. \quad (12.1)$$

The charge density $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}$ and current density $\mathbf{J} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the real four-current paravector:

$$\mathcal{J} := \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \quad \mathcal{J}^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.$$

The paravector derivative of (11.1) is used, with spatial gradient $\partial_{\mathbf{r}}$ and d'Alembertian $\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2$. The symbols ∂ , ∂^* , and $\square$ denote differential operators.

The dot/cross product (2.8) combines metric and orientation. With a spatial derivative it gives divergence and curl, packing the six real electromagnetic components into one complex field $\mathcal{F}$. The product explains the compact form; Maxwell dynamics remains physical input, derived from an action in Section 12.3.

12.1 Maxwell's equations and wave propagation

Maxwell's four equations are the single field law (8, Eq. (177)):

$$\boxed{\partial \mathcal{F} = \mathcal{J}^*}. \quad (12.2)$$

The product rule expands its left side into

$$\begin{aligned} \partial \mathcal{F} = & \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i \partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \\ & + \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \\ & + \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \end{aligned}$$

Equating the four parts gives (7, Eqs. (1.4)–(1.7)):

$$\text{Real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (12.3)$$

$$\text{Imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (12.4)$$

$$\text{Real vector: } \partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} = -\mathbf{J}, \quad (12.5)$$

$$\text{Imaginary vector: } \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} = 0. \quad (12.6)$$

These are Gauss' electric and magnetic laws, Ampère–Maxwell's law, and Faraday's law.

SI units: Maxwell's equation

With t in seconds, $\mathbf{r}$ in metres, electric field in volts per metre, magnetic field in teslas, charge density in coulombs per cubic metre, and current density in amperes per square metre,

$$\begin{aligned} \mathcal{F} &= (\mathbf{E} + i c \mathbf{B}) \cdot \boldsymbol{\sigma}, & \partial &= \frac{1}{c} \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \\ \partial \mathcal{F} &= \frac{1}{\epsilon_0} \left(\rho - \frac{\mathbf{J}}{c} \cdot \boldsymbol{\sigma} \right). \end{aligned}$$

The electric and magnetic terms now have the same units. By separating the scalar and vector parts, the four SI Maxwell equations are recovered; the vacuum constants satisfy $\epsilon_0 \mu_0 c^2 = 1$.

In natural units, applying ∂^* gives the sourced wave equation:

$$\partial^*(\partial \mathcal{F}) = \square \mathcal{F} = \partial^* \mathcal{J}^*.$$

Taking its star yields (8, Eq. (152)):

$$\boxed{\square \mathcal{F}^* = \partial \mathcal{J}}. \quad (12.7)$$

By expansion of the source derivative,

$$\partial \mathcal{J} = (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (\partial_t \mathbf{J} + \partial_{\mathbf{r}} \rho + i \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.$$

Because $\mathcal{F}$ is scalar-free, its real and imaginary scalar parts vanish. Matching all four parts gives (7, Eq. (1.1)):

$$\text{Real scalar: } 0 = \partial_t \rho + \partial_r \cdot \mathbf{J}, \quad (12.8)$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_r \rho, \quad (12.9)$$

$$\text{Imaginary vector: } \square \mathbf{B} = \partial_r \times \mathbf{J}. \quad (12.10)$$

Equation (12.8) is charge conservation; (12.9) and (12.10) are the sourced field waves. In source-free vacuum, $\mathcal{J} = 0$ and $\square \mathcal{F} = 0$: both fields obey the free wave equation.

12.2 Potentials and gauge freedom

The scalar potential $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and vector potential $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ are combined into the real four-potential; its four-divergence is also defined:

$$\mathcal{A} := V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \quad S := \text{sc}(\partial \mathcal{A}) \in \mathbb{R}.$$

By the product rule,

$$\partial \mathcal{A} = (\partial_t V + \partial_r \cdot \mathbf{A}) + (\partial_t \mathbf{A} + \partial_r V) \cdot \boldsymbol{\sigma} + i(\partial_r \times \mathbf{A}) \cdot \boldsymbol{\sigma}.$$

Subtracting its scalar part gives the field (7, p. 105):

$$\boxed{\mathcal{F}^* = \partial \mathcal{A} - S.} \quad (12.11)$$

Thus $\partial \mathcal{A} = S + \mathcal{F}^* \in \mathbb{R} \oplus \mathbb{C}^3$, with components:

$$\text{Real scalar: } S = \partial_t V + \partial_r \cdot \mathbf{A},$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \mathbf{E} = -\partial_t \mathbf{A} - \partial_r V, \quad (12.12)$$

$$\text{Imaginary vector: } \mathbf{B} = \partial_r \times \mathbf{A}. \quad (12.13)$$

Equation (12.11) separates the four-divergence from the field; (12.12) and (12.13) satisfy (12.4) and (12.6) identically.

Applying ∂^* gives the source equations in any gauge:

$$\square \mathcal{A} = \partial^*(\partial \mathcal{A}) = \partial^*(S + \mathcal{F}^*) = \partial^* S + (\partial \mathcal{F})^*.$$

Using Maxwell's equation (12.2) gives (7, Eq. (6.1)):

$$\boxed{\square \mathcal{A} = \partial^* S + \mathcal{J}.}$$

With scalar and vector parts separated,

$$\text{Real scalar: } \square V - \partial_t S = \rho,$$

$$\text{Imaginary scalar: } 0 = 0,$$

$$\text{Real vector: } \square \mathbf{A} + \partial_r S = \mathbf{J},$$

$$\text{Imaginary vector: } \mathbf{0} = \mathbf{0}.$$

These source equations hold before gauge fixing. For a real gauge function $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$, the transformation is defined by (7, Eq. (6.2)):

$$\mathcal{A}' := \mathcal{A} + \partial^* \lambda.$$

By differentiation, $\partial \mathcal{A}' = \partial \mathcal{A} + \square \lambda$. Only the scalar changes, so the field is gauge invariant:

$$S' = S + \square \lambda, \quad \mathcal{F}' = \mathcal{F}.$$

In the Lorenz gauge $S = 0$, the potential wave equations are (7, Eq. (6.4)):

$$\text{Real scalar: } \square V = \rho,$$

$$\text{Real vector: } \square \mathbf{A} = \mathbf{J}.$$

12.3 The electromagnetic field action

For real scalar potential $V(t, \mathbf{r})$ and vector potential $\mathbf{A}(t, \mathbf{r})$, the notation $\mathcal{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ is used. With prescribed conserved current $\mathcal{J} = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$, Maxwell dynamics follows from the scalar Lagrangian density $L : \mathbb{R}^4 \rightarrow \mathbb{R}$ (7, Eq. (5.42)) (7, Eq. (5.43))

$$L = \frac{1}{2} \text{re}(\mathcal{F}^2) - \text{sc}(\mathcal{A}^* \mathcal{J}) = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho V + \mathbf{J} \cdot \mathbf{A}. \quad (12.14)$$

For boundary-vanishing variations $\delta V \in \mathbb{R}$ and $\delta \mathbf{A} \in \mathbb{R}^3$, the induced field variations $\delta \mathbf{E}, \delta \mathbf{B} \in \mathbb{R}^3$ satisfy

$$\delta \mathbf{E} = -\partial_{\mathbf{r}} \delta V - \partial_t \delta \mathbf{A},$$

$$\delta \mathbf{B} = \partial_{\mathbf{r}} \times \delta \mathbf{A}.$$

By integration by parts,

$$\delta \int L \, dt \, d^3 \mathbf{r} = \int ((\partial_{\mathbf{r}} \cdot \mathbf{E} - \rho) \delta V + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} + \mathbf{J}) \cdot \delta \mathbf{A}) \, dt \, d^3 \mathbf{r}.$$

From independent variations, (12.3) and (12.5); the potential identities supply (12.4) and (12.6). Together they recover (12.2).

12.4 Field invariants

Since $\mathcal{F} \in \{0\} \oplus \mathbb{C}^3$, its adjugate is $-\mathcal{F}$. Its square is a complex scalar, $\mathcal{F}^2 \in \mathbb{C}$ (3, p. 106)

$$\mathcal{F}^2 = -\det \mathcal{F} = \mathbf{E}^2 - \mathbf{B}^2 + 2i \mathbf{E} \cdot \mathbf{B}. \quad (12.15)$$

For a complex invertible paravector $\mathcal{T}$, the field similarity $\mathcal{F}' = \mathcal{T} \mathcal{F} \mathcal{T}^{-1}$ preserves this scalar by (5.1). Physical rotations and boosts use $\mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, respectively, with real unit vector $\mathbf{u}$ and real angle or rapidity θ ; their field laws are derived in Section 18. The real and imaginary parts of (12.15) are therefore observer-independent. At a point with a nonzero field, they give the classification

Condition	Consequence
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 > \mathbf{B}^2$	Some frame has $\mathbf{B}' = 0$.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 < \mathbf{B}^2$	Some frame has $\mathbf{E}' = 0$.
$\mathbf{E} \cdot \mathbf{B} \neq 0$	Neither field can be eliminated.
$\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 = \mathbf{B}^2$	Null field: equal, perpendicular fields.

For the first two rows, the following factors are chosen, respectively:

$$\tanh \theta \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{E^2}, \quad \tanh \theta \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{B^2}.$$

Both speeds are less than one. Substitution in (18.6) and (18.7) gives $\mathbf{B}' = 0$ and $\mathbf{E}' = 0$, respectively.

12.5 Lorentz force from the action

A dot is used for a total derivative along the particle's path with respect to coordinate time t , for example $\dot{\mathbf{P}} = d\mathbf{P}/dt$. The field derivatives ∂_t are partial derivatives at fixed position.

Notation: the real four-velocity $\mathbf{u} = d\mathbf{x}/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$, an invertible paravector with $\mathbf{u}\mathbf{u}^{-1} = 1$. Charge and constant mass are denoted by $q \in \mathbb{R}$ and $m > 0$, respectively. The real momentum is $\mathbf{P} = m\mathbf{u} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, with $E = m\gamma$ and $\mathbf{P} = m\gamma\mathbf{v}$. The real particle Lagrangian L is fixed by the action (7, Eq. (5.41))

$$\int L dt = -m \int ds - q \int \text{sc}(\mathbf{A}^* \mathbf{u}) ds. \quad (12.16)$$

Since $\text{sc}(\mathbf{A}^* \mathbf{u}) = \gamma(V - \mathbf{v} \cdot \mathbf{A})$, the Lagrangian is

$$L = -m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}.$$

Its Euler–Lagrange terms are

$$\begin{aligned} \partial_{\mathbf{v}} L &= \mathbf{P} + q\mathbf{A}, \\ \frac{d(\partial_{\mathbf{v}} L)}{dt} &= \dot{\mathbf{P}} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \\ \partial_{\mathbf{r}} L &= -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Here spatial differentiation holds the independent velocity argument fixed. Using $\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives (7, Eq. (5.32))

$$\dot{\mathbf{P}} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (12.17)$$

Here (12.12) and (12.13) turn the potential derivatives into the fields. The Hamiltonian is

$$H = \mathbf{v} \cdot (\mathbf{P} + q\mathbf{A}) - L = E + qV. \quad (12.18)$$

Using (7.17) in (12.18),

$$\dot{E} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V = -\partial_t L = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}.$$

Canceling $q\partial_t V$ and using (12.12) gives (7, p. 116)

$$\dot{E} = q\mathbf{v} \cdot \mathbf{E}. \quad (12.19)$$

This is the mechanical work rate. Thus $\mathbf{P} + q\mathbf{A}$ and $H = E + qV$ are canonical momentum and energy. The momentum $\mathbf{P}$ is kinetic; the particle energy E includes rest energy and excludes the potential energy qV . Canonical energy is conserved for static potentials in the chosen gauge. A time-dependent gauge transformation can change H while leaving E and the force unchanged.

The complex product of the field and four-velocity is expanded:

$$\begin{aligned} \frac{\mathcal{F}\mathcal{U}}{\gamma} &= \mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &+ i(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} - \mathbf{v} \times \mathbf{E}) \cdot \boldsymbol{\sigma}). \end{aligned} \quad (12.20)$$

The real scalar and vector parts reproduce (12.19) and (12.17). Since $\mathcal{P}$ is real and $d/ds = \gamma d/dt$, they combine into (10, Eq. (2))

$$\boxed{\frac{d\mathcal{P}}{ds} = q \operatorname{re}(\mathcal{F}\mathcal{U}) = \frac{q}{2}(\mathcal{F}\mathcal{U} + \mathcal{U}\mathcal{F}^H)}. \quad (12.21)$$

Hermitian extraction enforces real kinetic four-momentum and reproduces the action-derived force. The imaginary part need not vanish for an electric charge.

12.6 Acceleration and the mass shell

Notation: the real four-velocity $\mathcal{U} = d\mathcal{X}/ds$, an invertible paravector with $\mathcal{U}\mathcal{U}^{-1} = 1$ and $\mathcal{U}^{-1} = \mathcal{U}^*$, real momentum $\mathcal{P} = m\mathcal{U}$, and coordinate acceleration $\dot{\mathbf{v}} = d\mathbf{v}/dt$. Scalar-part cyclicity proves tangency to the mass shell:

$$\begin{aligned} \operatorname{sc}(\mathcal{U}^* d\mathcal{P}/ds) &= \frac{q}{2} \operatorname{sc}(\mathcal{U}^* \mathcal{F}\mathcal{U} + \mathcal{F}^H) \\ &= \frac{q}{2} (\operatorname{sc}(\mathcal{F}) + \operatorname{sc}(\mathcal{F}^H)) = 0. \end{aligned}$$

This agrees with $\operatorname{sc}(\mathcal{U}^* d\mathcal{U}/ds) = 0$ in Section 8.1. Combining (8.2) with the force components in (12.20) gives

$$\begin{aligned} m\gamma^3 \mathbf{v} \cdot \dot{\mathbf{v}} &= q\mathbf{v} \cdot \mathbf{E}, \\ m\gamma (\dot{\mathbf{v}} + \gamma^2(\mathbf{v} \cdot \dot{\mathbf{v}})\mathbf{v}) &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

The coordinate acceleration is obtained by eliminating $\mathbf{v} \cdot \dot{\mathbf{v}}$

$$\boxed{\dot{\mathbf{v}} = \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v})}.$$

The magnetic term changes momentum without doing work.

Parallel and perpendicular components. For $\mathbf{v} \neq 0$, all projections here are relative to the instantaneous velocity, as in Section 8.1:

$$\mathbf{E}_{\parallel} = \frac{(\mathbf{v} \cdot \mathbf{E})\mathbf{v}}{v^2}, \quad \mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}.$$

Since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$, projecting (12.17) and using (8.3) gives

$$\begin{aligned} \dot{\mathbf{P}}_{\parallel} &= m\gamma^3 \dot{\mathbf{v}}_{\parallel} = q\mathbf{E}_{\parallel}, \\ \dot{\mathbf{P}}_{\perp} &= m\gamma \dot{\mathbf{v}}_{\perp} = q(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

The subscripts denote projections of the force along the instantaneous velocity axis. Thus

$$\begin{aligned}\dot{\mathbf{v}}_{\parallel} &= \frac{q}{m\gamma^3} \mathbf{E}_{\parallel}, \\ \dot{\mathbf{v}}_{\perp} &= \frac{q}{m\gamma} (\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}).\end{aligned}\tag{12.22}$$

The parallel component changes speed; the perpendicular component changes direction. The magnetic force contributes only to the latter. At $\mathbf{v} = 0$, the velocity axis is undefined and the unsplit equation gives $\dot{\mathbf{v}} = q\mathbf{E}/m$.

Proper acceleration and the rest-frame field. Taking the determinant of the real force gives

$$\begin{aligned}-\det\left(\frac{d\mathcal{P}}{ds}\right) &= -m^2 \det\left(\frac{d\mathcal{U}}{ds}\right) \\ &= q^2 \gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2).\end{aligned}$$

The invariant squared proper acceleration is obtained by division by m^2 . For $\mathbf{v} \neq 0$, orthogonality gives

$$(\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2 = \gamma^{-2} \mathbf{E}_{\parallel}^2 + \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2.$$

Equivalently, substituting (12.22) into the acceleration determinant from Section 8.1 gives

$$\begin{aligned}-\det\left(\frac{d\mathcal{U}}{ds}\right) &= \gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2 \\ &= \frac{q^2}{m^2} \left(\mathbf{E}_{\parallel}^2 + \gamma^2 \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2 \right).\end{aligned}\tag{12.23}$$

The boost into the instantaneous rest frame gives $\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B})$, as derived in (18.6). Thus the squared proper acceleration is $q^2 \|\mathbf{E}'\|^2 / m^2$: the particle experiences only electric force at that instant in its rest frame.

SI units: Lorentz force and acceleration

Here t is in seconds, $\mathbf{v}$ in metres per second, m in kilograms, q in coulombs, $\mathbf{E}$ in volts per metre, and $\mathbf{B}$ in teslas. Dots denote derivatives with respect to this t , and $\gamma = (1 - \mathbf{v}^2/c^2)^{-1/2}$. For $\mathbf{v} \neq 0$, the same projections give

$$\begin{aligned}\dot{\mathbf{P}}_{\parallel} &= q\mathbf{E}_{\parallel}, & \dot{\mathbf{P}}_{\perp} &= q(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}), \\ \dot{\mathbf{v}}_{\parallel} &= \frac{q}{m\gamma^3} \mathbf{E}_{\parallel}, & \dot{\mathbf{v}}_{\perp} &= \frac{q}{m\gamma} (\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}).\end{aligned}$$

The force is in newtons and acceleration in ms^{-2} . The unsplit form, also valid at rest, is

$$\dot{\mathbf{v}} = \frac{q}{m\gamma} \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{\mathbf{v} \cdot \mathbf{E}}{c^2} \mathbf{v} \right).$$

SI units: Lorentz force and acceleration (continued)

With τ in seconds, $\mathbf{r}$ in metres, $s = c\tau$, $\mathcal{X} = ct + \mathbf{r} \cdot \boldsymbol{\sigma}$, and dimensionless $\mathcal{U} = d\mathcal{X}/ds$, the squared proper-acceleration magnitude is

$$\begin{aligned} -c^4 \det \left(\frac{d\mathcal{U}}{ds} \right) &= \gamma^6 \dot{\mathbf{v}}_{\parallel}^2 + \gamma^4 \dot{\mathbf{v}}_{\perp}^2 \\ &= \frac{q^2}{m^2} \left(\mathbf{E}_{\parallel}^2 + \gamma^2 \|\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}\|^2 \right), \end{aligned}$$

in $\text{m}^2 \text{s}^{-4}$. At rest it reduces to $q^2 \|\mathbf{E}\|^2 / m^2$ without choosing a projection axis.

12.7 Energy and momentum flux

Notation: $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$. Its Hermitian product is (8, Eq. (180))

$$\frac{1}{2} \mathcal{F} \mathcal{F}^{\text{H}} = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

The energy density $u = (\mathbf{E}^2 + \mathbf{B}^2)/2 \in \mathbb{R}_{\geq 0}$ and Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B} \in \mathbb{R}^3$ are defined. These coefficients are frame-dependent, whereas $\mathcal{F}^2$ contains the Lorentz invariants.

SI units: Field energy and flux

Using the SI field $\mathcal{F} = (\mathbf{E} + ic\mathbf{B}) \cdot \boldsymbol{\sigma}$,

$$\begin{aligned} u &= \frac{\epsilon_0}{2} (\mathbf{E}^2 + c^2 \mathbf{B}^2), & \mathbf{S} &= \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}, \\ \frac{\epsilon_0}{2} \mathcal{F} \mathcal{F}^{\text{H}} &= u + \frac{\mathbf{S}}{c} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here u is energy per unit volume in J m^{-3} , and $\mathbf{S}$ is energy flux in W m^{-2} .

In natural units, the full stress-energy tensor is represented by the real linear map (8, Eq. (167))

$$T(\mathcal{X}) := \frac{1}{2} \mathcal{F} \mathcal{X} \mathcal{F}^{\text{H}}, \quad (12.24)$$

Here $T : \mathbb{R} \oplus \mathbb{R}^3 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$ is a real-linear map, with $\mathcal{X} \in \mathbb{R} \oplus \mathbb{R}^3$; T denotes the stress map, not a frame factor. Its time-direction value is $T(1) = u + \mathbf{S} \cdot \boldsymbol{\sigma}$. It is future causal because $u \geq 0$ and

$$\det(T(1)) = u^2 - \mathbf{S}^2 = \frac{1}{4} \|\det \mathcal{F}\|^2 \geq 0.$$

Hence $\|\mathbf{S}\| \leq u$, with equality for null fields.

SI units: Field determinant and flux bound

In SI units, the determinant and flux bound are

$$\begin{aligned} u^2 - \frac{\mathbf{S}^2}{c^2} &= \frac{\epsilon_0^2}{4} \|\det \mathcal{F}\|^2 \geq 0, \\ \|\mathbf{S}\| &\leq cu. \end{aligned}$$

Equality holds for null fields.

Returning to natural units, reversal of the order gives $\mathcal{F}^H \mathcal{F} = (\mathcal{F} \mathcal{F}^H)^*$: the same energy density and opposite flux. The transformation of the full map is derived in Section 18.

Using Maxwell's vector equations,

$$\begin{aligned}\partial_t u &= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \\ &= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E},\end{aligned}$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Thus Poynting's energy-balance law is (7, Eq. (5.45))

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (12.25)$$

Equivalently, the left side is $\frac{1}{2} \partial_t \text{sc}(\mathcal{F} \mathcal{F}^H) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(\mathcal{F} \mathcal{F}^H)$. The term $\mathbf{J} \cdot \mathbf{E}$ is the power delivered to matter per unit volume.

13 Quantum amplitudes and differential operators

Here $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ has real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$, $m > 0$ is mass, $q \in \mathbb{R}$ is charge, and $\hbar = c = 1$. Amplitudes are complex functions of these real coordinates; their scalar or paravector type is specified below.

13.1 Spinor amplitudes and probability density

All products and norms here act on paravector values at one real event $\mathcal{X}$. They sum algebra components only; no spatial integration is implied. For $\varphi, \psi \in \mathbb{C} \oplus \mathbb{C}^3$, their algebraic inner product is $\langle \varphi | \psi \rangle := \text{sc}(\varphi^H \psi) \in \mathbb{C}$. Two entries separated by a vertical bar denote this inner product; one entry in angle brackets denotes a normalized expectation below.

For arbitrary constant perpendicular real unit vectors $\mathbf{u}, \mathbf{u}'$, a constant complex paravector ψ_+ of unit norm with $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_+ = \psi_+$ is chosen, and $\psi_- = (\mathbf{u}' \cdot \boldsymbol{\sigma})\psi_+$ is defined. Then

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_{\pm} = \pm \psi_{\pm}. \quad (13.1)$$

The pair is orthonormal:

$$\begin{aligned}\text{sc}(\psi_+^H \psi_+) &= 1, \\ \text{sc}(\psi_-^H \psi_-) &= 1, \\ \text{sc}(\psi_+^H \psi_-) &= \text{sc}(\psi_-^H \psi_+) = 0.\end{aligned}$$

The two cross terms are complex conjugates. A Pauli amplitude or either Dirac entry is (11, Eq. (69a))

$$\psi(t, \mathbf{r}) = c_+(t, \mathbf{r})\psi_+ + c_-(t, \mathbf{r})\psi_-. \quad (13.2)$$

Here $c_+, c_- : \mathbb{R}^4 \rightarrow \mathbb{C}$, and pointwise

$$\psi \in \text{span}_{\mathbb{C}}\{\psi_+, \psi_-\} \subset \mathbb{C} \oplus \mathbb{C}^3.$$

This fixed two-dimensional complex subspace is preserved by left multiplication by any paravector. Its coordinates are in $\mathbb{C}^2$, not the full four-complex-dimensional paravector space. Here $\psi_{\pm}$ denote

constant basis elements; in the Dirac sections they denote fields taking values in this same subspace. Left multiplication by $\mathbf{u} \cdot \boldsymbol{\sigma}$ or $\mathbf{u}' \cdot \boldsymbol{\sigma}$ preserves this two-component form; these generate all paravector left actions. The matrix verification is in Appendix A.5.

The local squared norm $\rho : \mathbb{R}^4 \rightarrow \mathbb{R}_{\geq 0}$ is (12, Eq. (1.1a))

$$\rho := \|\boldsymbol{\psi}\|^2 = \text{sc}(\boldsymbol{\psi}^H \boldsymbol{\psi}) \geq 0. \quad (13.3)$$

This ρ denotes probability density, distinct from charge density in the electromagnetic sections. Orthonormality gives $\rho = \|c_+\|^2 + \|c_-\|^2$. For a Pauli wavefunction normalized to unit total probability, this is its position density; Dirac uses the sum over its two entries. Total spatial normalization is separate from the pointwise norm. This probability interpretation does not apply to an arbitrary scalar KG field.

13.2 Weighted densities and local expectations

$\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is taken to be real ($\mathcal{Z}^H = \mathcal{Z}$), with $c \in \mathbb{R}$ and $\mathbf{Z} \in \mathbb{R}^3$. It acts by multiplication, not differentiation. For $\mathbf{Z} \neq 0$, the real unit vector $\mathbf{u} = \mathbf{Z}/\|\mathbf{Z}\|$ is used with the projectors $\boldsymbol{\Pi}_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ from Section 3.2. Their spectral decomposition gives

$$\text{sc}(\boldsymbol{\psi}^H \mathcal{Z} \boldsymbol{\psi}) = (c + \|\mathbf{Z}\|)\|\boldsymbol{\Pi}_+ \boldsymbol{\psi}\|^2 + (c - \|\mathbf{Z}\|)\|\boldsymbol{\Pi}_- \boldsymbol{\psi}\|^2. \quad (13.4)$$

For $\mathbf{Z} = 0$, this reduces to $c\|\boldsymbol{\psi}\|^2$. Thus the product is nonnegative for every $\boldsymbol{\psi}$ exactly when $c \geq \|\mathbf{Z}\|$. For such a *nonnegative weight*, the following is defined:

$$\|\boldsymbol{\psi}\|_{\mathcal{Z}}^2 := \text{sc}(\boldsymbol{\psi}^H \mathcal{Z} \boldsymbol{\psi}) = \|\sqrt{\mathcal{Z}} \boldsymbol{\psi}\|^2 \geq 0. \quad (13.5)$$

The square root has nonnegative eigenvalues, as in Section 3. For $\mathcal{Z} = 1$, this is ρ . It is a true squared norm when $c > \|\mathbf{Z}\|$; at equality it can vanish for nonzero $\boldsymbol{\psi}$, so the notation then denotes a weighted density. Projectors are an important example. Reality alone is insufficient: $\mathcal{Z} = -1$ gives $\text{sc}(\boldsymbol{\psi}^H \mathcal{Z} \boldsymbol{\psi}) = -\rho$.

Local expectations. For $\rho > 0$ and $\mathcal{Z} \in \mathbb{C} \oplus \mathbb{C}^3$ acting by multiplication, the scalar expectation $\langle \mathcal{Z} \rangle_{\boldsymbol{\psi}} \in \mathbb{C}$ is defined by (13, Eq. (6.1))

$$\langle \mathcal{Z} \rangle_{\boldsymbol{\psi}} := \frac{\text{sc}(\boldsymbol{\psi}^H \mathcal{Z} \boldsymbol{\psi})}{\rho}. \quad (13.6)$$

For real $\mathcal{Z}$, this is real and may have either sign. Dividing (13.4) by ρ shows its meaning: it averages the two eigenvalues with their projected probabilities. For fields this mean is conditional on position; it is not a spatial average. A whole-wavefunction expectation additionally requires spatial weighting. The same formula extends linearly to complex $\mathcal{Z}$, including commutators. For a nonnegative weight, $\langle \mathcal{Z} \rangle_{\boldsymbol{\psi}} = \|\boldsymbol{\psi}\|_{\mathcal{Z}}^2 / \rho$. For a projector, this ratio is a Born probability.

Transforming the amplitude. For a constant complex invertible paravector $\mathcal{T}$, the transformation $\boldsymbol{\psi}' = \mathcal{T} \boldsymbol{\psi}$ is applied. For a nonnegative weight $\mathcal{Z}$, the following is obtained by substitution:

$$\|\boldsymbol{\psi}'\|_{\mathcal{Z}}^2 = \text{sc}(\boldsymbol{\psi}^H \mathcal{T}^H \mathcal{Z} \mathcal{T} \boldsymbol{\psi}) = \|\boldsymbol{\psi}\|_{\mathcal{T}^H \mathcal{Z} \mathcal{T}}^2. \quad (13.7)$$

This also proves that the transformed weight is nonnegative. The same substitution in (13.6) applies to any complex $\mathcal{Z}$. With $\rho' = \|\psi'\|^2 > 0$, it gives

$$\langle \mathcal{Z} \rangle_{\psi'} = \frac{\rho}{\rho'} \langle \mathcal{T}^H \mathcal{Z} \mathcal{T} \rangle_{\psi}. \quad (13.8)$$

For $\mathcal{T} = \mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real angle θ and real unit $\mathbf{u}$, unitarity preserves both $\rho' = \rho$ and $\langle \varphi' | \psi' \rangle = \langle \varphi | \psi \rangle$. For $\mathcal{T} = \mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real rapidity θ , this one-sided law instead gives $\rho' = \|\psi\|_{\mathcal{L}^2}^2$; the norm and inner product are generally not invariant. The two Dirac entries use opposite boost factors, as derived in Section 18.

13.3 Energy and momentum as differential operators

Notation: $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with real coordinates $(t, \mathbf{r}) \in \mathbb{R}^4$.

SI units: Wave energy and momentum

In the following wave relations, SI units are restored and Planck's constant $h > 0$ and $\hbar = h/(2\pi)$ are retained. For a free wave, angular frequency in radians per second is denoted by $\omega \in \mathbb{R}$, and the wavenumber vector in cycles per metre by $\mathbf{k} \in \mathbb{R}^3$. For $\mathbf{k} \neq 0$, the wavelength $\lambda > 0$ satisfies $\|\mathbf{k}\| = 1/\lambda$.

The physical input is the Planck–Einstein energy relation and de Broglie momentum relation (14, p. 248) (14, p. 249):

$$E = \hbar\omega, \quad (13.9)$$

$$\mathbf{P} = \hbar\mathbf{k}. \quad (13.10)$$

Here $E \in \mathbb{R}$ is total energy in joules, including rest energy, and $\mathbf{P} \in \mathbb{R}^3$ is momentum in kg m s^{-1} . Photoelectric measurements test the energy relation (15, p. 62); electron diffraction tests the momentum relation (14, p. 255).

For constant E and $\mathbf{P}$, with time t in seconds and position $\mathbf{r}$ in metres, the complex scalar plane wave is

$$\begin{aligned} f(t, \mathbf{r}) &:= \exp(-i(\omega t - 2\pi\mathbf{k} \cdot \mathbf{r})), \\ f(t, \mathbf{r}) &= \exp\left(-\frac{i}{\hbar}(Et - \mathbf{P} \cdot \mathbf{r})\right). \end{aligned}$$

The second line substitutes (13.9) and (13.10). By differentiation,

$$\begin{aligned} +i\hbar\partial_t f &= E f, \\ -i\hbar\partial_{\mathbf{r}} f &= \mathbf{P} f. \end{aligned}$$

These operators represent energy and momentum on each mode and extend to superpositions by linearity.

Returning to natural units $\hbar = c = 1$ gives $h = 2\pi$. With $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ and $\mathcal{P} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$, the following is obtained (16, Eq. (4.104))

$$f(\mathcal{X}, \mathcal{P}) = \exp(-i(Et - \mathbf{P} \cdot \mathbf{r})). \quad (13.11)$$

The two eigenvalue equations combine into

$$\mathbf{i}\partial^* f = \mathcal{P}f. \quad (13.12)$$

The canonical energy and momentum operators are defined by

$$\begin{aligned} \hat{E} &:= +\mathbf{i}\partial_t, \\ \hat{\mathcal{P}} &:= -\mathbf{i}\partial_{\mathbf{r}}, \end{aligned} \quad (13.13)$$

which combine to form the canonical four-momentum operator:

$$\hat{\mathcal{P}} := \mathbf{i}\partial^* = \hat{E} + \hat{\mathcal{P}} \cdot \boldsymbol{\sigma}. \quad (13.14)$$

Here $\hat{E}$ is a scalar differential operator, $\hat{\mathcal{P}}$ is a triple of scalar differential operators, and $\hat{\mathcal{P}}$ is a paravector differential operator. These act on fields, not elements of the numerical coefficient spaces. On a smooth scalar field ψ , the pointwise outputs satisfy $\hat{E}\psi \in \mathbb{C}$, $\hat{\mathcal{P}}\psi \in \mathbb{C}^3$, and $\hat{\mathcal{P}}\psi \in \mathbb{C} \oplus \mathbb{C}^3$. On spinor fields, $\hat{\mathcal{P}}$ preserves the fixed subspace of (13.2). The coupled operators have the same types. In a field, the following notation is retained: the real potential paravector $\mathcal{A}(t, \mathbf{r}) = V(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$: $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, and $\mathcal{A} : \mathbb{R}^4 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$. Within the coupling formulas, subscript 0 labels the uncoupled operators of (13.13):

$$\begin{aligned} \hat{E}_0 &:= +\mathbf{i}\partial_t, \\ \hat{\mathcal{P}}_0 &:= -\mathbf{i}\partial_{\mathbf{r}}, \\ \hat{\mathcal{P}}_0 &:= \mathbf{i}\partial^*. \end{aligned}$$

The charge is $q \in \mathbb{R}$. The particle action gives canonical energy $E + qV$ and momentum $\mathbf{P} + q\mathbf{A}$; identifying them with translation operators defines the gauge-covariant energy and kinetic momentum operators (12, Eqs. (2.11)–(2.12)):

$$\hat{E}_q := \hat{E}_0 - qV, \quad (13.15)$$

$$\hat{\mathcal{P}}_q := \hat{\mathcal{P}}_0 - q\mathbf{A}, \quad (13.16)$$

which combine to form the kinetic four-momentum operator:

$$\hat{\mathcal{P}}_q := \hat{\mathcal{P}}_0 - q\mathcal{A} = \hat{E}_q + \hat{\mathcal{P}}_q \cdot \boldsymbol{\sigma}. \quad (13.17)$$

Here $\hat{E}_q$ includes rest energy and excludes the potential energy qV . Hats distinguish named differential operators from physical values and mode eigenvalues $E, \mathbf{P}$. The derivative symbols ∂ , $\partial_{\mathbf{r}}$, and $\square$ remain unhatted. Fields obtained by differentiating potentials are coefficient fields, not differential operators. The operators act to the right. The adjugate of $\hat{\mathcal{P}}_q$ reverses the vector part without conjugating $\mathbf{i}$. Operator order matters in a field; an ordinary paravector determinant cannot replace their product. The canonical operators commute, giving

$$\hat{\mathcal{P}} \operatorname{adj}(\hat{\mathcal{P}}) = \det \hat{\mathcal{P}} = \hat{E}^2 - \hat{\mathcal{P}}^2 = -\square. \quad (13.18)$$

The same result is obtained by reversal of the order.

13.4 Field strength from noncommuting derivatives

Notation: the real electric and magnetic fields from (12.12) and (12.13), and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real scalar in (12.11). The real potential is $\mathcal{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$; $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex vector paravector.

The operators are applied to a differentiable complex amplitude. The product rule cancels all derivatives of the amplitude in the commutators; writing $\hat{P}_{q,j} := (\hat{\mathbf{P}}_q)_j$ for Cartesian components $j, k = 1, 2, 3$,

$$\begin{aligned} [\hat{E}_q, \hat{P}_{q,j}] &= -iq(\partial_t \mathcal{A}_j + \partial_j V) = iqE_j, \\ [\hat{P}_{q,j}, \hat{P}_{q,k}] &= iq(\partial_j \mathcal{A}_k - \partial_k \mathcal{A}_j), \\ \hat{\mathbf{P}}_q \times \hat{\mathbf{P}}_q &= iq\mathbf{B}. \end{aligned}$$

Sigma multiplication and the energy commutator consequently give (12, Eq. (2.13)) (12, p. 6)

$$(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (13.19)$$

$$[\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}. \quad (13.20)$$

Here E_j is an electric-field component. Reversing the commutator order and combining the electric and magnetic terms also gives

$$\begin{aligned} [\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q] &= -iq\mathbf{E} \cdot \boldsymbol{\sigma}, \\ \hat{\mathbf{P}}_q^2 + iq\mathcal{F} &= (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q]. \end{aligned}$$

Using (13.19) and (13.20) in the ordered products gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{P}}_q^2 - iq\mathcal{F}, \end{aligned} \quad (13.21)$$

$$\begin{aligned} (\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 + [\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{P}}_q^2 + iq\mathcal{F}^*. \end{aligned} \quad (13.22)$$

Their common scalar part gives KG; the vector terms give the Dirac spin coupling.

14 The Klein–Gordon equation

14.1 The mass shell as a wave equation

A scalar field $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ of constant mass $m \in \mathbb{R}_{>0}$ is considered. The derivation in Section 7.5 gives the real momentum $\mathcal{P} = m\mathcal{U} \in \mathbb{R} \oplus \mathbb{R}^3$ and its mass shell $\det \mathcal{P} = E^2 - \mathbf{P}^2 = m^2$. Represent $E \in \mathbb{R}$ and $\mathbf{P} \in \mathbb{R}^3$ by the translation operators $\hat{E} = i\partial_t$ and $\hat{\mathbf{P}} = -i\partial_{\mathbf{r}}$ from (13.13). Imposing the mass shell on the scalar wavefunction gives

$$\begin{aligned} 0 &= (\det \hat{\mathcal{P}} - m^2)\psi \\ &= (\hat{E}^2 - \hat{\mathbf{P}}^2 - m^2)\psi \\ &= (-\partial_t^2 + \partial_{\mathbf{r}}^2 - m^2)\psi \\ &= -(\square + m^2)\psi. \end{aligned}$$

Here the free coefficients commute, so the operator determinant is the scalar operator in (13.18). This yields the Klein–Gordon equation (16, Eq. (1.13)):

$$\boxed{(\square + m^2)\psi = 0.} \quad (14.1)$$

14.2 Positive and negative frequency

Notation: the complex scalar field $\psi(t, \mathbf{r})$ and real mass $m > 0$. Here the branch fields $\psi_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$ are scalars, not Dirac entries. The positive scalar operator is defined by

$$\hat{\Omega} := \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \quad \hat{\Omega}^2 = \det(m + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}).$$

Since $\hat{\Omega}$ commutes with ∂_t , KG factors as

$$\square + m^2 = \partial_t^2 + \hat{\Omega}^2 = (\partial_t - i\hat{\Omega})(\partial_t + i\hat{\Omega}).$$

The reverse product is the same. Its two frequency branches obey

$$(\partial_t \pm i\hat{\Omega})\psi_{\pm} = 0, \quad \psi_{\pm}(t, \mathbf{r}) = e^{\mp i\hat{\Omega}t} \psi_{\pm}(0, \mathbf{r}).$$

Their sum gives the general solution:

$$\boxed{\psi(t, \mathbf{r}) = e^{-i\hat{\Omega}t} \psi_+(0, \mathbf{r}) + e^{i\hat{\Omega}t} \psi_-(0, \mathbf{r}).} \quad (14.2)$$

The two initial data determine its complex branch fields by

$$\psi_{\pm}(0, \mathbf{r}) = \frac{1}{2} \left(\psi(0, \mathbf{r}) \pm i\hat{\Omega}^{-1} \partial_t \psi(0, \mathbf{r}) \right).$$

14.3 Square roots from Fourier series

For spatially periodic free KG solutions, the following discrete Fourier series is used:

$$\psi = \sum_n c_n f_n, \quad f_n(t, \mathbf{r}) = \exp(i(\omega_n t + 2\pi \mathbf{k}_n \cdot \mathbf{r})).$$

Here n labels modes, $c_n \in \mathbb{C}$ are constant coefficients, $\omega_n \in \mathbb{R}$ are signed angular phase frequencies, and $\mathbf{k}_n \in \mathbb{R}^3$ are wavenumber vectors in cycles per unit length. For $\mathbf{k}_n \neq 0$, its real wavelength $\lambda_n > 0$ satisfies $\|\mathbf{k}_n\| = 1/\lambda_n$. By (13.10), the mode momentum is $2\pi \mathbf{k}_n$ in these units. The $+\omega_n t$ convention gives energy eigenvalue $-\omega_n$. By differentiation,

$$\begin{aligned} \square \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2) \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n. \end{aligned}$$

Thus each nonzero KG coefficient requires

$$\omega_n^2 = 4\pi^2 \mathbf{k}_n^2 + m^2, \quad \omega_n = \pm \sqrt{4\pi^2 \mathbf{k}_n^2 + m^2}.$$

Both signs are independent modes (16, pp. 33–34); no common time period is assumed. A square-root operator acts on the same coefficients by

$$\sqrt{\square}\psi = \sum_n \sqrt{4\pi^2 \mathbf{k}_n^2 - \omega_n^2} c_n f_n.$$

One square-root branch is chosen consistently. Squaring restores $\square$, giving

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square}).$$

For the positive spatial root, the same series gives

$$\begin{aligned}\hat{\Omega}^2\psi &= \sum_n (m^2 + 4\pi^2 \mathbf{k}_n^2) c_n f_n, \\ \hat{\Omega}\psi &= \sum_n \sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} c_n f_n.\end{aligned}$$

Thus $e^{\mp i\hat{\Omega}t}$ multiplies each spatial mode by $e^{\mp i\sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} t}$. Unlike $\hat{\Omega}$, $\sqrt{\square}$ has imaginary eigenvalues on KG modes; its branch does not select the frequency sign.

14.4 Electromagnetic coupling

Notation: the real potentials V , $\mathbf{A}$, charge q , and coupled operators $\hat{E}_q = i\partial_t - qV$, $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The real potential paravector is $\mathbf{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, with real scalar $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. The scalar equation in a field is (17, Eq. (299))

$$\boxed{(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi = 0.} \quad (14.3)$$

The mass-shell equation uses the scalar part $\hat{E}_q^2 - \hat{\mathbf{P}}_q^2$ of (13.21) or (13.22). By the product rule,

$$\hat{E}_q^2 = -\partial_t^2 - iq(\partial_t V) - 2iqV\partial_t + q^2 V^2, \quad (14.4)$$

$$\hat{\mathbf{P}}_q^2 = -\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2 \mathbf{A}^2. \quad (14.5)$$

Substituting (14.4) and (14.5) into (14.3) gives

$$\boxed{(\square + iqS - q^2 \det \mathbf{A} + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}))\psi = 0.}$$

All potential derivatives are retained.

15 The Dirac equation

15.1 Block form and free factorization

Notation: $m > 0$ and real space–time coordinates $(t, \mathbf{r})$. In the Weyl representation, a Dirac field is the two-entry block

$$\boldsymbol{\psi}(t, \mathbf{r}) = \begin{Bmatrix} \boldsymbol{\psi}_+(t, \mathbf{r}) \\ \boldsymbol{\psi}_-(t, \mathbf{r}) \end{Bmatrix}.$$

Each entry takes values in the same fixed two-dimensional complex subspace of $\mathbb{C} \oplus \mathbb{C}^3$ from (13.2). Each has coordinates in $\mathbb{C}^2$, so the block has coordinates in $\mathbb{C}^4$. On blocks, Hermitian conjugation also transposes entries; sigma conjugation acts entrywise. Summing the two local squared norms gives the nonnegative real density (16, Eqs. (4.97)–(4.99))

$$\|\boldsymbol{\psi}\|^2 = \text{sc}(\boldsymbol{\psi}_+^H \boldsymbol{\psi}_+) + \text{sc}(\boldsymbol{\psi}_-^H \boldsymbol{\psi}_-). \quad (15.1)$$

For $\boldsymbol{Z} \in \mathbb{C} \oplus \mathbb{C}^3$, the complex-linear block map is defined by

$$\boldsymbol{W}(\boldsymbol{Z}) := \begin{Bmatrix} 0 & \text{adj}(\boldsymbol{Z}) \\ \boldsymbol{Z} & 0 \end{Bmatrix}. \quad (15.2)$$

In spinor coordinates, $\boldsymbol{W}(\boldsymbol{Z})$ and $\boldsymbol{I}$ are matrices in $\mathbb{C}^{4 \times 4}$. For differential-operator arguments, $\boldsymbol{W}(\hat{\boldsymbol{P}})$, $\hat{\boldsymbol{D}}$, and $\hat{\boldsymbol{D}}_q$ are instead matrix differential operators on four-component fields. Here $\boldsymbol{I}$ is the two-entry block identity, represented by $\boldsymbol{I}_4$. Applying $\boldsymbol{W}(\boldsymbol{Z})$ twice multiplies the first entry by $\text{adj}(\boldsymbol{Z}) \boldsymbol{Z}$ and the second by $\boldsymbol{Z} \text{adj}(\boldsymbol{Z})$. Both products are the paravector determinant, so

$$\boldsymbol{W}(\boldsymbol{Z})^2 = \det(\boldsymbol{Z}) \boldsymbol{I}.$$

Sigma conjugation on each block entry gives $\boldsymbol{W}(\boldsymbol{Z})^* = \boldsymbol{W}(\boldsymbol{Z}^*)$; Hermitian conjugation also transposes the block:

$$\boldsymbol{W}(\boldsymbol{Z})^H = \boldsymbol{W}(\boldsymbol{Z}^*).$$

Thus $\boldsymbol{W}(\boldsymbol{Z})$ is not generally Hermitian. For operator arguments, adj reverses the vector part without reordering derivatives. For the free field ($q = 0$), the canonical operators in (13.13) are used. The block map and Dirac operator are

$$\begin{aligned} \boldsymbol{W}(\hat{\boldsymbol{P}}) &= \begin{Bmatrix} 0 & \text{adj}(\hat{\boldsymbol{P}}) \\ \hat{\boldsymbol{P}} & 0 \end{Bmatrix}, \\ \hat{\boldsymbol{D}}(m) &:= \boldsymbol{W}(\hat{\boldsymbol{P}}) - m \boldsymbol{I}, \\ \hat{\boldsymbol{D}}(-m) &= \boldsymbol{W}(\hat{\boldsymbol{P}}) + m \boldsymbol{I}. \end{aligned} \quad (15.3)$$

The Dirac equation is (16, Eq. (4.51))

$$\boxed{\hat{\boldsymbol{D}}(m) \boldsymbol{\psi} = 0.} \quad (15.4)$$

Using $\boldsymbol{W}(\hat{\boldsymbol{P}})^2 = (\det \hat{\boldsymbol{P}}) \boldsymbol{I} = -\square \boldsymbol{I}$ from (13.18), multiplication by the opposite-mass operator gives (16, Eqs. (4.53)–(4.54))

$$\begin{aligned} \hat{\boldsymbol{D}}(m) \hat{\boldsymbol{D}}(-m) &= \hat{\boldsymbol{D}}(-m) \hat{\boldsymbol{D}}(m) \\ &= \boldsymbol{W}(\hat{\boldsymbol{P}})^2 - m^2 \boldsymbol{I} = -(\square + m^2) \boldsymbol{I}. \end{aligned} \quad (15.5)$$

Thus every free Dirac solution satisfies the wave form of the mass shell $\det \hat{\mathcal{P}} = E^2 - \mathbf{P}^2 = m^2$ in (7.11):

$$(\det \hat{\mathcal{P}} - m^2)\psi = 0 \iff (\square + m^2)\psi = 0.$$

The original first-order constraint is the coupled pair (16, Eq. (4.108))

$$i\partial\psi_- = (\hat{E} - \hat{\mathbf{P}} \cdot \boldsymbol{\sigma})\psi_- = m\psi_+, \quad (15.6)$$

$$i\partial^*\psi_+ = (\hat{E} + \hat{\mathbf{P}} \cdot \boldsymbol{\sigma})\psi_+ = m\psi_-. \quad (15.7)$$

On one Fourier mode, the eigenvalue $s \in \mathbb{C}$ is defined by $\hat{\mathbf{D}}(m)\psi = s\psi$ for a nonzero amplitude. Here $\square$ denotes its scalar mode eigenvalue. Since $\hat{\mathbf{D}}(m) + m\mathbf{I} = \mathbf{W}(\hat{\mathcal{P}})$, (15.5) gives

$$(\hat{\mathbf{D}}(m) + m\mathbf{I})^2\psi = -\square\psi \implies ((s + m)^2 + \square)\psi = 0.$$

Thus the eigenvalues are

$$s = -m \pm i\sqrt{\square}.$$

When distinct, each allows two independent complex coefficients in ψ_+ , with $\psi_- = i\partial^*\psi_+/(s + m)$. Each root therefore occurs twice, giving the scalar characteristic equation

$$\boxed{((s + m)^2 + \square)^2 = 0.} \quad (15.8)$$

At $\square = 0$ the roots coalesce with total algebraic multiplicity four. For KG modes, $\square = -m^2$, so $s = 0, -2m$; the Dirac equation selects $s = 0$. The matrix verification is (A.14).

15.2 Reconstructing free solutions from Klein–Gordon

A KG solution $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$ is combined with the constant basis amplitude ψ_+ . For constants $a, b \in \mathbb{C}$, a two-constant construction is

$$\psi = \frac{1}{m} \left\{ \begin{matrix} ma + ib\partial \\ ia\partial^* + mb \end{matrix} \right\} \psi' \psi_+. \quad (15.9)$$

Substituting (15.9) into (15.6) and (15.7) gives

$$\begin{aligned} i\partial\psi_- - m\psi_+ &= -\frac{a}{m}(\square + m^2)\psi' \psi_+ = 0, \\ i\partial^*\psi_+ - m\psi_- &= -\frac{b}{m}(\square + m^2)\psi' \psi_+ = 0. \end{aligned}$$

Taking $a = 1, b = 0$ gives one scalar-seeded solution. To span every Dirac field, ψ is instead expressed in the form (13.2), with both scalar coefficients solving KG. Then

$$\boxed{\psi = \left\{ \begin{matrix} \psi \\ i\partial^*\psi/m \end{matrix} \right\}.} \quad (15.10)$$

The second Dirac equation holds by construction; the first is KG. Conversely, every massive free solution has this form with $\psi = \psi_+$. Equation (14.2) evolves its two coefficients.

15.3 Potential and paravector components

Notation: the real $\mathcal{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, real charge q , and $\hat{E}_q = i\partial_t - qV$, $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$. The kinetic four-momentum is $\hat{\mathcal{P}}_q = \hat{\mathcal{P}}_0 - q\mathcal{A}$ from (13.17). For the fixed potential $\mathcal{A}$ and charge q , the following interacting Dirac operator is used:

$$\begin{aligned} \hat{D}_q(m) &:= \mathbf{W}(\hat{\mathcal{P}}_q) - m\mathbf{I} \\ &= \begin{Bmatrix} -m & \text{adj}(\hat{\mathcal{P}}_q) \\ \hat{\mathcal{P}}_q & -m \end{Bmatrix}. \end{aligned} \quad (15.11)$$

Minimal coupling gives (8, Eq. (191))

$$\boxed{\hat{D}_q(m)\boldsymbol{\psi} = (\mathbf{W}(\hat{\mathcal{P}}_q) - m\mathbf{I})\boldsymbol{\psi} = 0.} \quad (15.12)$$

Its entries are

$$(\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\boldsymbol{\psi}_- = m\boldsymbol{\psi}_+, \quad (15.13)$$

$$(\hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\boldsymbol{\psi}_+ = m\boldsymbol{\psi}_-. \quad (15.14)$$

With the time derivative isolated, $i\partial_t\boldsymbol{\psi} = \hat{H}\boldsymbol{\psi}$, where the Hamiltonian is (17, Eq. (412))

$$\hat{H} := qV\mathbf{I} + \mathbf{W}(1)(m\mathbf{I} - \mathbf{W}(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})). \quad (15.15)$$

For real potentials, the Dirac equation conserves total probability when the spatial boundary flux vanishes. The pointwise density (15.1) can still vary through probability flux.

Each coefficient is exposed by the decomposition $\boldsymbol{\psi}_{\pm} = a_{\pm} + \mathbf{b}_{\pm} \cdot \boldsymbol{\sigma}$, with complex scalar functions $a_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$ and vector functions $\mathbf{b}_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Their pairs $(a_{\pm}, \mathbf{b}_{\pm})$ are restricted to the fixed spinor space, so their four complex coefficients per entry are not independent. By sigma multiplication,

$$\begin{aligned} \hat{E}_q a_- - \hat{\mathbf{P}}_q \cdot \mathbf{b}_- &= m a_+, \\ \hat{E}_q \mathbf{b}_- - \hat{\mathbf{P}}_q a_- - i\hat{\mathbf{P}}_q \times \mathbf{b}_- &= m \mathbf{b}_+, \\ \hat{E}_q a_+ + \hat{\mathbf{P}}_q \cdot \mathbf{b}_+ &= m a_-, \\ \hat{E}_q \mathbf{b}_+ + \hat{\mathbf{P}}_q a_+ + i\hat{\mathbf{P}}_q \times \mathbf{b}_+ &= m \mathbf{b}_-. \end{aligned}$$

Every operator acts on the following coefficient. Each entry still has the two independent complex components of (13.2). Appendix A.7 converts the system to the conventional four-component Dirac equation.

15.4 Interacting factorization and reconstruction

Notation: the real potential $\mathcal{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the complex field $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$, and $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$. For any field $\boldsymbol{\psi} : \mathbb{R}^4 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$, the product rule gives

$$\partial^*(\mathcal{A}^*\boldsymbol{\psi}) + \mathcal{A}\partial\boldsymbol{\psi} = (S + \mathcal{F})\boldsymbol{\psi} + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\boldsymbol{\psi}, \quad (15.16)$$

$$\partial(\mathcal{A}\boldsymbol{\psi}) + \mathcal{A}^*\partial^*\boldsymbol{\psi} = (S + \mathcal{F}^*)\boldsymbol{\psi} + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\boldsymbol{\psi}. \quad (15.17)$$

The difference of (15.16) and (15.17) is $(\mathcal{F} - \mathcal{F}^*)\psi = 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi$.

Applying the block map $\mathbf{W}$ twice gives

$$\begin{aligned}\mathbf{W}(\partial^*)\mathbf{W}(\mathcal{A})\psi &= \left\{ \begin{array}{c} \partial(\mathcal{A}\psi_+) \\ \partial^*(\mathcal{A}^*\psi_-) \end{array} \right\}, \\ \mathbf{W}(\mathcal{A})\mathbf{W}(\partial^*)\psi &= \left\{ \begin{array}{c} \mathcal{A}^*\partial^*\psi_+ \\ \mathcal{A}\partial\psi_- \end{array} \right\}, \\ \mathbf{W}(\mathcal{A})^2 &= \det \mathcal{A} \mathbf{I}.\end{aligned}$$

Differentiating a parenthesized product includes the potential derivative. The argument $-m$ changes only the mass sign, keeping q and $\mathcal{A}$ fixed. Multiplying and using the preceding mixed-derivative identities gives

$$\hat{\mathbf{D}}_q(-m)\hat{\mathbf{D}}_q(m) = (\hat{E}_q^2 - \hat{\mathbf{P}}_q^2 - m^2)\mathbf{I} - iq \begin{Bmatrix} \mathcal{F}^* & 0 \\ 0 & \mathcal{F} \end{Bmatrix}. \quad (15.18)$$

The derivatives act on potentials as well as amplitudes. For every Dirac solution,

$$\hat{\mathbf{D}}_q(-m)\hat{\mathbf{D}}_q(m)\psi = 0.$$

At $q = 0$, this recovers (15.5) and hence the free Klein–Gordon equation. With coupling, its entries obey

$$\begin{aligned}((\partial + iq\mathcal{A}^*)(\partial^* + iq\mathcal{A}) + m^2)\psi_+ &= 0, \\ ((\partial^* + iq\mathcal{A})(\partial + iq\mathcal{A}^*) + m^2)\psi_- &= 0.\end{aligned}$$

The exact second-order equations are obtained by expansion (18, Eq. (6))

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2 + iq\mathcal{F}^*)\psi_+ = 0, \quad (15.19)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2 + iq\mathcal{F})\psi_- = 0. \quad (15.20)$$

For the spin-term components, the notation $\mathbf{Z} = \mathbf{E} + i\mathbf{B}$ is used, with $\psi = a + \mathbf{b} \cdot \boldsymbol{\sigma}$, with complex functions $a : \mathbb{R}^4 \rightarrow \mathbb{C}$ and $\mathbf{b}, \mathbf{Z} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$. Here $\mathbf{Z}^*$ conjugates its scalar coefficients. By sigma multiplication,

$$\begin{aligned}iq\mathcal{F}\psi &= iq\mathbf{Z} \cdot \mathbf{b} + (iq\mathbf{a}\mathbf{Z} - q\mathbf{Z} \times \mathbf{b}) \cdot \boldsymbol{\sigma}, \\ iq\mathcal{F}^*\psi &= -iq\mathbf{Z}^* \cdot \mathbf{b} + (-iq\mathbf{a}\mathbf{Z}^* + q\mathbf{Z}^* \times \mathbf{b}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Solutions are constructed from a complex paravector amplitude ψ of (13.2) satisfying (15.20) with ψ in place of ψ_- . The Dirac field is defined by

$$\boxed{\psi = \left\{ \begin{array}{c} (\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi/m \\ \psi \end{array} \right\}.} \quad (15.21)$$

The first Dirac equation holds by construction; the second is the seed equation. Every massive solution has this form with $\psi = \psi_-$. That seed equation must be solved for the chosen potential: a scalar KG seed lacks the spin term, and independently solved squared entries need not satisfy the first-order coupling.

15.5 Gauge phase

Notation: the real scalar $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$ and gauge transformation $\mathcal{A}' = \mathcal{A} + \partial^* \lambda$, or $V' = V + \partial_t \lambda$, $\mathcal{A}' = \mathcal{A} - \partial_r \lambda$. For $\hat{E}'_q := i\partial_t - qV'$, $\hat{\mathcal{P}}'_q := -i\partial_r - q\mathcal{A}'$, and $\hat{\mathcal{P}}'_q := \hat{E}'_q + \hat{\mathcal{P}}'_q \cdot \sigma$, the product rule gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathcal{P}}_q \cdot \sigma)(e^{iq\lambda} \psi'_\pm) &= e^{iq\lambda} (\hat{E}_q - q\partial_t \lambda + \hat{\mathcal{P}}_q \cdot \sigma + q(\partial_r \lambda) \cdot \sigma) \psi'_\pm \\ &= e^{iq\lambda} (\hat{E}'_q + \hat{\mathcal{P}}'_q \cdot \sigma) \psi'_\pm. \end{aligned}$$

The adjugate block follows with the vector signs reversed. Thus the new amplitude is

$$\psi' = e^{-iq\lambda} \psi.$$

Equivalently,

$$\begin{aligned} \hat{E}'_q \psi' &= e^{-iq\lambda} \hat{E}_q \psi, \\ \hat{\mathcal{P}}'_q \psi' &= e^{-iq\lambda} \hat{\mathcal{P}}_q \psi. \end{aligned}$$

Consequently (16, Eqs. (6.68)–(6.69))

$$\begin{aligned} (\mathcal{W}(\hat{\mathcal{P}}'_q) - m\mathcal{I}) \psi' \\ = e^{-iq\lambda} (\mathcal{W}(\hat{\mathcal{P}}_q) - m\mathcal{I}) \psi. \end{aligned} \tag{15.22}$$

Equation (15.22) preserves the Dirac equation; the phase sign is fixed by the potential shift. The factor $e^{-iq\lambda} \in \mathbb{C}$ has modulus one and preserves the spinor space, so the norm is unchanged. The same product rules preserve the KG and Pauli equations.

16 Low-energy limits

Notation: $m \in \mathbb{R}_{>0}$, $q \in \mathbb{R}$, and potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$, $\mathcal{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The coupled operators are $\hat{E}_q = i\partial_t - qV$ and $\hat{\mathcal{P}}_q = -i\partial_r - q\mathcal{A}$; the fields are $\mathcal{E} = -\partial_r V - \partial_t \mathcal{A}$ and $\mathcal{B} = \partial_r \times \mathcal{A}$. The rest oscillation is removed and the leading nonrelativistic terms are retained. Schrödinger is obtained from KG, and Pauli from Dirac.

16.1 Klein–Gordon to Schrödinger

Schrödinger dynamics is obtained from the scalar Klein–Gordon equation after removal of the rest oscillation. The coupled operators $\hat{E}_q, \hat{\mathcal{P}}_q$ defined above are retained. For the full scalar KG field $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$, its scalar envelope $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ is defined by (16, Eqs. (2.103)–(2.104))

$$\psi'(t, \mathbf{r}) = e^{-imt} \psi(t, \mathbf{r}). \tag{16.1}$$

The product rule makes the rest-energy shift explicit:

$$\begin{aligned} \hat{E}_q \psi' &= (i\partial_t - qV)(e^{-imt} \psi) \\ &= e^{-imt} (m\psi + i\partial_t \psi - qV\psi) \\ &= e^{-imt} (m + \hat{E}_q) \psi, \\ \hat{\mathcal{P}}_q \psi' &= e^{-imt} \hat{\mathcal{P}}_q \psi. \end{aligned}$$

Since m is constant, applying $\hat{E}_q$ again gives $\hat{E}_q^2\psi' = e^{-imt}(\hat{E}_q + m)^2\psi$; spatial derivatives leave the phase untouched. Substitution into (14.3) therefore gives

$$\begin{aligned} 0 &= (-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi' \\ &= e^{-imt}(-(\hat{E}_q + m)^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi \\ &= e^{-imt}(-\hat{E}_q^2 - 2m\hat{E}_q + \hat{\mathbf{P}}_q^2)\psi. \end{aligned}$$

On the envelope, $\hat{E}_q$ measures energy above rest, excluding qV . The rest terms cancel. By division by the nonzero phase, the following is obtained:

$$\hat{E}_q\psi = \frac{\hat{\mathbf{P}}_q^2 - \hat{E}_q^2}{2m}\psi. \quad (16.2)$$

For weak, slowly varying fields and a slow positive-energy envelope, a real expansion parameter $0 < v \ll 1$ is chosen with $\hat{E}_q/m = O(v^2)$. Then $\hat{E}_q^2\psi$ is smaller than $2m\hat{E}_q\psi$ by $O(v^2)$. Dropping it at leading order gives (19, Eq. (21.3))

$$\boxed{i\partial_t\psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV\right)\psi.} \quad (16.3)$$

With $\mathbf{A} = 0$, hence $\mathbf{B} = 0$, the kinetic-energy operator is $-\partial_{\mathbf{r}}^2/(2m)$, and the conventional scalar form is

$$i\partial_t\psi = \left(-\frac{\partial_{\mathbf{r}}^2}{2m} + qV\right)\psi.$$

Schrödinger and Pauli occur at the same leading order $1/m$. The following Dirac limit retains the magnetic spin term; when $\mathbf{B} = 0$, each complex coefficient of the Pauli amplitude obeys this same scalar Schrödinger equation.

16.2 Dirac to Pauli

The fields $\psi_{\pm}, \psi'_{\pm}, \phi_{\pm}$ all take values in the same fixed two-dimensional complex subspace of $\mathbb{C} \oplus \mathbb{C}^3$, with coordinate columns in $\mathbb{C}^2$. The phase and change of blocks below preserve this domain.

The Dirac envelopes are defined by the same rest-energy phase shift (12, Eq. (2.1)):

$$\psi'_{\pm} := e^{imt}\psi_{\pm}. \quad (16.4)$$

By the product rule,

$$\hat{E}_q\psi_{\pm} = e^{-imt}(m + \hat{E}_q)\psi'_{\pm}.$$

On the envelopes, $\hat{E}_q$ measures energy above rest, excluding qV . The two rows of the Dirac equation, (15.13) and (15.14), become

$$\begin{aligned} (m + \hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi'_{-} &= m\psi'_{+}, \\ (m + \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})\psi'_{+} &= m\psi'_{-}. \end{aligned}$$

At free positive-energy rest, $\psi'_{+} = \psi'_{-}$. The large and small components are defined by

$$\phi_{+} := \frac{\psi'_{+} + \psi'_{-}}{\sqrt{2}}, \quad (16.5)$$

$$\phi_{-} := \frac{\psi'_{-} - \psi'_{+}}{\sqrt{2}}. \quad (16.6)$$

The exact pair is obtained by addition and subtraction (12, Eqs. (2.15)–(2.16)):

$$\hat{E}_q \phi_+ = (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \phi_-, \quad (16.7)$$

$$(2m + \hat{E}_q) \phi_- = (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \phi_+. \quad (16.8)$$

For nonrelativistic positive-energy states in weak, slowly varying fields, $\hat{E}_q \phi_-$ is neglected relative to $2m \phi_-$. The small component is substituted into the first equation, and (13.19) is applied:

$$\begin{aligned} \phi_- &\simeq \frac{\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}}{2m} \phi_+, \\ \hat{E}_q \phi_+ &\simeq \frac{(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2}{2m} \phi_+ = \frac{\hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \phi_+. \end{aligned}$$

Restoring $\hat{E}_q = i\partial_t - qV$ gives Pauli's equation through order $1/m$ (12, Eq. (2.13)):

$$\boxed{i\partial_t \phi_+ = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi_+.} \quad (16.9)$$

SI units: Pauli's equation

With m in kilograms, q in coulombs, V in volts, $\mathbf{A}$ in T m, and $\mathbf{B}$ in teslas, restoring $\hbar$ gives

$$i\hbar\partial_t \phi_+ = \left(\frac{(-i\hbar\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q\hbar}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi_+.$$

Each term in parentheses has energy units. The same nonrelativistic approximation as in (16.9) is retained.

Returning to natural units, the change of entries preserves the density exactly:

$$\|\boldsymbol{\psi}\|^2 = \|\phi_+\|^2 + \|\phi_-\|^2.$$

The leading probability density is $\|\phi_+\|^2$. Its two complex components are the Pauli amplitudes, as in Appendix A.5. This is the amplitude used in the spin discussion. For a real expansion parameter $0 < v \ll 1$, the scaling $\hat{\mathbf{P}}_q/m = O(v)$, $\hat{E}_q/m = O(v^2)$, $qV/m = O(v^2)$, $q\mathbf{B}/m^2 = O(v^2)$, $q\mathbf{E}/m^2 = O(v^3)$, with fields varying on spatial scales $1/(mv)$ and time scales $1/(mv^2)$ or longer, gives $\phi_- = O(v\phi_+)$ and omitted energy terms of order mv^4 .

16.3 Spin components in a uniform magnetic field

For a constant field $\mathbf{B} \in \mathbb{R}^3 \setminus \{0\}$, its **unit direction** and **spin projectors** are defined by:

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|} \in \mathbb{R}^3, \quad \boldsymbol{\Pi}_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \in \mathbb{R} \oplus \mathbb{R}^3.$$

Spectral decomposition gives

$$\mathbf{B} \cdot \boldsymbol{\sigma} = \|\mathbf{B}\|(\boldsymbol{\Pi}_+ - \boldsymbol{\Pi}_-), \quad \boldsymbol{\Pi}_{\pm}(\mathbf{B} \cdot \boldsymbol{\sigma}) = \pm \|\mathbf{B}\| \boldsymbol{\Pi}_{\pm}.$$

The two spin amplitudes $\phi_{++}, \phi_{+-} : \mathbb{R}^4 \rightarrow \mathbb{C}$ are encoded in an aligned paravector $\phi_+ : \mathbb{R}^4 \rightarrow \text{span}_{\mathbb{C}}\{1, \mathbf{u} \cdot \boldsymbol{\sigma}\} \subset \mathbb{C} \oplus \mathbb{C}^3$ by spectral decomposition:

$$\begin{aligned}\phi_+ &= \phi_{++} \mathbf{\Pi}_+ + \phi_{+-} \mathbf{\Pi}_-, \\ \mathbf{\Pi}_{\pm} \phi_+ &= \phi_+ \mathbf{\Pi}_{\pm} = \phi_{\pm\pm} \mathbf{\Pi}_{\pm}.\end{aligned}$$

Constant $\mathbf{\Pi}_{\pm}$ commute with $\hat{E}_q$ and $\hat{\mathbf{P}}_q^2$, so the Pauli equation gives

$$\begin{aligned}0 &= \mathbf{\Pi}_{\pm} \left(2m\hat{E}_q - \hat{\mathbf{P}}_q^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} \right) \phi_+ \\ &= \left(\left(2m\hat{E}_q - \hat{\mathbf{P}}_q^2 \pm q\|\mathbf{B}\| \right) \phi_{\pm\pm} \right) \mathbf{\Pi}_{\pm}.\end{aligned}$$

Since $\mathbf{\Pi}_{\pm} \neq 0$, the scalar coefficients vanish. Restoring $\hat{E}_q = i\partial_t - qV$ gives

$$\boxed{i\partial_t \phi_{\pm\pm} = \left(\frac{\hat{\mathbf{P}}_q^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi_{\pm\pm}.}$$

States with spin projections $\pm \frac{1}{2}$ along $\mathbf{u}$ have opposite Zeeman shifts. The kinetic momentum $\hat{\mathbf{P}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ retains the orbital coupling.

The separation ΔE of the two spin-energy contributions, excluding orbital energy, is

$$\Delta E = \frac{|q|}{m} \|\mathbf{B}\|.$$

SI units: Zeeman splitting
Restoring SI units gives $\Delta E = \frac{ q \hbar}{m} \ \mathbf{B}\ .$ This is the splitting from the derived Dirac magnetic coupling; anomalous magnetic-moment corrections are not included.

This representation applies to uniform-field Pauli evolution; the full Dirac equation and spin measurements use the fixed spinor space.

The four complex amplitudes of the Dirac spinor, after removal of the rest-energy phase, are arranged as:

$$\phi := \begin{pmatrix} \phi_{++} & \phi_{+-} \\ \phi_{-+} & \phi_{--} \end{pmatrix} \in \mathbb{C}^{2 \times 2}.$$

Dirac blocks are indexed by rows and spin projections by columns; the upper row is governed by the Pauli limit.

In a basis diagonalizing $\mathbf{u} \cdot \boldsymbol{\sigma}$, the aligned paravector is $\text{diag}(\phi_{++}, \phi_{+-})$. The corresponding spinor in the fixed subspace of (13.2) is $\phi_{++} \mathbf{Z}_+ + \phi_{+-} \mathbf{Z}_-$, where constant orthonormal $\mathbf{Z}_{\pm}$ satisfy $\mathbf{\Pi}_{\pm} \mathbf{Z}_{\pm} = \mathbf{Z}_{\pm}$ and $\mathbf{\Pi}_{\pm} \mathbf{Z}_{\mp} = 0$. Both representations give the same scalar Pauli equations. The full Dirac coupling $\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$ need not preserve alignment.

For the original spinor blocks, the lower block still satisfies $\phi_- \simeq (\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \phi_+ / (2m)$. For a nonzero field with varying direction, its projectors depend on $(t, \mathbf{r})$. With $\hat{H}$ the Pauli Hamiltonian in (16.9),

the product rule for the original spinor gives

$$\begin{aligned} i\partial_t(\mathbf{\Pi}_\pm \phi_+) &= i(\partial_t \mathbf{\Pi}_\pm) \phi_+ + \mathbf{\Pi}_\pm \hat{H} \phi_+ \\ &= \hat{H}(\mathbf{\Pi}_\pm \phi_+) + \left(i\partial_t \mathbf{\Pi}_\pm + \frac{[\mathbf{\Pi}_\pm, \hat{P}_q^2]}{2m} \right) \phi_+. \end{aligned}$$

Only the projector derivatives and kinetic commutator couple the sectors; the magnetic term commutes with its own projectors.

16.4 Evolution about an alternate spin axis

A constant unit vector $\mathbf{u}' \in \mathbb{R}^3$ and $0 \leq \theta \leq \pi$ are chosen. Initial scalar wavefunctions $\phi_{++}, \phi_{+-} : \mathbb{R}^3 \rightarrow \mathbb{C}$ are chosen along this axis:

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= \cos \theta, \quad \mathbf{\Pi}'_\pm := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}) \in \mathbb{R} \oplus \mathbb{R}^3, \\ \phi_+(0) &= \phi_{++} \mathbf{\Pi}'_+ + \phi_{+-} \mathbf{\Pi}'_- \in \mathbb{C} \oplus \mathbb{C}^3. \end{aligned}$$

The magnetic term has both diagonal and off-diagonal parts in these projectors:

$$\begin{aligned} \mathbf{\Pi}'_\pm (\mathbf{u} \cdot \boldsymbol{\sigma}) \mathbf{\Pi}'_\pm &= \pm \cos \theta \mathbf{\Pi}'_\pm, \\ \mathbf{\Pi}'_\mp (\mathbf{u} \cdot \boldsymbol{\sigma}) \mathbf{\Pi}'_\pm &= ((\mathbf{u} - \cos \theta \mathbf{u}') \cdot \boldsymbol{\sigma}) \mathbf{\Pi}'_\pm. \end{aligned}$$

By projection of the Pauli equation,

$$\begin{aligned} 2m \hat{E}_q \mathbf{\Pi}'_\pm \phi_+ &= \left(\hat{P}_q^2 \mp q \|\mathbf{B}\| \cos \theta \right) \mathbf{\Pi}'_\pm \phi_+ \\ &\quad - q \|\mathbf{B}\| \mathbf{\Pi}'_\pm (\mathbf{u} \cdot \boldsymbol{\sigma}) \mathbf{\Pi}'_\mp \phi_+. \end{aligned}$$

Initially $\mathbf{\Pi}'_\mp \phi_+(0) \mathbf{\Pi}'_\pm = 0$, but

$$i\partial_t(\mathbf{\Pi}'_\mp \phi_+ \mathbf{\Pi}'_\pm) \big|_{t=0} = -\frac{q \|\mathbf{B}\|}{2m} \phi_{+\pm} ((\mathbf{u} - \cos \theta \mathbf{u}') \cdot \boldsymbol{\sigma}) \mathbf{\Pi}'_\pm.$$

For $q \neq 0$ and $\sin \theta \neq 0$, evolution generates off-diagonal components: two scalar multiples of the fixed $\mathbf{\Pi}'_\pm$ no longer suffice. At $\theta = 0$ the earlier equations return; at $\theta = \pi$ the signs exchange. For perpendicular axes, the magnetic term is entirely off-diagonal.

For time-independent $V : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $\mathbf{A} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the orbital operator commutes with the constant magnetic term. The full Pauli evolution of this initial paravector is

$$\begin{aligned} \phi_+(t) &= e^{-it(\hat{P}_q^2/(2m) + qV)} e^{iq \|\mathbf{B}\| t \mathbf{u} \cdot \boldsymbol{\sigma} / (2m)} \phi_+(0) \\ &= e^{-it(\hat{P}_q^2/(2m) + qV)} \left(\cos \frac{q \|\mathbf{B}\| t}{2m} + i \mathbf{u} \cdot \boldsymbol{\sigma} \sin \frac{q \|\mathbf{B}\| t}{2m} \right) \phi_+(0). \end{aligned}$$

The exponential follows from $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$. Its transverse contribution produces the off-diagonal terms above.

16.5 The next free order

For the free complex scalar envelope $\psi(t, \mathbf{r})$, the conditions $qV = 0$ and $q\mathbf{A} = 0$ are imposed, so $\hat{E}_q = \hat{E}$, $\hat{\mathbf{P}}_q = \hat{\mathbf{P}}$, and the operators commute. On the positive-frequency KG branch, solving (16.2) gives

$$\hat{E}\psi = \left(\sqrt{m^2 + \hat{\mathbf{P}}^2} - m \right) \psi.$$

The same energy acts on either normalized positive-energy Dirac spin component. Iteration starts with $\hat{E}\psi \simeq \hat{\mathbf{P}}^2\psi/(2m)$; its square supplies the next term (12, Eq. (2.19)):

$$\hat{E}\psi = \left(\frac{\hat{\mathbf{P}}^2}{2m} - \frac{\hat{\mathbf{P}}^4}{8m^3} + O\left(\frac{\hat{\mathbf{P}}^6}{m^5}\right) \right) \psi.$$

For the charged scalar equation, the same iteration applies whenever $\hat{E}_q$ commutes with $\hat{\mathbf{P}}_q^2$. On the positive-energy branch,

$$\begin{aligned} \hat{E}_q\psi &= \left(\sqrt{m^2 + \hat{\mathbf{P}}_q^2} - m \right) \psi \\ &= \left(\frac{\hat{\mathbf{P}}_q^2}{2m} - \frac{\hat{\mathbf{P}}_q^4}{8m^3} + O\left(\frac{\hat{\mathbf{P}}_q^6}{m^5}\right) \right) \psi. \end{aligned}$$

This expansion applies when the kinetic momenta in the state are small compared with m .

17 Spin and probability

Here ψ takes values in the fixed spinor space of (13.2), with coordinates in $\mathbb{C}^2$. Axes and polarizations $\mathbf{u}, \mathbf{u}', \mathbf{a} \in \mathbb{R}^3$ are unit vectors; angles $\theta, \varphi \in \mathbb{R}$, projectors $\Pi_\pm, \Pi'_\pm \in \mathbb{R} \oplus \mathbb{R}^3$, and probabilities lie between 0 and 1, inclusive. The density $\rho \in \mathbb{R}_{\geq 0}$ is a probability density.

Spin is specified by the intrinsic rotation of an amplitude. Here the state is rotated at fixed coordinates. A scalar KG amplitude stays unchanged under this intrinsic action and has spin zero; a two-component amplitude has the one-sided law below. The wave equation alone does not fix spin: free Dirac components also satisfy KG.

17.1 Spin one-half and rotations

Notation: the complex paravector amplitude $\psi(t, \mathbf{r})$ of (13.2). For a real unit axis $\mathbf{u}$ and real angle θ , the following unitary rotation factor is used: $\mathcal{R} = \exp(-i\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$ of (4.5). The spinor law is (11, p. 4)

$$\boxed{\psi' = \mathcal{R}\psi.} \tag{17.1}$$

Writing its infinitesimal action as $\psi' = \psi - i\theta\mathcal{S}(\mathbf{u})\psi + O(\theta^2)$ defines the spin operator along $\mathbf{u}$, given by left multiplication by the real pure-vector paravector $\mathcal{S}(\mathbf{u}) \in \{0\} \oplus \mathbb{R}^3$ (13, p. 62):

$$\mathcal{S}(\mathbf{u})\psi = \frac{1}{2}(\mathbf{u} \cdot \boldsymbol{\sigma})\psi. \tag{17.2}$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$, its two spin projections are $\pm 1/2$ in units $\hbar = 1$. A 2π rotation gives $\mathcal{R} = -1$, reversing the amplitude's sign; a 4π rotation restores it. Probabilities are unchanged by the overall

sign, while a relative sign can be detected by interference (20, Eq. (6.18)). Each Dirac entry obeys this same active rotation law.

For a real paravector observable $\mathcal{Z}$, the state law (17.1) gives, by (13.8),

$$\langle \mathcal{Z} \rangle_{\psi'} = \langle \mathcal{R}^H \mathcal{Z} \mathcal{R} \rangle_{\psi}.$$

The observable inside the original state's pairing rotates through $-\theta$. If the observable rotates with the state, $\mathcal{Z}' = \mathcal{R} \mathcal{Z} \mathcal{R}^H$, then $\langle \mathcal{Z}' \rangle_{\psi'} = \langle \mathcal{Z} \rangle_{\psi}$.

17.2 Projectors and probabilities

Notation: a nonzero Pauli amplitude ψ of (13.2). For a real unit measurement axis $\mathbf{u}$, the real projectors are

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).$$

Their algebra in Section 3.2 gives

$$\begin{aligned} \psi &= \Pi_+ \psi + \Pi_- \psi, \\ \mathcal{S}(\mathbf{u})(\Pi_{\pm} \psi) &= \pm \frac{1}{2} \Pi_{\pm} \psi, \\ \text{sc}\left((\Pi_+ \psi)^H (\Pi_- \psi)\right) &= 0, \\ \|\Pi_{\pm} \psi\|^2 &= \text{sc}\left(\psi^H \Pi_{\pm} \psi\right) = \|\psi\|_{\Pi_{\pm}}^2. \end{aligned}$$

By (13.5), these are nonnegative weighted densities. The Born probability is this density divided by $\rho = \|\psi\|^2$. In the fixed orthonormal basis (13.1), $\psi = c_+ \psi_+ + c_- \psi_-$ with complex scalars $c_{\pm}$, so $\rho = \|c_+\|^2 + \|c_-\|^2$. When the measurement axis is that basis axis, the probabilities are $\|c_{\pm}\|^2/\rho$. The same-side and cross-side projector products are already derived in Section 3.2.

The real spin-polarization vector $\mathbf{u}' \in \mathbb{R}^3$ is defined by its projection along every real unit measurement axis $\mathbf{u}$:

$$\mathbf{u} \cdot \mathbf{u}' := \frac{\text{sc}(\psi^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi)}{\rho}. \quad (17.3)$$

The right side is real and linear in $\mathbf{u}$, so it uniquely specifies $\mathbf{u}'$. For the arbitrary perpendicular basis axes $\mathbf{u}, \mathbf{a}$ in (13.2), expanding the same pairing gives

$$\begin{aligned} \rho \mathbf{u}' &= (\|c_+\|^2 - \|c_-\|^2) \mathbf{u} \\ &\quad + 2 \text{re}(c_+^* c_-) \mathbf{a} + 2 \text{im}(c_+^* c_-) (\mathbf{u} \times \mathbf{a}). \end{aligned}$$

These three directions are orthonormal, hence

$$\rho^2 (\mathbf{u}')^2 = (\|c_+\|^2 - \|c_-\|^2)^2 + 4 \|c_+^* c_-\|^2 = \rho^2.$$

Thus $\mathbf{u}'$ is a real unit vector, with mean spin $\mathbf{u}'/2$. For any real unit measurement axis, inserting $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ directly into the pairing gives

$$\|\Pi_{\pm} \psi\|^2 = \text{sc}\left(\psi^H \Pi_{\pm} \psi\right) = \frac{1}{2} \rho (1 \pm \mathbf{u} \cdot \mathbf{u}').$$

Therefore (11, Eqs. (66)–(67))

$$\boxed{\text{prob}(\pm | \mathbf{u}) = \frac{\|\Pi_{\pm} \psi\|^2}{\rho} = \frac{1}{2} (1 \pm \mathbf{u} \cdot \mathbf{u}').} \quad (17.4)$$

The mean spin along $\mathbf{u}$ is $(\text{prob}(+) - \text{prob}(-))/2 = \mathbf{u} \cdot \mathbf{u}'/2$. For a spatial wavefunction, (17.4) is the spin probability conditional on position, not a position density.

17.3 Measurement along a second axis

The prepared spin direction and measurement axis are denoted by the real unit vectors $\mathbf{u}'$ and $\mathbf{u}$, respectively. A nonzero spin-up amplitude $\psi = \Pi'_+ \psi$ is prepared, where $\Pi'_+ = (1 + \mathbf{u}' \cdot \boldsymbol{\sigma})/2$. The angle between the axes is denoted by $0 \leq \theta \leq \pi$, and the following notation is used:

$$K' := \mathbf{u}' \cdot \boldsymbol{\sigma}, \quad K := \mathbf{u} \cdot \boldsymbol{\sigma}, \quad c' := \mathbf{u} \cdot \mathbf{u}' = \cos \theta.$$

Here $K, K' \in \{0\} \oplus \mathbb{R}^3$ and $-1 \leq c' \leq 1$. The Pauli product gives $(K')^2 = 1$ and $K'K + KK' = 2c'$. Multiplication by K' gives $K'KK' = 2c'K' - K$, hence

$$\begin{aligned} \Pi'_+ K \Pi'_+ &= \frac{1}{4}(1 + K')K(1 + K') \\ &= \frac{1}{4}(K + K'K + KK' + K'KK') \\ &= \frac{1}{4}(2c' + 2c'K') = c'\Pi'_+. \end{aligned}$$

The measurement projectors $\Pi_{\pm} = (1 \pm K)/2$ therefore satisfy

$$\Pi'_+ \Pi_{\pm} \Pi'_+ = \frac{1}{2} (\Pi'_+ \pm \Pi'_+ K \Pi'_+) = \frac{1 \pm c'}{2} \Pi'_+. \quad (17.5)$$

Using $\|\mathbf{z}\|^2 = \text{sc}(\mathbf{z}\mathbf{z}^H)$ for $\mathbf{z} \in \mathbb{C} \oplus \mathbb{C}^3$, the denominator is

$$\|\psi\|^2 = \text{sc}(\psi\psi^H) > 0.$$

For the prepared state, the projected squared norm is

$$\begin{aligned} \|\Pi_{\pm} \psi\|^2 &= \text{sc}((\Pi_{\pm} \psi)(\Pi_{\pm} \psi)^H) \\ &= \text{sc}(\Pi_{\pm} \psi \psi^H) \\ &= \text{sc}(\Pi'_+ \Pi_{\pm} \Pi'_+ \psi \psi^H) \\ &= \frac{1 \pm \cos \theta}{2} \text{sc}(\psi \psi^H) = \frac{1 \pm \cos \theta}{2} \|\psi\|^2. \end{aligned}$$

The Born rule divides by the nonzero squared norm, so no prior normalization is needed:

$$\text{prob}(\pm) := \frac{\|\Pi_{\pm} \psi\|^2}{\|\psi\|^2} = \frac{1 \pm \cos \theta}{2}, \quad (17.6)$$

$$\boxed{\text{prob}(+) = \cos^2 \frac{\theta}{2}, \quad \text{prob}(-) = \sin^2 \frac{\theta}{2}.}$$

Here $+$ and $-$ label measured spin projections $+1/2$ and $-1/2$ along $\mathbf{u}$. The probabilities sum to one. Aligned axes give spin up with certainty, perpendicular axes give equal probabilities, and opposite axes give spin down with certainty. Preparing $\psi = \Pi'_- \psi$ interchanges the probabilities. For a spatial wavefunction with the same spin preparation at every position, integrating both norm densities gives the same result.

17.4 Rotating the polarization

For $\mathcal{R} = \exp(-i\varphi \mathbf{a} \cdot \boldsymbol{\sigma}/2)$, with real unit rotation axis $\mathbf{a}$ and real angle φ , the polarization of the rotated amplitude $\psi' = \mathcal{R}\psi$ is denoted by $\mathbf{u}''$. The amplitude law (17.1) gives

$$\mathbf{u} \cdot \mathbf{u}'' = \langle \mathcal{R}^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \mathcal{R} \rangle_\psi.$$

This is the inverse of Rodrigues' rotation (5.5): the measurement axis rotates through $-\varphi$, so $\mathbf{u}''$ rotates through $+\varphi$ about $\mathbf{a}$. Thus the polarization rotates as a vector although its amplitude has spin one-half. Its overlap with the original polarization is

$$\mathbf{u}' \cdot \mathbf{u}'' = (\mathbf{a} \cdot \mathbf{u}')^2 + (1 - (\mathbf{a} \cdot \mathbf{u}')^2) \cos \varphi.$$

For $\mathbf{a} \cdot \mathbf{u}' = 0$, the probability to retain the initial spin is $\cos^2(\varphi/2)$.

18 Lorentz covariance and electromagnetic transformations

At corresponding events, the domains are

$$\begin{aligned} \mathcal{X}, \mathcal{U}, \mathcal{P}, \mathcal{A}, \mathcal{J} &\in \mathbb{R} \oplus \mathbb{R}^3, & \mathcal{F} &\in \{0\} \oplus \mathbb{C}^3, \\ \mathcal{T} &\in \mathbb{C} \oplus \mathbb{C}^3, \quad \det \mathcal{T} = 1, & \mathcal{R} &\in \mathbb{R} \oplus i\mathbb{R}^3, \quad \mathcal{L} \in \mathbb{R} \oplus \mathbb{R}^3, \\ t, s, E, V, \rho, \theta, \beta, \gamma &\in \mathbb{R}, & \mathbf{r}, \mathbf{v}, \mathbf{P}, \mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J}, \mathbf{u} &\in \mathbb{R}^3. \end{aligned}$$

Here $\mathbf{u}^2 = 1$, $|\beta| < 1$, $\gamma \geq 1$, and ρ is charge density. Transformed quantities retain their types. Scalar KG fields take values in $\mathbb{C}$; Dirac entries remain in their fixed two-dimensional complex spinor space.

The algebraic actions of Section 5 are now applied to the physical variables already defined. Appendix C compares alternative exponent signs and sandwich orders. Primed fields are evaluated at $\mathcal{X}'$ and unprimed fields at $\mathcal{X}$. The frame factors here are constant; varying factors are treated in Section 11.4.

18.1 Transformation laws for fields and derivatives

$\mathcal{T}$ is taken to be a complex invertible paravector with $\det \mathcal{T} = 1$. It may be any ordered product of $\mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ and $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, with real unit vector $\mathbf{u}$ and real angle or rapidity θ . Notation: the real event $\mathcal{X}$, momentum $\mathcal{P}$, potential $\mathcal{A}$, and electric current $\mathcal{J}$. Each transforms by (5, Eq. (24))

$$\mathcal{Z}' = \mathcal{T} \mathcal{Z} \mathcal{T}^H, \quad \mathcal{Z} = \mathcal{X}, \mathcal{P}, \mathcal{A}, \mathcal{J}. \quad (18.1)$$

Since $\det \mathcal{T} = 1$, sigma conjugation gives $\mathcal{T}^* = (\mathcal{T}^{-1})^H$. The derivative and complex field $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ use (5, Eq. (33))

$$\partial' = \mathcal{T}^* \partial \mathcal{T}^{-1}, \quad (18.2)$$

$$\mathcal{F}' = \mathcal{T} \mathcal{F} \mathcal{T}^{-1}. \quad (18.3)$$

Equation (18.2) follows from the chain rule. For a differentiable scalar $f : \mathbb{R}^4 \rightarrow \mathbb{C}$ with $f'(\mathcal{X}') = f(\mathcal{X})$,

$$df = \text{sc}((\partial f) d\mathcal{X}) = \text{sc}((\mathcal{T}^*(\partial f) \mathcal{T}^{-1}) d\mathcal{X}').$$

Since this holds for every real increment $d\mathcal{X}'$, the coefficient is $\partial' f'$ (7, p. 101).

18.2 Maxwell covariance

The field law follows from the potential law. The identity (12.11), with real scalar $S = \text{sc}(\partial\mathcal{A})$, gives

$$\partial'\mathcal{A}' = \mathcal{T}^*(S + \mathcal{F}^*)\mathcal{T}^H.$$

By scalar-part cyclicity, $S' = S$. Since $\mathcal{T}^* = (\mathcal{T}^H)^{-1}$, sigma conjugation of the vector part gives (18.3). Maxwell covariance now follows by cancellation:

$$\partial'\mathcal{F}' = \mathcal{T}^*(\partial\mathcal{F})\mathcal{T}^{-1} = \mathcal{T}^*\mathcal{J}^*\mathcal{T}^{-1} = \mathcal{J}'^*.$$

Thus the transformed potentials and fields satisfy the same Maxwell equation (12.2); the scalar wave operator also obeys $\square' = \det \partial' = \det \partial = \square$.

18.3 Wave phases and spinor covariance

For real $\mathcal{P}, \mathcal{X}$, the metric pairing (5.3) is the wave phase. Adjunction of $\mathcal{P}'$ and cyclicity give

$$\text{sc}(\mathcal{P}'^* \mathcal{X}') = \text{sc}(\mathcal{T}^* \mathcal{P}^* \mathcal{X} \mathcal{T}^H) = \text{sc}(\mathcal{P}^* \mathcal{X}).$$

Consequently $E't' - \mathbf{P}' \cdot \mathbf{r}' = Et - \mathbf{P} \cdot \mathbf{r}$, and the plane wave (13.11) satisfies $f(\mathcal{X}', \mathcal{P}') = f(\mathcal{X}, \mathcal{P})$.

A scalar Klein–Gordon field uses $\psi'(\mathcal{X}') = \psi(\mathcal{X})$; the derivative and potential laws above preserve its wave equation. For the two complex paravector Dirac entries, a change of coordinates $\mathcal{X}' = \mathcal{T}\mathcal{X}\mathcal{T}^H$ uses

$$\begin{aligned}\psi'_+(\mathcal{X}') &= \mathcal{T}^* \psi_+(\mathcal{X}), \\ \psi'_-(\mathcal{X}') &= \mathcal{T} \psi_-(\mathcal{X}).\end{aligned}$$

Primed coupled operators use the primed coordinates and potentials. The first interacting equation (15.13) transforms by cancellation:

$$(\hat{E}'_q - \hat{\mathbf{P}}'_q \cdot \boldsymbol{\sigma}) \psi'_- = \mathcal{T}^*(\hat{E}_q - \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}) \psi_- = m \psi'_+.$$

Equation (15.14) transforms in the same way with left factor $\mathcal{T}$; $q = 0$ gives the free case. For $\mathcal{T} = \mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, both entries transform by $\mathcal{R}$, consistently with the active rotation (17.1); the spatial argument also rotates. A passive rotation of axes uses the inverse factor $\mathcal{T} = \mathcal{R}^{-1}$. For $\mathcal{T} = \mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, the two entries instead use $\mathcal{L}^{-1}$ and $\mathcal{L}$, respectively (16, Eq. (4.31)). The factor $\mathcal{L}$ already contains half the rapidity; no further square root is needed for either entry. Their cross-pairing $\text{sc}(\psi_+^H \psi_-)$ is invariant: the two transformation factors cancel. Their summed positive density (15.1) is observer-dependent under boosts. The full Dirac pair is used for boosts before the Pauli low-energy limit is taken.

18.4 Boosted fields and energy flux

For $\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ with the parameters above, (18.2) becomes $\partial' = \mathcal{L}^{-1} \partial \mathcal{L}^{-1}$. Its scalar and vector parts are

$$\begin{aligned}\partial_{t'} &= \gamma(\partial_t + \beta \mathbf{u} \cdot \partial_{\mathbf{r}}), \\ \partial_{\mathbf{r}'} &= \partial_{\mathbf{r}} + (\gamma - 1) \mathbf{u}(\mathbf{u} \cdot \partial_{\mathbf{r}}) + \gamma \beta \mathbf{u} \partial_t.\end{aligned}$$

These inverse signs express the chain rule. The real potential $\mathcal{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ and current $\mathcal{J} = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$ follow the same component formulas as (7.12)–(7.13), with $(E, \mathbf{P})$ replaced by $(V, \mathbf{A})$ or $(\rho, \mathbf{J})$.

At corresponding events, (18.3) reduces to

$$\mathcal{F}' = \mathcal{L} \mathcal{F} \mathcal{L}^{-1}. \quad (18.4)$$

The similarity preserves the field's zero scalar part, while its imaginary vector coefficients represent the magnetic field. As shown algebraically in (5.14), a real vector can acquire an imaginary part under this action. In contrast, a nonzero boost congruence would give a scalar part to a nonzero field parallel to $\mathbf{u}$. Thus the field uses similarity, whereas events and the real four-potential use congruence. Both the field coefficients and their spacetime arguments transform:

$$\mathcal{F}'(\mathcal{X}') = \mathcal{L} \mathcal{F} (\mathcal{L}^{-1} \mathcal{X}' \mathcal{L}^{-1}) \mathcal{L}^{-1}. \quad (18.5)$$

Here $\mathcal{X} = \mathcal{L}^{-1} \mathcal{X}' \mathcal{L}^{-1}$. To expand (18.4), (5.14) is applied to $\mathbf{E} + i\mathbf{B}$ and separate its real and imaginary parts (7, Eqs. (5.17)–(5.18)):

$$\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (18.6)$$

$$\mathbf{B}' = \mathbf{B}_{\parallel} + \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (18.7)$$

Parallel components are unchanged. These are the usual boosted fields along any real unit axis. The inverse field law is obtained by reversing β .

Similarity preserves $\mathcal{F}'^2 = \mathcal{L} \mathcal{F}^2 \mathcal{L}^{-1} = \mathcal{F}^2$, since $\mathcal{F}^2$ is scalar. Consequently (12.15) gives

$$\begin{aligned} \mathbf{E}'^2 - \mathbf{B}'^2 &= \mathbf{E}^2 - \mathbf{B}^2, \\ \mathbf{E}' \cdot \mathbf{B}' &= \mathbf{E} \cdot \mathbf{B}. \end{aligned}$$

In contrast, $\|\mathcal{F}\|^2 = \mathbf{E}^2 + \mathbf{B}^2$ is twice the observer-dependent energy density. For example, if $\mathbf{B} = 0$ and $\mathbf{u} \cdot \mathbf{E} = 0$, (18.6) and (18.7) give $\|\mathcal{F}'\|^2 = \gamma^2(1 + \beta^2)\mathbf{E}^2$, while $\mathbf{E}'^2 - \mathbf{B}'^2 = \mathbf{E}^2$.

Notation: the stress map $T(\mathcal{X}) = \mathcal{F} \mathcal{X} \mathcal{F}^H / 2$ from (12.24), with real paravector input $\mathcal{X}$; here T denotes this map. Since $\mathcal{X}' = \mathcal{L} \mathcal{X} \mathcal{L}$,

$$T'(\mathcal{X}') = \frac{1}{2} \mathcal{F}' \mathcal{X}' \mathcal{F}'^H = \mathcal{L} T(\mathcal{X}) \mathcal{L}.$$

Here T' is the map formed from $\mathcal{F}'$. To measure the primed energy density and Poynting vector, $\mathcal{X}' = 1$ is used, hence $\mathcal{X} = \mathcal{L}^{-2}$:

$$T'(1) = \mathcal{L} T(\mathcal{L}^{-2}) \mathcal{L} = \frac{1}{2} \mathcal{F}' \mathcal{F}'^H.$$

The full map transforms covariantly; its value at 1 depends on the observer's time direction.

Finally, the following notation is retained: the real four-velocity $\mathbf{u} = d\mathcal{X}/ds$, an invertible paravector with $\mathbf{u} \mathbf{u}^{-1} = 1$, and real momentum $\mathcal{P} = m\mathbf{u}$, with real charge q and constant mass $m > 0$. With $\mathbf{u}' = \mathcal{L} \mathbf{u} \mathcal{L}$ and $\mathcal{P}' = \mathcal{L} \mathcal{P} \mathcal{L}$, the force law (12.21) transforms as

$$\frac{d\mathcal{P}'}{ds} = \mathcal{L} \frac{q}{2} (\mathcal{F} \mathbf{u} + \mathbf{u} \mathcal{F}^H) \mathcal{L} = \frac{q}{2} (\mathcal{F}' \mathbf{u}' + \mathbf{u}' \mathcal{F}'^H).$$

Thus the electric and magnetic components above reproduce the same real four-force in the moving frame; the Hermitian extraction is preserved.

19 Position-dependent paravector bases

Section 11.3 starts with an inertial coordinate map $\mathcal{X} = a(t, \mathbf{r}) + \mathbf{X}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$. Its exact differential gives

$$d\mathcal{X} = da + d\mathbf{X} \cdot \boldsymbol{\sigma}, \quad ds^2 = da^2 - d\mathbf{X}^2.$$

This is flat spacetime in generalized coordinates. In the block notation used below,

$$\begin{Bmatrix} da \\ d\mathbf{X} \end{Bmatrix} = \mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix};$$

so $\mathbf{M}^\top$ is the Jacobian of the same functions $a, \mathbf{X}$.

Here we assign a local displacement directly:

$$d\mathcal{X} = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r, \quad ds^2 = \det(d\mathcal{X}).$$

A smooth real invertible $\mathbf{M}(t, \mathbf{r})$ specifies the basis elements, without requiring $\mathbf{M}^\top$ to be a coordinate Jacobian. The notation $d\mathcal{X}$ now denotes that local assignment; it need not be an exact differential, even locally. For fixed coordinate increments $dt, d\mathbf{r}$, the basis specifies their metric. This is the usual local-frame (tetrad) construction, expressed in paravector form (24, Eq. (3.119)). The product, conjugations, and determinant retain their earlier definitions.

Generalized coordinates are a flat special case. A general basis can describe curvature, although position dependence or failure of exactness alone does not establish it: a varying Lorentz basis can still give a flat metric. We use the metric's unique torsion-free, metric-compatible parallel transport to determine free fall and curvature. This is the standard geometric input; Einstein's field equation is additional physical input selecting metrics from matter and boundary conditions.

19.1 The basis and its interval

Let $\mathbf{M}(t, \mathbf{r})$ be a smooth real invertible 4×4 matrix, partitioned as

$$\mathbf{M} = \begin{Bmatrix} M_{tt} & \mathbf{M}_{tr} \\ \mathbf{M}_{rt} & \mathbf{M}_{rr} \end{Bmatrix}.$$

Here M_{tt} is a scalar, $\mathbf{M}_{tr}$ a 1×3 row, $\mathbf{M}_{rt}$ a 3×1 column, and $\mathbf{M}_{rr}$ a 3×3 matrix. The blocks are independent: no symmetry of $\mathbf{M}$ is assumed, and $\mathbf{M}_{rt} = \mathbf{M}_{tr}^\top$ is not required. Let $\mathbf{I}$ be the 3×3 identity and $\mathbf{0}$ the zero three-vector. The local basis is

$$\begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix} = \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \quad (19.1)$$

Here σ'_t is one real paravector and $\boldsymbol{\sigma}'_r$ is a triple of real paravectors, all Hermitian. Their scalar and vector parts are

$$\sigma'_t = M_{tt} + \mathbf{M}_{tr}^\top \cdot \boldsymbol{\sigma}, \quad \boldsymbol{\sigma}'_r = \mathbf{M}_{rt} + \mathbf{M}_{rr} \boldsymbol{\sigma}.$$

In the second expression, $\mathbf{M}_{rt}$ collects the three scalar parts and each row of $\mathbf{M}_{rr}$ gives the corresponding vector part. The labels t, r identify basis entries, not algebraic scalar and vector grades. Sigma conjugation acts componentwise and leaves the real entries of $\mathbf{M}$ unchanged:

$$\begin{aligned} \begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix}^* &= \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}^* = \mathbf{M} \begin{Bmatrix} 1 \\ -\boldsymbol{\sigma} \end{Bmatrix} \\ &= \mathbf{M} \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & -\mathbf{I} \end{Bmatrix} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \end{aligned} \quad (19.2)$$

Write the local displacement as $d\mathbf{x} = dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r$. Its interval is

$$\boxed{ds^2 = \det(dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r).} \quad (19.3)$$

Its scalar–vector coefficient column in the fixed basis is $\mathbf{M}^\top \{dt, d\mathbf{r}\}^\top$, as derived below. Taking the scalar part squared minus the vector part squared gives the metric matrix

$$\begin{aligned} \mathbf{g} &= \mathbf{M} \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & -\mathbf{I} \end{Bmatrix} \mathbf{M}^\top = \begin{Bmatrix} g_{tt} & \mathbf{g}_{tr} \\ \mathbf{g}_{tr}^\top & \mathbf{g}_{rr} \end{Bmatrix}, \\ ds^2 &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \mathbf{g} \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} \\ &= g_{tt} dt^2 + 2 dt \mathbf{g}_{tr} d\mathbf{r} + d\mathbf{r}^\top \mathbf{g}_{rr} d\mathbf{r}. \end{aligned} \quad (19.4)$$

Here g_{tt} is a real scalar, $\mathbf{g}_{tr}$ a real 1×3 row, and $\mathbf{g}_{rr}$ a real symmetric 3×3 matrix. Explicitly,

$$\begin{aligned} g_{tt} &= M_{tt}^2 - \mathbf{M}_{tr} \mathbf{M}_{tr}^\top, \\ \mathbf{g}_{tr} &= M_{tt} \mathbf{M}_{rt}^\top - \mathbf{M}_{tr} \mathbf{M}_{rr}^\top, \\ \mathbf{g}_{rr} &= \mathbf{M}_{rt} \mathbf{M}_{rt}^\top - \mathbf{M}_{rr} \mathbf{M}_{rr}^\top. \end{aligned}$$

Invertibility of $\mathbf{M}$ ensures signature $(+, -, -, -)$ and $\det \mathbf{g} = -(\det \mathbf{M})^2 < 0$. Every smooth Lorentzian metric has such a representation locally (24, Eq. (3.119)). Along a time-like path with $dt/ds > 0$,

$$\frac{ds}{dt} = \sqrt{g_{tt} + 2\mathbf{g}_{tr} \dot{\mathbf{r}} + \dot{\mathbf{r}}^\top \mathbf{g}_{rr} \dot{\mathbf{r}}}. \quad (19.5)$$

The earlier similarity basis $\boldsymbol{\sigma}'_r = \boldsymbol{\tau} \boldsymbol{\sigma} \boldsymbol{\tau}^{-1}$ preserves

$$\det(dt + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r) = dt^2 - d\mathbf{r}^2,$$

so it retains the flat interval. The general real matrix $\mathbf{M}$ allows the metric blocks above to vary independently of that similarity law.

Local measurements and velocity. The primed basis represents coordinate directions; it is generally not an observer's orthonormal time and space basis. The matrix $\mathbf{M}$ converts coordinate increments to components in an orthonormal local frame (24, Eqs. (3.119)–(3.123)):

$$\begin{aligned} \begin{Bmatrix} dt' \\ d\mathbf{r}' \end{Bmatrix} &= \mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}, \\ d\mathbf{x} &= dt \sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r = dt' + d\mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned} \quad (19.6)$$

Here dt' , $d\mathbf{r}'$ denote local measurement increments. They need not be exact differentials of coordinate functions t' , $\mathbf{r}'$ on a region. The fixed algebraic symbols $1, \boldsymbol{\sigma}$ describe orthonormal axes at each point, not a single global inertial frame. Their interval is

$$ds^2 = dt'^2 - d\mathbf{r}'^2.$$

The observer at rest in this frame has $d\mathbf{r}' = \mathbf{0}$ and $dt' = ds$. Choose the time orientation so that future-directed time-like displacements have $dt' > 0$. For a massive particle with proper time s , the earlier four-velocity relation becomes

$$\begin{aligned} \mathbf{u} &= \frac{d\mathbf{x}}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ \boxed{\mathbf{v} = \frac{d\mathbf{r}'}{dt'}, \quad \gamma = \frac{dt'}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.} \end{aligned} \quad (19.7)$$

These are measurements relative to that observer. Writing $\dot{\mathbf{r}} = d\mathbf{r}/dt$ and $\dot{s} = ds/dt > 0$, with $\dot{s}$ given by (19.5), gives

$$\boxed{\begin{aligned} \gamma &= \frac{M_{tt} + \mathbf{M}_{rt} \cdot \dot{\mathbf{r}}}{\dot{s}}, \\ \mathbf{v} &= \frac{\mathbf{M}_{tr}^\top + \mathbf{M}_{rr}^\top \dot{\mathbf{r}}}{M_{tt} + \mathbf{M}_{rt} \cdot \dot{\mathbf{r}}}. \end{aligned}} \quad (19.8)$$

Thus γ is generally not dt/ds , and $\mathbf{v}$ is generally not $\dot{\mathbf{r}}$. A local Lorentz change of observer preserves $\mathbf{g}$ but changes these measurements; the metric alone does not select an observer. Nor does choosing a frame imply that its observers are in free fall.

19.2 Algebra and differentiation

For the real local displacement $d\mathbf{x} = dt\sigma'_t + d\mathbf{r} \cdot \boldsymbol{\sigma}'_r$, the coefficient row multiplies the basis column. Substituting (19.1) gives

$$\begin{aligned} d\mathbf{x} &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \begin{Bmatrix} \sigma'_t \\ \boldsymbol{\sigma}'_r \end{Bmatrix} \\ &= \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}^\top \mathbf{M} \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix} = \left(\mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} \right)^\top \begin{Bmatrix} 1 \\ \boldsymbol{\sigma} \end{Bmatrix}. \end{aligned}$$

Thus expansion in the fixed basis uses $\mathbf{M}^\top$ on the coefficient column. The transpose introduces no complex conjugation. Transposition swaps and transposes the off-diagonal blocks; it does not equate them:

$$\begin{Bmatrix} dt' \\ d\mathbf{r}' \end{Bmatrix} = \mathbf{M}^\top \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix} = \begin{Bmatrix} M_{tt} & \mathbf{M}_{rt}^\top \\ \mathbf{M}_{tr}^\top & \mathbf{M}_{rr}^\top \end{Bmatrix} \begin{Bmatrix} dt \\ d\mathbf{r} \end{Bmatrix}. \quad (19.9)$$

Equivalently, the paravector is

$$d\mathbf{x} = (M_{tt} dt + \mathbf{M}_{rt} \cdot d\mathbf{r}) + \left(dt \mathbf{M}_{tr}^\top + \mathbf{M}_{rr}^\top d\mathbf{r} \right) \cdot \boldsymbol{\sigma}.$$

The first term is dt' ; the coefficient of $\boldsymbol{\sigma}$ is $d\mathbf{r}'$. In general these are not dt and $d\mathbf{r}$. To multiply paravectors, apply the existing rule (2.9) to their scalar–vector parts, then apply $\mathbf{M}^{-\top}$ to the resulting coefficient column to return to the local basis. The product may have complex coefficients even when both factors are real.

Differentiation. Holding the coordinate increments $dt, d\mathbf{r}$ fixed while varying the base point gives

$$\begin{aligned} \partial_t(d\mathbf{x}) &= ((\partial_t M_{tt}) dt + (\partial_t \mathbf{M}_{rt}) \cdot d\mathbf{r}) \\ &\quad + \left(dt \partial_t \mathbf{M}_{tr}^\top + (\partial_t \mathbf{M}_{rr}^\top) d\mathbf{r} \right) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Refactoring into the local basis gives

$$\partial_t(\mathrm{d}\boldsymbol{\chi}) = \left\{ \sigma'_t \quad \boldsymbol{\sigma}'_r{}^\top \right\} \boldsymbol{M}^{-\top} (\partial_t \boldsymbol{M}^\top) \left\{ \frac{\mathrm{d}t}{\mathrm{d}\boldsymbol{r}} \right\}. \quad (19.10)$$

The same rule holds for any coordinate or fixed-direction spatial derivative. If the increments also vary, their derivatives contribute the usual product-rule terms. Refactoring an ordinary derivative changes its representation; it does not supply the metric's torsion-free parallel transport.

19.3 Expansion, free fall, and curvature

For a smooth positive scale factor $\lambda(t)$, choose

$$\boldsymbol{M} = \begin{Bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \lambda \boldsymbol{I} \end{Bmatrix}, \quad \sigma'_t = 1, \quad \boldsymbol{\sigma}'_r = \lambda \boldsymbol{\sigma}. \quad (19.11)$$

The metric blocks are

$$g_{tt} = 1, \quad \boldsymbol{g}_{tr} = \mathbf{0}^\top, \quad \boldsymbol{g}_{rr} = -\lambda(t)^2 \boldsymbol{I}. \quad (19.12)$$

For real vector increments $\mathrm{d}\boldsymbol{X}$, $\mathrm{d}\boldsymbol{Y}$, the algebra gives

$$(\mathrm{d}\boldsymbol{X} \cdot \boldsymbol{\sigma}'_r)(\mathrm{d}\boldsymbol{Y} \cdot \boldsymbol{\sigma}'_r) = \lambda^2 \mathrm{d}\boldsymbol{X} \cdot \mathrm{d}\boldsymbol{Y} + i\lambda(\mathrm{d}\boldsymbol{X} \times \mathrm{d}\boldsymbol{Y}) \cdot \boldsymbol{\sigma}'_r.$$

For $\mathrm{d}\boldsymbol{\chi} = \mathrm{d}t \sigma'_t + \mathrm{d}\boldsymbol{r} \cdot \boldsymbol{\sigma}'_r$, the interval is

$$\boxed{\mathrm{d}s^2 = \det(\mathrm{d}\boldsymbol{\chi}) = \mathrm{d}t^2 - \lambda(t)^2 \mathrm{d}\boldsymbol{r}^2,} \quad (19.13)$$

the spatially flat Robertson–Walker metric (24, Eqs. (8.7)–(8.9)). Its constant-time spatial slices are flat; its spacetime need not be.

Free fall in an expanding universe. Observers at fixed $\boldsymbol{r}$ are comoving: their clocks measure t , and the scale factor $\lambda(t)$ converts coordinate separations into physical distances. Here (19.6) reduces to $\mathrm{d}t' = \mathrm{d}t$ and $\mathrm{d}\boldsymbol{r}' = \lambda \mathrm{d}\boldsymbol{r}$, so the local frame's rest observers are exactly the comoving observers. For this metric, the quadratic geodesic action reduces to

$$\frac{1}{2} \int \left(\left(\frac{\mathrm{d}t}{\mathrm{d}s} \right)^2 - \lambda(t)^2 \left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} \right)^2 \right) \mathrm{d}s.$$

Varying t and $\boldsymbol{r}$ with fixed endpoints gives the affinely parameterized geodesic equations (24, pp. 68–70)

$$\frac{\mathrm{d}^2 t}{\mathrm{d}s^2} = -\lambda \dot{\lambda} \left(\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} \right)^2, \quad \frac{\mathrm{d}}{\mathrm{d}s} \left(\lambda^2 \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} \right) = \mathbf{0}. \quad (19.14)$$

Here $\dot{\lambda} = \mathrm{d}\lambda/\mathrm{d}t$. For a massive particle, choose s to be proper time. Its velocity relative to a comoving observer is $\boldsymbol{v} = \lambda \mathrm{d}\boldsymbol{r}/\mathrm{d}t$, so $\gamma = \mathrm{d}t/\mathrm{d}s = (1 - \boldsymbol{v}^2)^{-1/2}$. For constant mass $m > 0$, the earlier momentum law $\boldsymbol{\mathcal{P}} = m\boldsymbol{\mathcal{U}}$ gives

$$E = m\gamma, \quad \boldsymbol{P} = m\lambda \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} = m\gamma \boldsymbol{v}.$$

The spatial geodesic equation therefore has the direct physical meaning

$$\boxed{\frac{d}{dt}(\lambda \mathbf{P}) = \mathbf{0}, \quad \frac{d\mathbf{P}}{dt} = -\frac{\dot{\lambda}}{\lambda} \mathbf{P}.} \quad (19.15)$$

The momentum measured by comoving observers decreases as $1/\lambda$ during expansion. This is the standard cosmological redshift of particle momentum (24, Eq. (8.64)). At low speeds, $\mathbf{P} \simeq m\mathbf{v}$, so the velocity relative to those observers also decreases as $1/\lambda$. A particle initially at fixed $\mathbf{r}$ remains there.

Parallel transport and redshift. Parallel transport carries a four-vector along a chosen path using the metric's torsion-free, metric-compatible connection. Write the real paravector in terms of its measured scalar and vector parts:

$$\mathbf{y} = b\sigma'_t + \frac{\mathbf{Y}}{\lambda} \cdot \sigma'_r = b + \mathbf{Y} \cdot \sigma, \quad b \in \mathbb{R}, \quad \mathbf{Y} \in \mathbb{R}^3.$$

The factor $1/\lambda$ converts the measured vector $\mathbf{Y}$ into coefficients of the scaled basis. The torsion-free transport law follows from the quadratic velocity terms in (19.14): replace one tangent by the carried vector and average the two placements. Converting to the measured components above and using t as the path parameter, with $\mathbf{v} = \lambda d\mathbf{r}/dt$, gives

$$\frac{db}{dt} = -\frac{\dot{\lambda}}{\lambda} \mathbf{v} \cdot \mathbf{Y}, \quad \frac{d\mathbf{Y}}{dt} = -\frac{\dot{\lambda}}{\lambda} b\mathbf{v}. \quad (19.16)$$

These equations preserve $\det \mathbf{y} = b^2 - \mathbf{Y}^2$. Choosing $\mathbf{y} = \mathbf{u}$, with $b = \gamma$ and $\mathbf{Y} = \gamma\mathbf{v}$, recovers free fall: a geodesic parallel-transport its own tangent.

For a freely propagating photon, take $\mathbf{y} = \mathbf{P} = E + \mathbf{P} \cdot \sigma$, with $\mathbf{P} = E\mathbf{v}$ and $\mathbf{v}^2 = 1$. The same transport rule gives

$$\boxed{\frac{dE}{dt} = -\frac{\dot{\lambda}}{\lambda} E, \quad \lambda E = \text{constant}.} \quad (19.17)$$

Photon energy and frequency scale as $1/\lambda$: expansion produces the familiar cosmological redshift (24, Eq. (8.66)). Both results follow from the assigned metric for any smooth positive $\lambda(t)$; determining $\lambda(t)$ from matter requires Einstein's field equation. Figure 1 illustrates the local light directions and these scalings for one simple choice of $\lambda(t)$.

Tidal acceleration and curvature. Two nearby comoving observers are both in free fall, with proper time t and fixed coordinate separation $\Delta\mathbf{r}$. Their physical separation ℓ therefore obeys

$$\ell = \lambda \|\Delta\mathbf{r}\|, \quad \dot{\ell} = \frac{\dot{\lambda}}{\lambda} \ell, \quad \boxed{\ddot{\ell} = \frac{\ddot{\lambda}}{\lambda} \ell.}$$

The middle relation is Hubble's law: recession speed is proportional to distance. The last relation is geodesic deviation, the tidal acceleration between neighboring freely falling observers (24, Eq. (3.113)). Each observer has zero proper acceleration, yet their separation can accelerate. The ratio $\ddot{\lambda}/\lambda$ measures this tidal effect.

Transport around infinitesimal loops gives curvature magnitudes $|\ddot{\lambda}/\lambda|$ in each of the three orthonormal time-space planes and $(\dot{\lambda}/\lambda)^2$ in each of the three spatial planes. These are spacetime transport

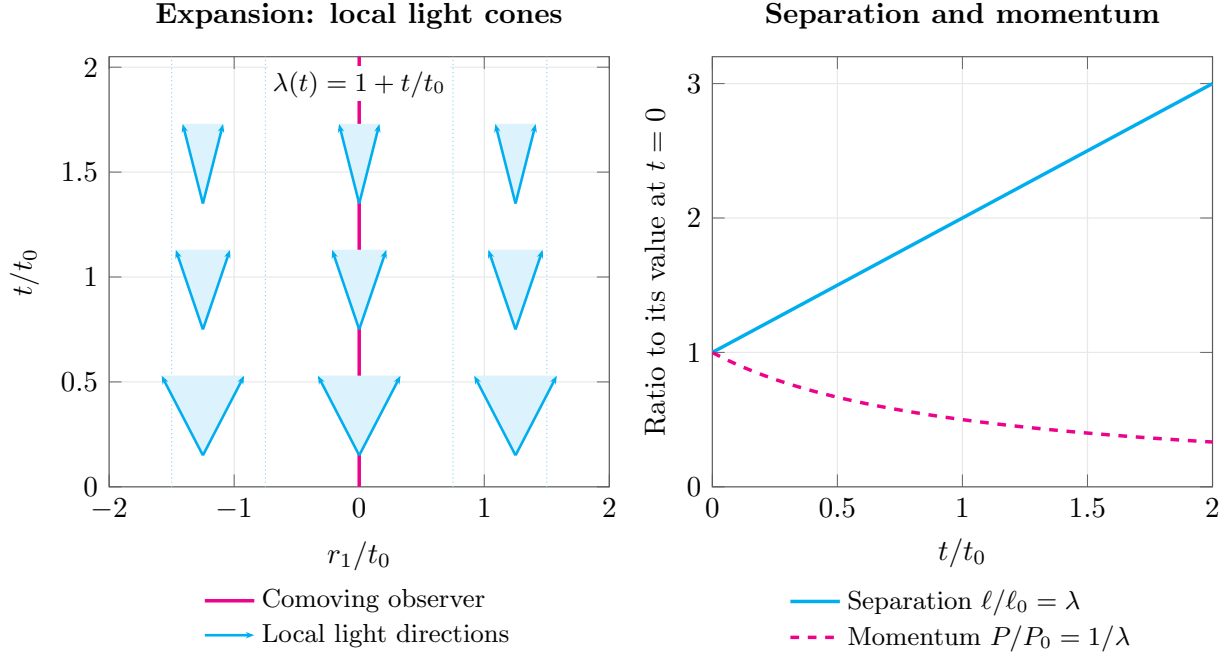


Figure 1: Expansion with $\lambda(t) = 1 + t/t_0$, where $t_0 > 0$. Left: a comoving coordinate slice, with the other spatial coordinates fixed. Cyan arrows are tangent light directions, $dr_1/dt = \pm 1/\lambda$; the magenta line is a comoving observer. Narrowing coordinate cones do not change the locally measured speed of light, which remains one. Right: the physical separation of comoving observers grows as $\ell/\ell_0 = \lambda$, while a freely moving particle's momentum magnitude $P = \|\mathbf{P}\|$ scales as $P/P_0 = 1/\lambda$. Subscript zero denotes $t = 0$. This illustrative scale factor is not selected by a matter model.

effects; each fixed-time spatial slice has zero intrinsic curvature. Contracting the Riemann tensor with itself counts each plane four times. The resulting coordinate-independent *Kretschmann scalar* is

$$12 \left(\left(\frac{\ddot{\lambda}}{\lambda} \right)^2 + \left(\frac{\dot{\lambda}}{\lambda} \right)^4 \right). \quad (19.18)$$

A nonzero value establishes spacetime curvature that no coordinate change can remove. In this spatially flat family it is nonzero wherever $\dot{\lambda}$ or $\ddot{\lambda}$ is nonzero; the spacetime is flat on a time interval precisely when λ is constant there. In particular, $\ddot{\lambda} = 0$ alone does not imply flat spacetime.

The role of the assigned displacement. The curved metric above uses $d\mathbf{x} = dt + \lambda d\mathbf{r} \cdot \boldsymbol{\sigma}$. An ordinary rescaling of inertial spatial coordinates instead gives

$$\begin{aligned} d(t + \lambda(t)\mathbf{r} \cdot \boldsymbol{\sigma}) &= dt + (\lambda d\mathbf{r} + \dot{\lambda}\mathbf{r} dt) \cdot \boldsymbol{\sigma}, \\ \det(d(t + \lambda(t)\mathbf{r} \cdot \boldsymbol{\sigma})) &= dt^2 - (\lambda d\mathbf{r} + \dot{\lambda}\mathbf{r} dt)^2. \end{aligned}$$

Since $\lambda > 0$, this is an invertible coordinate change of flat Minkowski spacetime. The product-rule term $\dot{\lambda}\mathbf{r} dt$ distinguishes this flat interval from (19.13): the two displacements assign different metrics to the same $dt, d\mathbf{r}$.

19.4 Newtonian motion and black-hole metrics

Schwarzschild and Kerr belong to a common family: their intervals can be written as the Minkowski interval plus a term proportional to the square of a null form. This is the Kerr–Schild construction (25, Eq. (1)). We now express it in our paravector basis. It is a special choice of $\mathbf{M}$, within the general metric construction of Section 19.1; the expansion example above uses a different choice.

A common basis. On an open coordinate region in $\mathbb{R} \times \mathbb{R}^3$, let $\lambda(t, \mathbf{r}) \in \mathbb{R}$ be a smooth dimensionless scalar field and $\mathbf{u}(t, \mathbf{r}) \in \mathbb{R}^3$ a smooth unit vector field, $\mathbf{u}^2 = 1$. Here λ denotes the Kerr–Schild scalar, distinct from the scale factor $\lambda(t)$ in the preceding example. We omit the field arguments $(t, \mathbf{r})$ below. The paravector $1 - \mathbf{u} \cdot \boldsymbol{\sigma}$ is null:

$$\det(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) = 1 - \mathbf{u}^2 = 0.$$

To change the displacement along this null direction, choose

$$\begin{aligned} \sigma'_t &= (1 - \lambda) + (\lambda\mathbf{u}) \cdot \boldsymbol{\sigma}, \\ \sigma'_r &= (-\lambda\mathbf{u}) + (\mathbf{I} + \lambda\mathbf{u}\mathbf{u}^\top) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (19.19)$$

In the second line, the dot contracts each row of the matrix with $\boldsymbol{\sigma}$: $-\lambda\mathbf{u}$ lists the three scalar parts, and the matrix rows give their vector parts. The assigned displacement is therefore

$$\begin{aligned} d\mathbf{x} &= dt \sigma'_t + d\mathbf{r} \cdot \sigma'_r \\ &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma} - \lambda(dt + \mathbf{u} \cdot d\mathbf{r})(1 - \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Its scalar and vector parts give the local measurement increments:

$$\begin{aligned} dt' &= dt - \lambda(dt + \mathbf{u} \cdot d\mathbf{r}), \\ d\mathbf{r}' &= d\mathbf{r} + \lambda(dt + \mathbf{u} \cdot d\mathbf{r})\mathbf{u}. \end{aligned} \quad (19.20)$$

Taking $\det(d\mathbf{r}) = dt'^2 - d\mathbf{r}'^2$, the terms quadratic in λ cancel because $\mathbf{u}^2 = 1$. The resulting metric blocks are

$$\begin{aligned} g_{tt} &= 1 - 2\lambda, & \mathbf{g}_{tr} &= -2\lambda\mathbf{u}^\top, \\ \mathbf{g}_{rr} &= -\mathbf{I} - 2\lambda\mathbf{u}\mathbf{u}^\top. \end{aligned} \quad (19.21)$$

Combining the three blocks gives

$$\boxed{ds^2 = dt^2 - d\mathbf{r}^2 - 2\lambda(dt + \mathbf{u} \cdot d\mathbf{r})^2.} \quad (19.22)$$

This is exactly the Kerr–Schild interval, with no weak-field approximation. Setting $\lambda = 0$ recovers Minkowski spacetime. The radial choice $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$ with $\lambda = Gm/\|\mathbf{r}\|$ gives Schwarzschild; the spin-dependent fields below give Kerr. Newtonian motion follows in the weak-field, slow-motion limit. The basis is invertible for every finite λ : its inverse change is obtained by replacing λ with $-\lambda$, and $\det \mathbf{M} = 1$, so $\det \mathbf{g} = -1$. The algebra converts the chosen fields into a metric. Einstein's equation is the additional physical law that constrains those fields for a given source; the basis identities alone do not determine the Schwarzschild or Kerr solutions.

Local velocity and observer. For a future-directed time-like path parametrized by t , the metric gives

$$\dot{s} = \sqrt{1 - \dot{\mathbf{r}}^2 - 2\lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})^2}. \quad (19.23)$$

Equations (19.20) and (19.7) then give

$$\boxed{\begin{aligned} \gamma &= \frac{1 - \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})}{\dot{s}}, \\ \mathbf{v} &= \frac{\dot{\mathbf{r}} + \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})\mathbf{u}}{1 - \lambda(1 + \mathbf{u} \cdot \dot{\mathbf{r}})}. \end{aligned}} \quad (19.24)$$

The observer at rest in this frame has $d\mathbf{r}' = \mathbf{0}$. For the black-hole examples below, $\lambda \geq 0$, and this observer follows

$$\frac{d\mathbf{r}_{\text{obs}}}{dt} = -\frac{\lambda}{1 + \lambda}\mathbf{u}, \quad \frac{dt}{ds} = 1 + \lambda.$$

Substitution gives $\mathbf{v} = \mathbf{0}$ and $\gamma = 1$, as required. This observer moves in the coordinate chart and need not be freely falling. Substituting the Schwarzschild or Kerr fields below into the basis therefore supplies both their exact metric and this choice of local measurement frame wherever the fields are regular. The increments dt' , $d\mathbf{r}'$ do not generally integrate to a coordinate system on a region. Their Minkowski-form interval expresses local measurements; the varying metric still determines curved-spacetime free fall and curvature.

Newtonian limit. For stationary λ , $\mathbf{u}$ with $|\lambda| \ll 1$ and $\|\dot{\mathbf{r}}\| \ll 1$, expand (19.23), retaining terms linear in λ and quadratic in velocity, dropping their products. The free-particle action per unit mass becomes

$$-\int ds \simeq \int \left(-1 + \frac{1}{2}\dot{\mathbf{r}}^2 + \lambda \right) dt.$$

The Euler–Lagrange equation gives

$$\ddot{\mathbf{r}} \simeq -\frac{1}{2}\partial_{\mathbf{r}}g_{tt} = \partial_{\mathbf{r}}\lambda. \quad (19.25)$$

Thus $-\lambda$ is the Newtonian gravitational potential (24, Eqs. (4.20)–(4.22)).

Schwarzschild. For stationary spherical symmetry, take $r = \|\mathbf{r}\| > 0$ and $\mathbf{u} = \mathbf{r}/r$, with λ depending on position only through r . For this radial ansatz, the vacuum Einstein equation with zero cosmological constant gives the angular condition

$$\frac{d}{dr}(r\lambda) = 0 \implies \lambda = \frac{Gm}{r}. \quad (19.26)$$

The remaining radial condition follows by differentiating this equation. Matching the Newtonian potential fixes the integration constant to Gm (24, Eqs. (7.23)–(7.29)). Here G is Newton’s gravitational constant and $m > 0$ is the source mass. Substitution into (19.21) gives

$$\begin{aligned} g_{tt} &= 1 - \frac{2Gm}{r}, & \mathbf{g}_{tr} &= -\frac{2Gm}{r^2}\mathbf{r}^\top, \\ \mathbf{g}_{rr} &= -\mathbf{I} - \frac{2Gm}{r^3}\mathbf{r}\mathbf{r}^\top. \end{aligned} \quad (19.27)$$

Since $\mathbf{u} \cdot d\mathbf{r} = dr$, these blocks give

$$ds^2 = dt^2 - d\mathbf{r}^2 - \frac{2Gm}{r}(dt + d\mathbf{r})^2. \quad (19.28)$$

This is the Schwarzschild metric in coordinates regular at the future horizon $r = 2Gm$ (26, Table 3). The weak-field acceleration is

$$\ddot{\mathbf{r}} \simeq -\frac{Gm}{r^3}\mathbf{r}.$$

For radial light, setting $ds^2 = 0$ gives the two coordinate velocities

$$\frac{dr}{dt} = -1, \quad \frac{dr}{dt} = \frac{r - 2Gm}{r + 2Gm}.$$

The outward branch has zero radial velocity at the horizon; inside, both future-directed radial light directions point toward smaller r . Figure 2 shows these directions alongside the scalar-field contours.

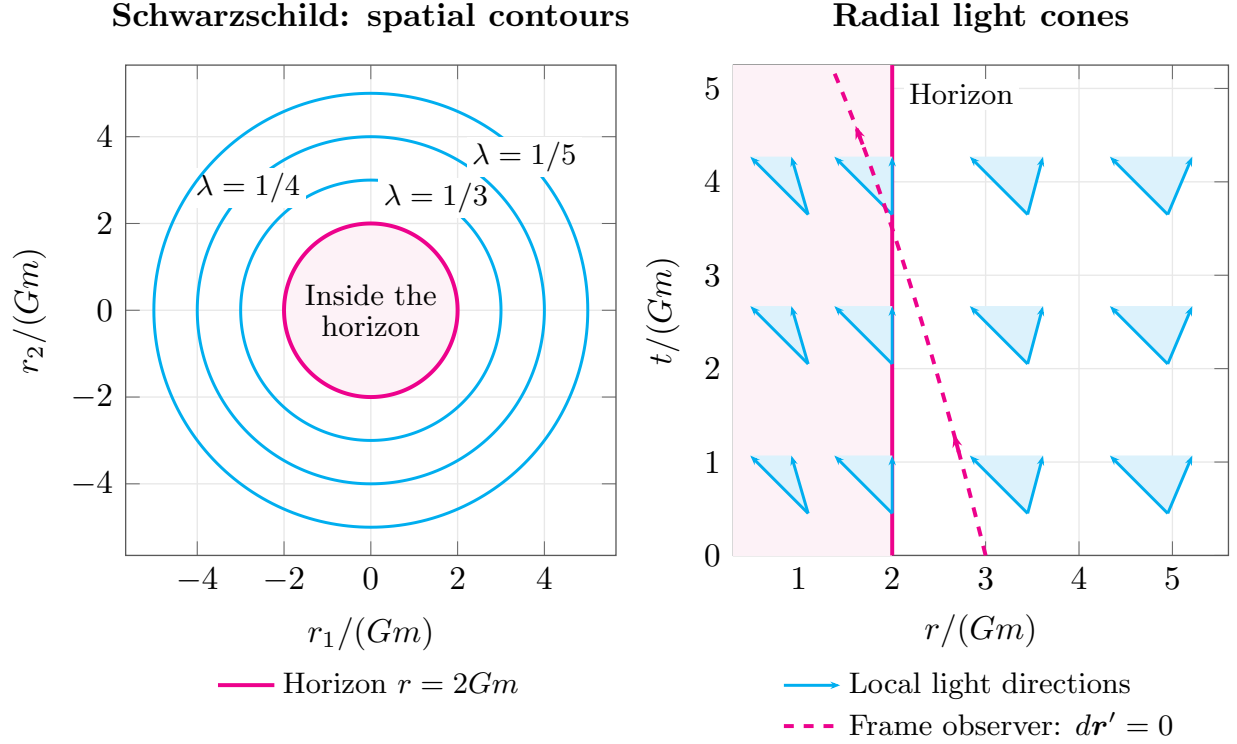


Figure 2: Schwarzschild in the coordinates of (19.28), with lengths and times in units of Gm . Left: constant- λ circles in the $r_3 = 0$ plane; the solid magenta circle is the horizon, with its interior shaded pale magenta. Right: future radial light cones, drawn as local tangent directions, with time increasing upward. The dashed magenta curve follows the chosen frame observer, $dr/dt = -Gm/(r + Gm)$, starting at $r = 3Gm$ when $t = 0$; it is not a free-fall trajectory. The spatial panel shows coordinate contours, not proper distances or an isometric embedding.

Kerr. Let the constant real vector ω be the outer horizon's angular velocity relative to a nonrotating observer at infinity. For $2Gm\|\omega\| \leq 1$, define the positive scalar

$$\rho = \frac{4(Gm)^2}{1 + 4(Gm)^2\omega^2}.$$

The angular momentum per unit mass is $\rho\omega$; ρ has dimensions of length squared. This scaling follows from the Kerr horizon angular-velocity relation (26, Eq. (145)). Use the positive ellipsoidal-radius branch $\epsilon > 0$, on the coordinate region where it is smooth. It is determined by r and the spin, and generally differs from $\|r\|$:

$$\epsilon^4 - \epsilon^2(r^2 - \rho^2\omega^2) - \rho^2(\omega \cdot r)^2 = 0. \quad (19.29)$$

The standard Kerr black-hole solution is represented by

$$\begin{aligned} u &= \frac{\epsilon r - \rho\omega \times r + \rho^2\omega(\omega \cdot r)/\epsilon}{\epsilon^2 + \rho^2\omega^2}, \\ \lambda &= \frac{Gm\epsilon^3}{\epsilon^4 + \rho^2(\omega \cdot r)^2}. \end{aligned} \quad (19.30)$$

The radius equation verifies the unit direction:

$$u^2 = \frac{r^2 + \rho^2(\omega \cdot r)^2/\epsilon^2}{\epsilon^2 + \rho^2\omega^2} = 1.$$

Substituting into (19.21) gives the explicit metric blocks

$$\begin{aligned} g_{tt} &= 1 - \frac{2Gm\epsilon^3}{\epsilon^4 + \rho^2(\omega \cdot r)^2}, \\ g_{tr} &= -\frac{2Gm\epsilon^3}{\epsilon^4 + \rho^2(\omega \cdot r)^2}u^\top, \\ g_{rr} &= -I - \frac{2Gm\epsilon^3}{\epsilon^4 + \rho^2(\omega \cdot r)^2}uu^\top, \end{aligned} \quad (19.31)$$

where u is given explicitly in (19.30). Thus the scalar-vector determinant gives

$$\begin{aligned} ds^2 &= \det(d\mathcal{X}) = g_{tt}dt^2 + 2dtg_{tr}dr + dr^\top g_{rr}dr \\ &= dt^2 - dr^2 - \frac{2Gm\epsilon^3}{\epsilon^4 + \rho^2(\omega \cdot r)^2}(dt + u \cdot dr)^2. \end{aligned} \quad (19.32)$$

This is the Cartesian Kerr-Schild vacuum solution with signature $(+, -, -, -)$ (25, Sec. 1, Eqs. (5)–(6)). Setting $\omega = 0$ gives $\epsilon = \|r\|$ and recovers Schwarzschild. The outer horizon is the larger root of $\epsilon^2 - 2Gm\epsilon + \rho^2\omega^2 = 0$:

$$\epsilon = \frac{\rho}{2Gm} = \frac{2Gm}{1 + 4(Gm)^2\omega^2}.$$

The metric remains regular there, with $\det g = -1$. At this radius, the tangent displacement $dr = (\omega \times r)dt$ gives $ds^2 = 0$, verifying that ω is the horizon's angular velocity. The outer stationary limit is instead the outer surface $g_{tt} = 0$, or $\lambda = 1/2$. Between it and the horizon lies the exterior ergoregion: observers cannot remain at fixed r , since that direction has $ds^2 = g_{tt}dt^2 < 0$ (26, Sec. 3.5). Figure 3 distinguishes these surfaces and shows the directions of our chosen frame observers.

These substitutions reproduce the known Kerr solution. A general paravector construction of transport, curvature, and Einstein's field equation remains to be developed.

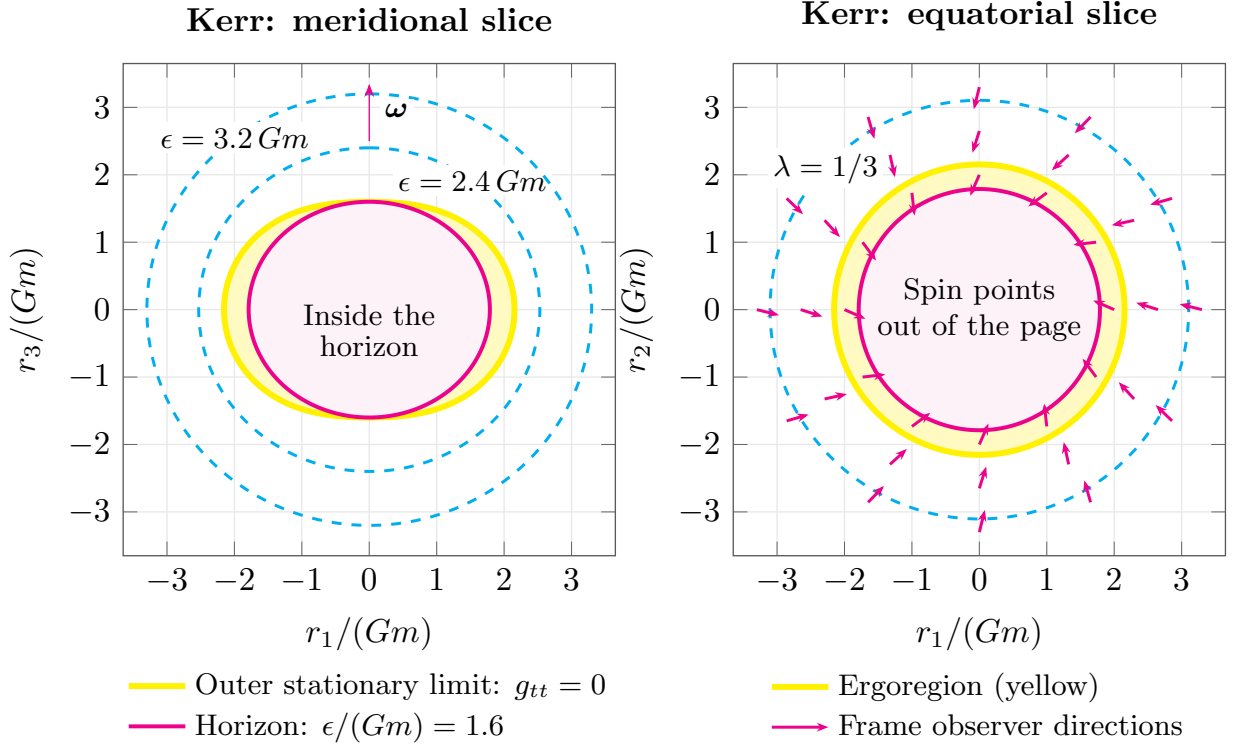


Figure 3: Kerr coordinate slices with spin along $+r_3$ and $Gm\|\omega\| = 1/4$, so $\rho\|\omega\| = 0.8Gm$ and the horizon is $\epsilon = 1.6Gm$. Left: the $r_2 = 0$ slice, with dashed cyan contours of the ellipsoidal radius ϵ . Right: the $r_3 = 0$ slice, with a dashed constant- λ contour and equal-length arrows showing the direction of $d\mathbf{r}/dt = -\lambda\mathbf{u}/(1 + \lambda)$. Solid magenta marks the horizon; yellow marks the outer stationary limit and shades the exterior ergoregion. Pale magenta marks the horizon's interior. The arrows describe the chosen frame observers, not general free fall; the coordinate shapes are not an isometric embedding.

A Pauli-matrix representation

The paravector formulas are verified in matrix form in this appendix. The decomposition $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ is used, with $c \in \mathbb{C}$ and $\mathbf{Z} \in \mathbb{C}^3$. The matrix map is $\text{mat} : \mathbb{C} \oplus \mathbb{C}^3 \rightarrow \mathbb{C}^{2 \times 2}$. The notation $\mathbf{M} = \text{mat}(\mathbf{Z}) \in \mathbb{C}^{2 \times 2}$ is used. Real paravectors map to Hermitian matrices, not necessarily real matrices. The identity matrices are $\mathbf{I}_2$ and $\mathbf{I}_4$ in their stated dimensions.

A.1 Sigmas and products

The sigmas are represented by the Pauli matrices (6, p. 11)

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Consequently,

$$\text{mat}(c + \mathbf{Z} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} c + Z_3 & Z_1 - iZ_2 \\ Z_1 + iZ_2 & c - Z_3 \end{Bmatrix}. \quad (\text{A.1})$$

The sigma product rule is satisfied by these matrices, which are linearly independent over $\mathbb{C}$ together with $\mathbf{I}_2$. Thus every complex 2×2 matrix corresponds to exactly one paravector. The map preserves addition, sends 1 to $\mathbf{I}_2$, and satisfies

$$\text{mat}(\mathbf{Z}' \mathbf{Z}) = \text{mat}(\mathbf{Z}') \text{mat}(\mathbf{Z}).$$

Here $\mathbf{Z}'$ is any second complex paravector. Thus algebra multiplication is ordinary matrix multiplication.

A.2 Conjugations and product order

For $\mathbf{M}, \mathbf{N} \in \mathbb{C}^{2 \times 2}$, transpose T exchanges row and column indices; entrywise complex conjugation C sends each $a + ib$ to $a - ib$ for real a, b . Their composition is $\mathbf{M}^H = (\mathbf{M}^C)^T$. Introduce the fixed matrix $\mathbf{S} \in \mathbb{R}^{2 \times 2}$:

$$\mathbf{S} = \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}, \quad \mathbf{S}^{-1} = -\mathbf{S}.$$

The paravector sigma conjugation has the matrix image

$$\text{mat}(\mathbf{Z}^*) = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} =: \mathbf{M}^*. \quad (\text{A.2})$$

Substituting the basis in (A.2) gives

Algebra basis	1	i	σ_k	$i\sigma_k$
Hermitian image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$\text{mat}(\sigma_k)$	$-i\text{mat}(\sigma_k)$
Sigma-conjugate image	$\mathbf{I}_2$	$-i\mathbf{I}_2$	$-\text{mat}(\sigma_k)$	$i\text{mat}(\sigma_k)$

Here $k = 1, 2, 3$. These are exactly the operations of Section 1.3, so $\text{mat}(\mathbf{Z}^H) = \mathbf{M}^H$. Their product-order rules are

$$\begin{aligned} (\mathbf{MN})^T &= \mathbf{N}^T \mathbf{M}^T, \\ (\mathbf{MN})^C &= \mathbf{M}^C \mathbf{N}^C, \\ (\mathbf{MN})^H &= \mathbf{N}^H \mathbf{M}^H. \end{aligned}$$

For sigma conjugation the middle factors cancel:

$$(\mathbf{M}\mathbf{N})^* = \mathbf{S}^{-1}\mathbf{M}^{\mathbf{C}}\mathbf{S}\mathbf{S}^{-1}\mathbf{N}^{\mathbf{C}}\mathbf{S}.$$

Thus sigma conjugation preserves product order:

$$(\mathbf{M}\mathbf{N})^* = \mathbf{M}^*\mathbf{N}^*.$$

This verifies (1.5) in matrices. Hermitian and sigma conjugation commute, and applying either twice restores the original matrix. The matrix-only operations can be recovered as

$$\begin{aligned}\mathbf{M}^{\mathbf{C}} &= \mathbf{S}\mathbf{M}^*\mathbf{S}^{-1}, \\ \mathbf{M}^{\mathbf{T}} &= \mathbf{S}(\mathbf{M}^*)^{\mathbf{H}}\mathbf{S}^{-1}.\end{aligned}$$

A.3 Scalar part, determinant, and norm

For $\mathbf{M} = (m_{jk}) \in \mathbb{C}^{2 \times 2}$, with $m_{jk} \in \mathbb{C}$, its trace is $m_{11} + m_{22}$ and its determinant is $m_{11}m_{22} - m_{12}m_{21}$. Substituting (A.1) gives

$$\text{sc}(\mathbf{Z}) = \frac{1}{2} \text{tr} \mathbf{M} = c, \tag{A.3}$$

$$\begin{aligned}(\text{vec}(\mathbf{Z}))_k &= \frac{1}{2} \text{tr}(\text{mat}(\sigma_k)\mathbf{M}) = Z_k, \\ \det \mathbf{Z} &= \det \mathbf{M} = c^2 - \|\mathbf{Z}\|^2.\end{aligned} \tag{A.4}$$

By (A.3), matrix-trace cyclicity is precisely the scalar-part cyclicity used in the main text. For $\mathbf{M}' = \text{mat}(\mathbf{Z}')$,

$$\text{sc}(\mathbf{Z}'\mathbf{Z}) = \frac{1}{2} \text{tr}(\mathbf{M}'\mathbf{M}) = \text{sc}(\mathbf{Z}\mathbf{Z}').$$

The remaining extraction and norm formulas are

$$\begin{aligned}\text{mat}(\text{vec}(\mathbf{Z}) \cdot \boldsymbol{\sigma}) &= \mathbf{M} - \text{sc}(\mathbf{Z}) \mathbf{I}_2, \\ \text{mat}(\text{re}(\mathbf{Z})) &= \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\mathbf{H}}),\end{aligned} \tag{A.5}$$

$$\text{mat}(\text{im}(\mathbf{Z})) = (\mathbf{M} - \mathbf{M}^{\mathbf{H}})/(2i), \tag{A.6}$$

$$\|\mathbf{Z}\|^2 = \frac{1}{2} \text{tr}(\mathbf{M}^{\mathbf{H}}\mathbf{M}) = \|c\|^2 + \|\mathbf{Z}\|^2. \tag{A.7}$$

Equations (A.5)–(A.6) extract Hermitian matrices, so a real paravector has a Hermitian image. The squared norm in (A.7) is half the sum of the squared magnitudes of the matrix entries. The determinant in (A.4) may be complex.

The matrix adjugate is

$$\text{adj}(\mathbf{M}) = (\text{tr} \mathbf{M})\mathbf{I}_2 - \mathbf{M} = \text{mat}(\text{adj}(\mathbf{Z})) = (\mathbf{M}^*)^{\mathbf{H}}.$$

It satisfies

$$\text{mat}(\text{adj}(\mathbf{Z}) \mathbf{Z}) = \det(\mathbf{M})\mathbf{I}_2.$$

Consequently $\mathbf{M}^{-1} = \text{adj}(\mathbf{M}) / \det \mathbf{M}$ when the determinant is nonzero. The determinant product and adjugate order rules are the same in both representations. The quadratic identity (3.1) is precisely Cayley–Hamilton for a 2×2 matrix.

A.4 Boost and rotation matrices

Notation: the real unit axis $\mathbf{u} = \{u_1, u_2, u_3\}$ and real rapidity θ . The matrix image of (4.6) is

$$\text{mat}(\mathcal{L}) = \cosh(\theta/2) \mathbf{I}_2 - \sinh(\theta/2) \begin{Bmatrix} u_3 & u_1 - iu_2 \\ u_1 + iu_2 & -u_3 \end{Bmatrix}. \quad (\text{A.8})$$

The displayed axis matrix is Hermitian, has square $\mathbf{I}_2$, and has eigenvalues ± 1 . Thus (A.8) has eigenvalues $e^{\mp\theta/2}$, determinant one, and squared norm $\frac{1}{2} \text{tr}(\text{mat}(\mathcal{L})^H \text{mat}(\mathcal{L})) = \cosh \theta$. For a real rotation angle θ , replacing the hyperbolic coefficients by $\cos(\theta/2)$ and $i \sin(\theta/2)$ gives $\text{mat}(\mathcal{R})$, which is unitary.

Multiplication preserves the distinct spacetime and field laws:

$$\begin{aligned} \text{mat}(\mathcal{X}') &= \text{mat}(\mathcal{L}) \text{mat}(\mathcal{X}) \text{mat}(\mathcal{L}), \\ \text{mat}(\mathcal{F}') &= \text{mat}(\mathcal{L}) \text{mat}(\mathcal{F}) \text{mat}(\mathcal{L})^{-1}. \end{aligned}$$

Here $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ is real and $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is complex, with $t \in \mathbb{R}$ and $\mathbf{r}, \mathbf{E}, \mathbf{B} \in \mathbb{R}^3$. Extracting scalar and vector coefficients by (A.3) and the preceding component rule gives (7.1)–(7.2) and (18.6)–(18.7) for the same arbitrary axis. The first matrix stays Hermitian; the second stays traceless.

A.5 Matrix amplitudes and column spinors

The matrix image of $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ is denoted by $\mathbf{M}_\psi \in \mathbb{C}^{2 \times 2}$ at one event $\mathcal{X} \in \mathbb{R} \oplus \mathbb{R}^3$. For a nonnegative weight $\mathcal{Z} \in \mathbb{R} \oplus \mathbb{R}^3$, the local densities (13.3) and (13.5) become

$$\begin{aligned} \|\psi\|^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^H \mathbf{M}_\psi), \\ \|\psi\|_{\mathcal{Z}}^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^H \text{mat}(\mathcal{Z}) \mathbf{M}_\psi). \end{aligned}$$

The first is one half the sum of the squared magnitudes of the matrix entries: the squared Frobenius norm with the factor $1/2$ fixed by $\text{sc} = \text{tr}/2$. The second is its weighted form. Both are pointwise; neither contains a spatial integral. Matrix multiplication gives exactly the congruence identity (13.7).

An arbitrary spin basis. Notation: the arbitrary constant perpendicular real unit vectors $\mathbf{u}, \mathbf{a} \in \mathbb{R}^3$ of (13.1). The projector $\Pi_+ = (1 + \mathbf{u} \cdot \boldsymbol{\sigma})/2$ is defined, and a constant complex paravector ψ' with $\Pi_+ \psi' \neq 0$ is chosen. Then

$$\begin{Bmatrix} \psi_+ \\ \psi_- \end{Bmatrix} := \frac{1}{\|\Pi_+ \psi'\|} \begin{Bmatrix} \Pi_+ \psi' \\ (\mathbf{a} \cdot \boldsymbol{\sigma}) \Pi_+ \psi' \end{Bmatrix}. \quad (\text{A.9})$$

This constructs (13.1) with $(\mathbf{u} \cdot \boldsymbol{\sigma}) \psi_\pm = \pm \psi_\pm$ and $\text{sc}(\psi_s^H \psi_{s'}) = \delta_{ss'}$ for $s, s' = +, -$. The two complex scalar components give

$$\psi = c_+ \psi_+ + c_- \psi_- = (c_+ + c_- \mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+. \quad (\text{A.10})$$

For fields, $c_\pm : \mathbb{R}^4 \rightarrow \mathbb{C}$, and (13.3) is $\rho = \|c_+\|^2 + \|c_-\|^2$.

The two-component column. Every column of $\mathbf{M}_{\psi_+}$ has eigenvalue $+1$ under $\text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma})$, whose $+1$ eigenspace is one-dimensional. Since $\|\psi_+\| = 1$, its matrix therefore factors as $\mathbf{M}_{\psi_+} = \sqrt{2} \psi_+ \mathbf{v}^H$ for constant columns $\psi_+, \mathbf{v} \in \mathbb{C}^2$ satisfying $\psi_+^H \psi_+ = \mathbf{v}^H \mathbf{v} = 1$.

By left multiplication, $\mathbf{M}_{\psi_-} = \sqrt{2} \psi_- \mathbf{v}^H$, where $\psi_- = \text{mat}(\mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+$. The columns ψ_+, ψ_- are orthonormal. Thus every amplitude in (A.10) has the conventional Pauli column

$$\psi := \frac{1}{\sqrt{2}} \mathbf{M}_{\psi} \mathbf{v} = c_+ \psi_+ + c_- \psi_- \in \mathbb{C}^2.$$

Conversely, $\mathbf{M}_{\psi} = \sqrt{2} \psi \mathbf{v}^H$, so this correspondence is one-to-one. For two amplitudes in this fixed basis and any complex multiplying paravector $\mathbf{Z}$, by substitution,

$$\begin{aligned} \text{sc}(\psi^H \psi) &= \psi^H \psi, \\ \text{sc}(\psi^H \mathbf{Z} \psi) &= \psi^H \text{mat}(\mathbf{Z}) \psi. \end{aligned}$$

For nonnegative $\mathbf{Z}$, the second line is $\|\psi\|_{\mathbf{Z}}^2$. Dividing by $\rho = \psi^H \psi > 0$ gives $\langle \mathbf{Z} \rangle_{\psi}$, the ordinary finite-component expectation. Left action $\mathbf{T} \psi$ maps to $\text{mat}(\mathbf{T}) \psi$ for any complex paravector $\mathbf{T}$.

$\mathbf{C} \in \mathbb{C}^{2 \times 2}$ is defined as the matrix with columns ψ_+, ψ_- . Then $\mathbf{C}^H \mathbf{C} = \mathbf{I}_2$ and $\psi = \mathbf{C} \{c_+, c_-\}^T$. For a complex paravector $\mathbf{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$, its matrix in this basis is

$$\text{mat}'(\mathbf{Z}) := \mathbf{C}^H \text{mat}(\mathbf{Z}) \mathbf{C} = \begin{Bmatrix} c + \mathbf{u} \cdot \mathbf{Z} & (\mathbf{a} - i \mathbf{u} \times \mathbf{a}) \cdot \mathbf{Z} \\ (\mathbf{a} + i \mathbf{u} \times \mathbf{a}) \cdot \mathbf{Z} & c - \mathbf{u} \cdot \mathbf{Z} \end{Bmatrix}. \quad (\text{A.11})$$

Here the prime denotes only this change of matrix basis. For nonzero squared norm $\rho = \psi^H \psi$, the polarization (17.3) and probabilities for $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ become

$$\mathbf{u} \cdot \mathbf{u}' = \psi^H \text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma}) \psi / \rho, \quad (\text{A.12})$$

$$\text{prob}(\pm | \mathbf{u}) = \psi^H \text{mat}(\Pi_{\pm}) \psi / \rho. \quad (\text{A.13})$$

Equations (A.12) and (A.13) are the ordinary spin expectation and Born rule.

A.6 Dirac columns

For a Dirac block $\psi = \{\psi_+, \psi_-\}^T$, the corresponding columns are denoted by $\mathbf{c}_+, \mathbf{c}_- \in \mathbb{C}^2$. Then

$$\mathbf{W}(\mathbf{Z}) \psi \longleftrightarrow \begin{Bmatrix} 0 & \text{mat}(\text{adj}(\mathbf{Z})) \\ \text{mat}(\mathbf{Z}) & 0 \end{Bmatrix} \begin{Bmatrix} \mathbf{c}_+ \\ \mathbf{c}_- \end{Bmatrix}.$$

The block squared norm in (15.1) becomes the squared norm of a four-component column:

$$\rho = \mathbf{c}_+^H \mathbf{c}_+ + \mathbf{c}_-^H \mathbf{c}_-.$$

Characteristic polynomial. On one Fourier mode, $s \in \mathbb{C}$ is the spectral parameter and $\square$ denotes the scalar eigenvalue of the wave operator. The two off-diagonal blocks multiply to $-\square \mathbf{I}_2$, so block elimination gives the ordinary matrix determinant

$$\begin{aligned} \det \begin{Bmatrix} -(s+m) \mathbf{I}_2 & \text{mat}(i \partial) \\ \text{mat}(i \partial^*) & -(s+m) \mathbf{I}_2 \end{Bmatrix} &= \det(((s+m)^2 + \square) \mathbf{I}_2) \\ &= ((s+m)^2 + \square)^2. \end{aligned} \quad (\text{A.14})$$

This verifies (15.8), including multiplicities. The polynomial identity also holds at $s = -m$.

A.7 Four components and the standard Dirac basis

The fixed basis elements of (A.9) are denoted by $\boldsymbol{\eta}_\pm \in \mathbb{C} \oplus \mathbb{C}^3$. Both entries are expressed in this complex span: $\boldsymbol{\psi}_+ = a\boldsymbol{\eta}_+ + b\boldsymbol{\eta}_-$ and $\boldsymbol{\psi}_- = c'\boldsymbol{\eta}_+ + d\boldsymbol{\eta}_-$, where $a, b, c', d : \mathbb{R}^4 \rightarrow \mathbb{C}$ are the four scalar components, forming a column in $\mathbb{C}^4$ at each event. Their columns are $\boldsymbol{c}'_+ = \{a, b\}^\top$ and $\boldsymbol{c}'_- = \{c', d\}^\top$; the primes on the columns denote the basis change $\boldsymbol{c}'_\pm = \boldsymbol{C}^H \boldsymbol{c}_\pm$. Notation: real mass $m > 0$, real charge q , and real potentials $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ and $\boldsymbol{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$. The gauge-covariant energy and kinetic momentum operators are $\hat{E}_q = i\partial_t - qV$ and $\hat{\boldsymbol{P}}_q = -i\partial_{\boldsymbol{r}} - q\boldsymbol{A}$. Their momentum components in this frame are

$$\hat{P}_{q,1} = \boldsymbol{a} \cdot \hat{\boldsymbol{P}}_q, \quad \hat{P}_{q,2} = (\boldsymbol{u} \times \boldsymbol{a}) \cdot \hat{\boldsymbol{P}}_q, \quad \hat{P}_{q,3} = \boldsymbol{u} \cdot \hat{\boldsymbol{P}}_q.$$

Using (A.11), the coupled paravector equations give

$$(\hat{E}_q - \hat{P}_{q,3})c' - (\hat{P}_{q,1} - i\hat{P}_{q,2})d = ma, \quad (\text{A.15})$$

$$-(\hat{P}_{q,1} + i\hat{P}_{q,2})c' + (\hat{E}_q + \hat{P}_{q,3})d = mb, \quad (\text{A.16})$$

$$(\hat{E}_q + \hat{P}_{q,3})a + (\hat{P}_{q,1} - i\hat{P}_{q,2})b = mc', \quad (\text{A.17})$$

$$(\hat{P}_{q,1} + i\hat{P}_{q,2})a + (\hat{E}_q - \hat{P}_{q,3})b = md. \quad (\text{A.18})$$

Operators in (A.15)–(A.18) act on the scalar immediately to their right. Their Hamiltonian on $\{a, b, c', d\}^\top$ is

$$\hat{H} = \begin{Bmatrix} qV - \hat{P}_{q,3} & -\hat{P}_{q,1} + i\hat{P}_{q,2} & m & 0 \\ -\hat{P}_{q,1} - i\hat{P}_{q,2} & qV + \hat{P}_{q,3} & 0 & m \\ m & 0 & qV + \hat{P}_{q,3} & \hat{P}_{q,1} - i\hat{P}_{q,2} \\ 0 & m & \hat{P}_{q,1} + i\hat{P}_{q,2} & qV - \hat{P}_{q,3} \end{Bmatrix}. \quad (\text{A.19})$$

This is the component image of (15.15).

Conventional gamma matrices are identified through $\boldsymbol{\sigma}_k = \text{mat}(\sigma_k)$ for $k = 1, 2, 3$. Here $\boldsymbol{\sigma}_k \in \mathbb{C}^{2 \times 2}$ and the gamma matrices below lie in $\mathbb{C}^{4 \times 4}$. The $\boldsymbol{\sigma}_k$ are also the images under mat' of $\boldsymbol{a} \cdot \boldsymbol{\sigma}$, $(\boldsymbol{u} \times \boldsymbol{a}) \cdot \boldsymbol{\sigma}$, and $\boldsymbol{u} \cdot \boldsymbol{\sigma}$, respectively. Their numerical entries apply to every choice of $\boldsymbol{u}, \boldsymbol{a}$. The following matrices are defined (16, Eqs. (4.16)–(4.17))

$$\gamma^0 = \begin{Bmatrix} 0 & \boldsymbol{I}_2 \\ \boldsymbol{I}_2 & 0 \end{Bmatrix}, \quad (\text{A.20})$$

$$\gamma^k = \begin{Bmatrix} 0 & \boldsymbol{\sigma}_k \\ -\boldsymbol{\sigma}_k & 0 \end{Bmatrix}. \quad (\text{A.21})$$

Equations (A.20)–(A.21) obey $(\gamma^0)^2 = \boldsymbol{I}_4$, $(\gamma^k)^2 = -\boldsymbol{I}_4$, and $\gamma^j \gamma^k + \gamma^k \gamma^j = 0$ for distinct $j, k = 0, 1, 2, 3$. The standard chiral Dirac operator in this basis, with signature $(+, -, -, -)$, is

$$\boldsymbol{W}(\hat{\boldsymbol{P}}_q) \longleftrightarrow \gamma^0 \hat{E}_q - \sum_{k=1}^3 \gamma^k \hat{P}_{q,k}.$$

Here $\boldsymbol{A} = V + \boldsymbol{A} \cdot \boldsymbol{\sigma}$ is the real potential paravector.

The unitary change to the standard Dirac basis commutes with the scalar rest-phase factor explained in (16.4). Together they define the envelope columns

$$\begin{Bmatrix} \phi_+ \\ \phi_- \end{Bmatrix} = \frac{e^{imt}}{\sqrt{2}} \begin{Bmatrix} \boldsymbol{I}_2 & \boldsymbol{I}_2 \\ -\boldsymbol{I}_2 & \boldsymbol{I}_2 \end{Bmatrix} \begin{Bmatrix} \boldsymbol{c}'_+ \\ \boldsymbol{c}'_- \end{Bmatrix}. \quad (\text{A.22})$$

Here $\phi_+, \phi_- \in \mathbb{C}^2$ are the coefficient columns in the same spin basis of the paravector fields $\phi_+ = e^{imt}(\psi_+ + \psi_-)/\sqrt{2}$ and $\phi_- = e^{imt}(\psi_- - \psi_+)/\sqrt{2}$. Before removing the rest phase, the basis change in (A.22) sends (A.19) to

$$\hat{H}' = \begin{Bmatrix} (qV + m)\mathbf{I}_2 & \sum_k \hat{P}_{q,k} \sigma_k \\ \sum_k \hat{P}_{q,k} \sigma_k & (qV - m)\mathbf{I}_2 \end{Bmatrix}. \quad (\text{A.23})$$

Multiplying the full column by e^{imt} subtracts m from its Hamiltonian: $i\partial_t(e^{imt}\psi) = e^{imt}(i\partial_t - m)\psi$. Thus (A.23) becomes $\hat{H}' - m\mathbf{I}_4$: its diagonal entries are qV and $qV - 2m$. Moving qV into $\hat{E}_q = i\partial_t - qV$ gives the exact envelope equations

$$\begin{aligned} \hat{E}_q \phi_+ &= \sum_k \hat{P}_{q,k} \sigma_k \phi_-, \\ (\hat{E}_q + 2m) \phi_- &= \sum_k \hat{P}_{q,k} \sigma_k \phi_+. \end{aligned}$$

These identify the large and small amplitudes used in the Pauli limit.

For real electric and magnetic components E_k, B_k in the same frame, the squared Dirac field coupling has the explicit image

$$\begin{aligned} \text{mat}'(iq\mathcal{F}) &= q \begin{Bmatrix} iE_3 - B_3 & iE_1 + E_2 - B_1 + iB_2 \\ iE_1 - E_2 - B_1 - iB_2 & -iE_3 + B_3 \end{Bmatrix}, \\ \text{mat}'(iq\mathcal{F}^*) &= \text{mat}'(iq\mathcal{F})^H. \end{aligned}$$

The indexed E_k are field components; whereas $\hat{E}_q$ is the gauge-covariant energy operator. The two couplings are $-q\mathbf{B} \cdot \sigma \pm iq\mathbf{E} \cdot \sigma$.

A.8 Quaternion matrix image

For the quaternion $\mathcal{Q} = a + \mathbf{Q} \cdot (-i\sigma) = a - i\mathbf{Q} \cdot \sigma$ of Section 5.4, with $a \in \mathbb{R}$, $\mathbf{Q} \in \mathbb{R}^3$, and $\mathcal{Q} \in \mathbb{R} \oplus i\mathbb{R}^3$, the explicit matrix image is (6, p. 10)

$$\text{mat}(\mathcal{Q}) = \begin{Bmatrix} a - iQ_3 & -Q_2 - iQ_1 \\ Q_2 - iQ_1 & a + iQ_3 \end{Bmatrix}. \quad (\text{A.24})$$

Here $\text{mat}((-i\sigma_k)) = -i\text{mat}(\sigma_k)$. The conjugation and norm identities become

$$\begin{aligned} \text{mat}(\mathcal{Q}^H) &= \text{mat}(\mathcal{Q})^H, \\ \text{mat}(\mathcal{Q})^H \text{mat}(\mathcal{Q}) &= (a^2 + \mathbf{Q}^2)\mathbf{I}_2, \\ \det(\text{mat}(\mathcal{Q})) &= a^2 + \mathbf{Q}^2. \end{aligned}$$

For a unit quaternion, $a^2 + \mathbf{Q}^2 = 1$, these identities give a unitary matrix of determinant one.

B Equation reference

Each displayed formula repeats a numbered result; its right-hand number links to the definition or derivation. Natural units are $\hbar = c = 1$. The linked text supplies domains, component expansions, and approximation conditions.

Algebra. Notation: complex paravectors $\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$ and $\mathcal{Z}' = c' + \mathbf{Z}' \cdot \boldsymbol{\sigma}$, with $c, c' \in \mathbb{C}$, $\mathbf{Z}, \mathbf{Z}' \in \mathbb{C}^3$, and $\mathcal{Z}, \mathcal{Z}' \in \mathbb{C} \oplus \mathbb{C}^3$. The prime here labels a second element. Their product is

$$\begin{aligned} \mathcal{Z}\mathcal{Z}' &= (cc' + \mathbf{Z} \cdot \mathbf{Z}') \\ &\quad + (c\mathbf{Z}' + c'\mathbf{Z} + i\mathbf{Z} \times \mathbf{Z}') \cdot \boldsymbol{\sigma}. \end{aligned} \quad (2.9)$$

The basic operations are

$$\mathcal{Z}^H = c^* + \mathbf{Z}^* \cdot \boldsymbol{\sigma}, \quad (1.7)$$

$$\mathcal{Z}^* = c^* - \mathbf{Z}^* \cdot \boldsymbol{\sigma}, \quad (1.6)$$

$$\text{adj } \mathcal{Z} = \mathcal{Z}^{*H} = c - \mathbf{Z} \cdot \boldsymbol{\sigma}, \quad (1.8)$$

$$\text{tr } \mathcal{Z} = 2 \text{sc}(\mathcal{Z}), \quad (1.9)$$

$$\text{re}(\mathcal{Z}) = \frac{1}{2}(\mathcal{Z} + \mathcal{Z}^H), \quad (1.14)$$

$$\text{im}(\mathcal{Z}) = (\mathcal{Z} - \mathcal{Z}^H)/(2i), \quad (1.15)$$

$$\det \mathcal{Z} = \text{adj}(\mathcal{Z}) \mathcal{Z} = c^2 - \mathbf{Z}^2, \quad (1.10)$$

$$\mathcal{Z}^{-1} = \text{adj}(\mathcal{Z}) / \det \mathcal{Z} \quad (\det \mathcal{Z} \neq 0), \quad (1.12)$$

$$\|\mathcal{Z}\|^2 = \text{sc}(\mathcal{Z}^H \mathcal{Z}). \quad (2.14)$$

The coefficient form is (2.1); product-order and cyclicity rules are (1.4), (1.5), and (1.13). Projectors and the spectral rule are (3.2) and (3.3). The quaternion component map is (5.6); matrix images are (A.1) and (A.24).

Rotations and boosts. Here $\mathbf{u} \in \mathbb{R}^3$ is a unit axis, $\theta \in \mathbb{R}$ an angle or rapidity, $\mathcal{R} \in \mathbb{R} \oplus i\mathbb{R}^3$, and $\mathcal{L} \in \mathbb{R} \oplus \mathbb{R}^3$; $\beta = \tanh \theta$ and $\gamma = \cosh \theta$ in the boost formulas. The factors and their actions are

$$\mathcal{R} = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.5)$$

$$\mathcal{L} = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cosh(\theta/2) - \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (4.6)$$

$$\mathcal{L} = \sqrt{\frac{\gamma+1}{2}} \left(1 - \frac{\gamma\beta}{\gamma+1} \mathbf{u} \cdot \boldsymbol{\sigma} \right), \quad (4.15)$$

$$\mathcal{L}^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (4.16)$$

$$\mathcal{Z}' = \mathcal{R} \mathcal{Z} \mathcal{R}^H, \quad (5.4)$$

$$\mathcal{Z}' = \mathcal{L} \mathcal{Z} \mathcal{L}, \quad (5.10)$$

$$\mathcal{F}' = \mathcal{L} \mathcal{F} \mathcal{L}^{-1}. \quad (18.4)$$

Both $\mathcal{Z}$ transformations use $\mathcal{Z}' = \mathcal{T} \mathcal{Z} \mathcal{T}^H$, with $\mathcal{R}^H = \mathcal{R}^{-1}$ and $\mathcal{L}^H = \mathcal{L}$. The $\mathcal{Z}$ boost acts on a real four-vector; the complex electromagnetic field $\mathcal{F}$ uses similarity. Both refer to the same frame. Rodrigues' rotation is (5.5); boost components are (5.11)–(5.12). The general determinant-one congruence is (5.2); interval preservation is (7.4). Physical field components are derived in Section 18, with matrix verification in Appendix A.4.

Spacetime and particles. Notation: $\mathbf{x}, \mathbf{u}, \mathbf{p} \in \mathbb{R} \oplus \mathbb{R}^3$, $s, t, E \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and constant $m \in \mathbb{R}_{>0}$. The real four-velocity $\mathbf{u} = d\mathbf{x}/ds$ is an **invertible paravector** with $\mathbf{u}\mathbf{u}^{-1} = 1$. The real kinetic four-momentum $\mathbf{p}$ has particle energy E (including rest energy, excluding potential energy) and spatial kinetic momentum $\mathbf{P}$. The kinematic identities are

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (11.1)$$

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2, \quad (11.2)$$

$$ds^2 = \det(d\mathbf{x}) = dt^2 - d\mathbf{r}^2, \quad (6.1)$$

$$\mathbf{u} = d\mathbf{x}/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad (6.7)$$

$$\mathbf{u}^* \mathbf{u} = 1, \quad (6.8)$$

$$\mathbf{p} = m\mathbf{u} = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad (7.10)$$

$$E^2 - \mathbf{P}^2 = m^2. \quad (7.11)$$

Here t is coordinate time, s is proper time along the time-like path, and $\gamma = dt/ds = \frac{1}{\sqrt{1-\mathbf{v}^2}}$. Four-velocity is real, with **determinant one**; Hermitian unitarity is the separate condition (2.4). For real proper times $s_0 < s_1$ and coordinate times $t_0 < t$,

$$\mathbf{x}(s_1) - \mathbf{x}(s_0) = \int_{s_0}^{s_1} \mathbf{u}(s) ds, \quad (6.9)$$

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'. \quad (6.5)$$

Clock comparison is (6.6); constant proper acceleration gives (8.8); the future cone is (6.2). Proper acceleration, inertia, and particle action are (8.2), (8.3), and (7.15). Moving-frame velocity and its causal bound are (7.6)–(7.7); clock dilation and measured contraction are (7.8) and (7.9); energy and momentum transform by (7.12)–(7.13).

Electromagnetism. Notation: the real four-velocity $\mathbf{u} = d\mathbf{x}/ds$, an **invertible paravector** with $\mathbf{u}\mathbf{u}^{-1} = 1$. Notation: $V, \rho, S, q \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\mathcal{A}, \mathcal{J} \in \mathbb{R} \oplus \mathbb{R}^3$, and $\mathcal{F} \in \{0\} \oplus \mathbb{C}^3$. The coefficient fields depend on real spacetime; q is constant charge. Their boosted arguments and components are (18.5) and (18.6)–(18.7). The field and particle equations are

$$\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (12.1)$$

$$\partial \mathcal{F} = \mathcal{J}^*, \quad (12.2)$$

$$\square \mathcal{F}^* = \partial \mathcal{J}, \quad \square \mathcal{F} = \partial^* \mathcal{J}^*, \quad (12.7)$$

$$\mathcal{F}^* = \partial \mathcal{A} - S, \quad (12.11)$$

$$\mathcal{F}^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}, \quad (12.15)$$

$$\frac{d\mathbf{p}}{ds} = q \operatorname{re}(\mathcal{F}\mathbf{u}), \quad (12.21)$$

$$T(\mathbf{x}) = \frac{1}{2} \mathcal{F} \mathbf{x} \mathcal{F}^H, \quad (12.24)$$

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (12.25)$$

Here $T : \mathbb{R} \oplus \mathbb{R}^3 \rightarrow \mathbb{R} \oplus \mathbb{R}^3$ is the stress map, $\mathbf{x} \in \mathbb{R} \oplus \mathbb{R}^3$ its argument, $u \in \mathbb{R}_{\geq 0}$ field energy density, and $\mathbf{S} \in \mathbb{R}^3$ energy flux. Potential components are (12.12)–(12.13); force components are (12.17) and (12.19). The action derivation uses (12.14) and (12.16).

Relativistic waves. Notation: $m \in \mathbb{R}_{>0}$, $q, V \in \mathbb{R}$, $\mathbf{A} \in \mathbb{R}^3$, and $\mathbf{A} = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$. Here $\hat{E}_q$ is a scalar differential operator and $\hat{\mathbf{P}}_q$ a triple of them; $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$ is a KG field. The Dirac field ψ has two entries in the same fixed spinor space, hence four complex coordinates. Here, subscript 0 labels the uncoupled operators of (13.13). The operators and wave equations are

$$\hat{E}_q = \hat{E}_0 - qV = i\partial_t - qV, \quad (13.15)$$

$$\hat{\mathbf{P}}_q = \hat{\mathbf{P}}_0 - q\mathbf{A} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (13.16)$$

$$\hat{\mathcal{P}}_q = \hat{\mathcal{P}}_0 - q\mathbf{A} = \hat{E}_q + \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}, \quad (13.17)$$

$$(\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (13.19)$$

$$[\hat{E}_q, \hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}, \quad (13.20)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{P}}_q^2 + m^2)\psi = 0, \quad (14.3)$$

$$\mathbf{W}(\mathbf{Z}) = \begin{Bmatrix} 0 & \text{adj}(\mathbf{Z}) \\ \mathbf{Z} & 0 \end{Bmatrix}. \quad (15.2)$$

Here $\mathbf{Z} \in \mathbb{C} \oplus \mathbb{C}^3$; $\mathbf{W}(\mathbf{Z})$ and $\mathbf{I}$ are represented by matrices in $\mathbb{C}^{4 \times 4}$, whereas $\hat{\mathbf{D}}$ and $\hat{\mathbf{D}}_q$ are matrix differential operators. For the free field ($q = 0$),

$$\hat{\mathcal{P}} \text{adj}(\hat{\mathcal{P}}) = \det \hat{\mathcal{P}} = \hat{E}^2 - \hat{\mathbf{P}}^2 = -\square, \quad (13.18)$$

$$\hat{\mathbf{D}}(m) = \mathbf{W}(\hat{\mathcal{P}}) - m\mathbf{I}, \quad (15.3)$$

$$\hat{\mathbf{D}}(m)\psi = 0, \quad (15.4)$$

$$\hat{\mathbf{D}}(-m)\hat{\mathbf{D}}(m) = -(\square + m^2)\mathbf{I}. \quad (15.5)$$

For fixed q and $\mathbf{A}$, the interacting definition and equation are

$$\hat{\mathbf{D}}_q(m) = \mathbf{W}(\hat{\mathcal{P}}_q) - m\mathbf{I}, \quad (15.11)$$

$$\hat{\mathbf{D}}_q(m)\psi = (\mathbf{W}(\hat{\mathcal{P}}_q) - m\mathbf{I})\psi = 0. \quad (15.12)$$

Both $\hat{\mathbf{D}}(-m)$ and $\hat{\mathbf{D}}_q(-m)$ change only the sign of the mass term. The free equations are (14.1) and (15.4); the two-branch solution is (14.2). Massive Dirac reconstruction is given by (15.10) and (15.21); the squared interacting entries are (15.19) and (15.20).

Low-energy limits. After removing rest energy by (16.4) and (16.1), the complex scalar envelope ψ obeys Schrödinger's equation to leading order:

$$i\partial_t\psi = \left(qV + \frac{\hat{\mathbf{P}}_q^2}{2m}\right)\psi. \quad (16.3)$$

The envelope ϕ_+ in the fixed two-dimensional complex spinor space of $\mathbb{C} \oplus \mathbb{C}^3$ obeys Pauli's equation:

$$i\partial_t\phi_+ = \left(qV + \frac{\hat{\mathbf{P}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\right)\phi_+. \quad (16.9)$$

The derivations and approximation conditions are in Section 16; these are the same leading order for scalar and spinor amplitudes respectively.

Spin and probabilities. Notation: a nonzero ψ in the fixed spinor space, $\rho \in \mathbb{R}_{>0}$, and polarization and measurement axes $\mathbf{u}', \mathbf{u} \in \mathbb{R}^3$ of unit norm. The real observable $\mathbf{Z} \in \mathbb{R} \oplus \mathbb{R}^3$ acts from the left. A rotation uses

$$\psi' = \mathcal{R}\psi. \quad (17.1)$$

It preserves the norm and inner product; the expectation transforms as $\langle \mathbf{Z} \rangle_{\psi'} = \langle \mathcal{R}^H \mathbf{Z} \mathcal{R} \rangle_{\psi}$ by (13.8). All formulas are pointwise. For a nonnegative real weight $\mathbf{Z}$ from (13.5), the densities and measurements are

$$\rho = \|\psi\|^2 = \text{sc}(\psi^H \psi), \quad (13.3)$$

$$\|\psi\|_{\mathbf{Z}}^2 = \text{sc}(\psi^H \mathbf{Z} \psi), \quad (13.5)$$

$$\mathbf{u} \cdot \mathbf{u}' = \text{sc}(\psi^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi) / \rho, \quad (17.3)$$

$$\text{prob}(\pm | \mathbf{u}) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{u}'), \quad (17.4)$$

$$\langle \mathbf{Z} \rangle_{\psi} = \text{sc}(\psi^H \mathbf{Z} \psi) / \rho. \quad (13.6)$$

The expectation formula also applies to signed real observables. The matrix verification is (A.12)–(A.13).

C Conventions

Unadorned source symbols below retain each source's meaning. In the coefficient notation, $\mathcal{Z}, \mathcal{G} \in \mathbb{C} \oplus \mathbb{C}^3$, $\mathcal{X}, \mathcal{Y}, \mathcal{A}, \mathcal{P}, \mathcal{J} \in \mathbb{R} \oplus \mathbb{R}^3$, and $\mathcal{F} \in \{0\} \oplus \mathbb{C}^3$. Physical factors $\mathcal{G}$ have determinant one; spinor amplitudes lie in the fixed ideal of (13.2).

Names and algebra classifications. The name *space-time algebra* is used in this paper for the formulation in complex paravectors defined by the sigma multiplication rules. In Clifford-algebra terminology, this eight-real-dimensional algebra is classified as $\text{Cl}_{3,0}$ and is commonly called the *algebra of physical space* (APS). It is isomorphic to the even subalgebra $\text{Cl}_{1,3}^+$ of the sixteen-real-dimensional algebra $\text{Cl}_{1,3}$, for which the name *spacetime algebra* (STA) is used by Hestenes (1, pp. 1–2). In $\text{Cl}_{p,q}$, p positive and q negative basis-vector squares are specified; the signature $(+, -, -, -)$ is denoted by $(1, 3)$, and the opposite signature by $(3, 1)$.

In this terminology, the sigmas are called *generators*. The basis elements $1, \sigma_k, i\sigma_k$, and i are assigned grades 0, 1, 2, and 3, respectively, and are called scalars, vectors, bivectors, and pseudoscalars. Thus scalar and pseudoscalar grades are combined in the complex scalar part, and vector and bivector grades in the complex vector term. Hermitian conjugation is called *reversion* in APS; sigma conjugation is called *grade involution*; and adjugation is called *Clifford conjugation* (1, pp. 4–6). These correspondences depend on the algebra: the four-dimensional comparison is given below.

A rotation factor is also called a *rotor*, a projector an *idempotent*, and an invertible element an *algebraic unit*; neither unit norm nor determinant one is implied by the last name. The fixed two-dimensional complex spinor space of (13.2) is called a *minimal left ideal*: it is preserved by left multiplication by every paravector. In particle dynamics, *mechanical momentum* is another name for kinetic momentum. The quantity γm in Section 8.2 is sometimes called *relativistic mass*; invariant mass is used throughout this paper.

Mapping to the literature. The conventions of Baylis and Sobczyk are compared below (1, pp. 4–6). The table translates the notation of the [Paravector article](#); A denotes a general element and x a real paravector.

Object or operation	This paper	APS notation
Spatial basis	σ_k	$\mathbf{e}_k$
Central imaginary unit	$i = \sigma_1 \sigma_2 \sigma_3$	$i = \mathbf{e}_{123}$
Real paravector	$\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$	$x = x^0 + \mathbf{x}$
General algebra element	$\mathcal{Z} = c + \mathbf{Z} \cdot \boldsymbol{\sigma}$	A (complex paravector)
Reversion	$\mathcal{Z}^H$	$A^\dagger$
Clifford conjugation	$\text{adj } \mathcal{Z}$	$\bar{A}$
Grade involution	$\mathcal{Z}^*$	$\bar{A}^\dagger$
Complex scalar part	$\text{sc}(\mathcal{Z})$	$\langle A \rangle_S$
Complex vector part	$\text{vec}(\mathcal{Z}) \cdot \boldsymbol{\sigma}$	$\langle A \rangle_V$
Real part	$\text{re}(\mathcal{Z})$	$\langle A \rangle_R$
Imaginary part	$i \text{im}(\mathcal{Z})$	$\langle A \rangle_I$
Spacetime quadratic form	$\det \mathcal{X} = t^2 - \mathbf{r}^2$	$x\bar{x}$
Projectors	Π_+, Π_-	$P_u, \bar{P}_u$

Real paravectors have four real components; complex $\mathcal{Z}$ packages all eight APS components. Here

$\text{sc}(\mathcal{Z})$ includes the pseudoscalar, and $\text{im}(\mathcal{Z})$ removes the imaginary part's factor i .

In particular, the bar is represented here by the adjugate, not by sigma conjugation:

$$\bar{A} \longleftrightarrow c - \mathcal{Z} \cdot \sigma, \quad \bar{A}^\dagger \longleftrightarrow c^* - \mathcal{Z}^* \cdot \sigma.$$

They coincide only for real paravectors. Complex coefficients are conjugated by sigma conjugation, which is identified with grade involution on the full algebra. Wikipedia's [paragradiant](#) is represented here by the starred derivative:

$$\partial_{\text{article}} = \partial_t - \nabla \longleftrightarrow \partial^*, \quad \bar{\partial}_{\text{article}} \longleftrightarrow \partial, \quad \square = \partial \partial^*.$$

Baylis's rotor action $x' = GxG^\dagger$ is written here as $\mathcal{X}' = \mathcal{G}\mathcal{X}\mathcal{G}^H$; Baylis's field $\mathbf{E} + i\mathbf{B}$ and law $F' = GF\bar{G}$ are written here as $\mathcal{F} = (\mathbf{E} + i\mathbf{B}) \cdot \sigma$ and $\mathcal{F}' = \mathcal{G}\mathcal{F}\mathcal{G}^{-1}$ for $\det \mathcal{G} = 1$ ($c = 1$) (5, Eq. (23)) (5, Eq. (24)) (5, Eq. (33)). Baylis's positively oriented spatial-plane generator is $\mathbf{e}_2\mathbf{e}_1 = -i\mathbf{e}_3$, giving $\mathcal{R} = e^{-i\theta\sigma_3/2}$ about $+\sigma_3$ (5, Eqs. (17)–(19)). Baylis's positive-rapidity boost factor, denoted here by $\mathcal{B}$, satisfies

$$\mathcal{B} = e^{+\theta \mathbf{u} \cdot \sigma / 2} = \mathcal{L}^{-1}, \quad \mathcal{B}^2 = \gamma(1 + \beta \mathbf{u} \cdot \sigma), \quad \mathcal{X}' = \mathcal{B}^{-1} \mathcal{X} (\mathcal{B}^{-1})^H = \mathcal{L} \mathcal{X} \mathcal{L}^H.$$

Thus the passive boost is obtained by inversion of Baylis's positive active boost (5, p. 23); the active/passive distinction is explicit in (1, Eq. (38)). The algebra is standard; exponent signs must be read with the transformation direction and sandwich order.

For translation to Hestenes's four-dimensional STA, an observer axis γ_0 is chosen, with $\gamma_0^2 = 1$ and $\gamma_k^2 = -1$. The identification with its even subalgebra is (1, Eq. (40)) (1, Eq. (41))

$$\sigma_k \longleftrightarrow \gamma_k \gamma_0, \quad \text{adj } \mathcal{Z} \longleftrightarrow \tilde{Z}_{\text{STA}}, \quad \mathcal{Z}^H \longleftrightarrow \gamma_0 \tilde{Z}_{\text{STA}} \gamma_0.$$

Here $\gamma_\mu \in \text{Cl}_{1,3}$ are spacetime basis vectors, not the numerical gamma matrices of Appendix A.7. The image Z_{STA} lies in the even subalgebra $\text{Cl}_{1,3}^+$, and the tilde reverses STA products: reversion in APS and reversion in STA are different operations.

Source comparisons for the cited equations. Body citations point to matching definitions or physical component results. The source locators below identify those comparisons; intervening algebra and the block-map packaging are derived in this paper.

Result here	Direct source comparison
Sigma product, conjugations, and extractions (1.1)–(1.15)	(1, pp. 4–6); (2, Eq. (3)).
Squared norm (2.14); inverse (1.12)	(2, Eqs. (46)–(47)).
Quaternions (5.6); projectors (3.2)	(3, p. 92); (3, p. 121).
Spectral rule (3.3); coalescence (3.5)	(4, pp. 3–4).
Interval and four-velocity (6.1)–(6.7)	(5, Eqs. (7)–(8)).
Momentum, acceleration, and world lines (7.10), (8.2), (8.8)	(9, Eq. (7.24)); (9, p. 141); (9, Eq. (7.33), p. 143).
Maxwell, stress map, and energy flux (12.2), (12.24)	(8, Eq. (177)); (8, Eq. (167)); (8, Eq. (180)).
Potential, gauge, and field action (12.12)–(12.13), (12.14)	(7, p. 105); (7, Eq. (5.42)); (7, Eq. (5.43)); (7, Eqs. (6.1)–(6.3)); (7, Eq. (6.4)).
Lorentz force (12.21); components (12.17), (12.19)	(10, Eq. (2)); (7, Eqs. (5.32)–(5.33)).
Boosted components (7.2), (18.7)	(7, Eq. (5.1)); (7, Eqs. (5.17)–(5.18)).
KG and minimally coupled Dirac (14.1), (14.3), (15.12)	(16, Eq. (1.13)); (17, Eq. (299)); (8, Eqs. (190)–(191)).
Chiral squared Dirac (15.19)	(18, Eq. (6)).
Kinetic operators, small amplitude, and Pauli (13.16), (16.8), (16.9)	(12, Eqs. (2.11)–(2.16)).
KG to Schrödinger (16.3)	(19, Eq. (21.3)); (16, Eqs. (2.103)–(2.104)); (16, Eq. (2.105)).
Spin amplitudes and probabilities (13.2), (17.4)	(11, Eqs. (66)–(67)); (11, Eq. (69a)); (13, Eq. (6.1)); (13, p. 62).

Translating the physical comparisons.

SI units: Comparison with Tong

SI units are used in Tong's component electrodynamics; $c = \epsilon_0 = \mu_0 = 1$ is used outside these boxes.

Tong's gauge parameter in Eq. (5.13) is $\chi = -\lambda$, whereas Eq. (6.2) is written with the same plus-sign convention (7, Eq. (5.13)) (7, Eq. (6.2)). Tong's positive-exponent spatial rotation is a passive axis rotation; the negative-exponent boost is identified with the passive boost used here (9, Eq. (7.38)) (9, Eq. (7.39)).

Chappell et al.'s scalar Z is represented here by c , and the vector–bivector F by $\mathbf{Z} \cdot \boldsymbol{\sigma}$, with $|F|^2 = -\mathbf{Z}^2$. Thus the trigonometric exponential is represented here by the hyperbolic form (2, Eq. (52)).

For the quaternion basis, the Hamilton units are identified by Baylis with $\mathbf{e}_{32}, \mathbf{e}_{13}, \mathbf{e}_{21}$, respectively (3, p. 92). With $\mathbf{e}_k \leftrightarrow \sigma_k$, the units are represented here by $(-i\sigma_1), (-i\sigma_2), (-i\sigma_3)$. Higham's spectral and repeated-root formulas apply through the faithful matrix image in Appendix A; no change of the intrinsic product is required (4, pp. 3–4).

Lenz's scalar potentials ϕ, Φ are represented here by V , with $\hbar = c = 1$ and $e = q$. The Dirac Hamiltonian is represented by the matrix form (A.23) (17, Eq. (412)).

For Hestenes's Maxwell equation, the spacetime derivative and current are denoted by ∇_{STA} and J_{STA} , respectively. Using the STA embedding above,

$$\nabla_{\text{STA}} = \gamma_0 \partial, \quad J_{\text{STA}} = \mathcal{J} \gamma_0 = \gamma_0 \mathcal{J}^*, \quad \partial \mathcal{F} = \mathcal{J}^* \iff \nabla_{\text{STA}} \mathcal{F} = J_{\text{STA}}.$$

Hestenes's explicit spacetime split is represented here by the component equation (8, Eqs. (175)–(177)). For a real spacetime vector $n \in \mathbb{R}^{1,3}$, $\mathcal{X} = n\gamma_0 \in \mathbb{R} \oplus \mathbb{R}^3$ is used in the observer's paravector identification. Hestenes's stress map $T_{\text{STA}}(n) = \frac{1}{2} F n \tilde{F}$, with $F^{\text{H}} = \gamma_0 \tilde{F} \gamma_0$, gives

$$T_{\text{STA}}(n)\gamma_0 = \frac{1}{2} \mathcal{F} \mathcal{X} \mathcal{F}^{\text{H}} = T(\mathcal{X}),$$

including the energy and flux components in Eq. (180) of that source (8, Eq. (167)) (8, Eq. (180)).

For Dirac and Pauli, the complex unit is represented by Hestenes and Gurtler–Hestenes through right multiplication of the real spinor by the spatial bivector $i\sigma_3$; it acts as ordinary i in the corresponding column spinor (8, Eqs. (187)–(188)) (8, Eqs. (192)–(193)). With $e = q$ and $\hbar = c = 1$, Gurtler–Hestenes's potential energy V is represented here by qV ; after removal of rest energy, the operators K and p are identified with $\hat{E}_q$ and $\hat{\mathbf{P}}_q \cdot \boldsymbol{\sigma}$. The same components are then obtained for the large/small equations and Pauli limit (12, Eqs. (2.11)–(2.19)).

Charge is absorbed into the potential by Feynman–Gell-Mann. With charge restored, Eq. (6) is identified with the first squared Dirac block because

$$-iq\mathcal{F}^* = q(\mathbf{B} + i\mathbf{E}) \cdot \boldsymbol{\sigma}, \quad -iq\mathcal{F} = q(\mathbf{B} - i\mathbf{E}) \cdot \boldsymbol{\sigma}.$$

The two blocks therefore have the same magnetic sign and opposite electric signs (18, Eq. (6)).

Baylis et al.'s RP_3 is a two-component minimal-ideal spinor; its rotations are represented by the left amplitude action. From the density-operator formula is obtained the same $(1 + \mathbf{u} \cdot \mathbf{u}')/2$ probability after basis normalization, extending to mixed states with $\|\mathbf{u}'\| \leq 1$ (11, Eqs. (66)–(67)). Passive basis rotations are used in Feynman's spin chapter; its 2π sign and probability statements agree, while an active rotor uses the inverse rotation (20, Eq. (6.19)).

The right-handed sigma product is fixed, and the following notation is used: $\mathcal{N} = \mathbf{u} \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{R}^3$, with $\mathbf{u} \in \mathbb{R}^3$, $\mathbf{u}^2 = 1$, and $\theta \in \mathbb{R}$. The alternatives below change notation or transformation direction, not the underlying physics. The main text uses $\mathcal{R} = \mathcal{R}_-$ and $\mathcal{L} = \mathcal{L}_-$, with

$$\mathcal{R}_\pm = e^{\pm i\theta \mathcal{N}/2}, \quad \mathcal{L}_\pm = e^{\pm \theta \mathcal{N}/2}, \quad \mathcal{R}_\pm^H = \mathcal{R}_\pm^{-1} = \mathcal{R}_\mp, \quad \mathcal{L}_\pm^H = \mathcal{L}_\pm, \quad \mathcal{L}_\pm^{-1} = \mathcal{L}_\mp.$$

Here $\mathcal{R}_\pm \in \mathbb{R} \oplus i\mathbb{R}^3$, $\mathcal{L}_\pm \in \mathbb{R} \oplus \mathbb{R}^3$, and θ is a rotation angle or rapidity respectively.

Exponent sign and sandwich order. For $\mathcal{X} = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, the four choices for each kind of factor are summarized by the two action columns:

Factor $\mathcal{T}$	$\mathcal{X}' = \mathcal{T}\mathcal{X}\mathcal{T}^H$	$\mathcal{X}' = \mathcal{T}^H\mathcal{X}\mathcal{T}$
$\mathcal{R}_-$	Active rotation $+\theta$	Active rotation $-\theta$
$\mathcal{R}_+$	Active rotation $-\theta$	Active rotation $+\theta$
$\mathcal{L}_-$	Frame velocity $+\beta\mathbf{u}$	Frame velocity $+\beta\mathbf{u}$
$\mathcal{L}_+$	Frame velocity $-\beta\mathbf{u}$	Frame velocity $-\beta\mathbf{u}$

For rotations $t' = t$. Both $+\theta$ entries give

$$\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (1 - \cos \theta)(\mathbf{u} \cdot \mathbf{r})\mathbf{u}.$$

This is the active right-handed Rodrigues formula (5.5); a passive rotation of axes uses the inverse map.

For boosts, $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. Writing $\mathbf{r}_\parallel = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ and $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$, both sandwiches with $\mathcal{L}_\pm$ give

$$t' = \gamma(t \pm \beta\mathbf{u} \cdot \mathbf{r}), \quad \mathbf{r}' = \mathbf{r}_\perp + \gamma(\mathbf{r}_\parallel \pm \beta t\mathbf{u}).$$

The minus choice sends $\mathbf{r} = \beta t\mathbf{u}$ to $\mathbf{r}' = 0$, fixing its frame velocity. Moving the adjoint reverses a rotation but has no effect on a boost; reversing the boost exponent gives the inverse frame transformation. The same $\mathcal{L}_+$ map can instead describe an active boost to velocity $+\beta\mathbf{u}$ in fixed coordinates.

Hamilton quaternions fix the basis orientation. With the sigma orientation held fixed, the two uniform sign choices obey

$$\begin{aligned} (-i\sigma_j)(-i\sigma_k) &= -\delta_{jk} + \epsilon_{jkl}(-i\sigma_l), \\ (+i\sigma_j)(+i\sigma_k) &= -\delta_{jk} - \epsilon_{jkl}(+i\sigma_l). \end{aligned}$$

Repeated l is summed. Hamilton's orientation is obtained with $(-i\sigma_k)$, and the opposite orientation with $(+i\sigma_k)$. For the Hamilton basis, Section 5.4 gives

$$\mathcal{Q} = e^{\theta\mathbf{u} \cdot (-i\boldsymbol{\sigma})/2} = \mathcal{R}_-, \quad \mathcal{Q}^H = \mathcal{Q}^{-1}, \quad \mathbf{r}' \cdot (-i\boldsymbol{\sigma}) = \mathcal{Q}(\mathbf{r} \cdot (-i\boldsymbol{\sigma}))\mathcal{Q}^H.$$

Since $\mathbf{r} \cdot (-i\boldsymbol{\sigma}) = -i\mathbf{r} \cdot \boldsymbol{\sigma}$ and i commutes with $\mathcal{Q}$, the same sandwich rotates $\mathbf{r} \cdot \boldsymbol{\sigma}$, with no extra sign. For example, $\mathcal{Q} = (1 - i\sigma_3)/\sqrt{2}$ sends σ_1 to σ_2 .

Using $(+i\sigma_k)$ with $\tilde{\mathcal{Q}} = e^{\theta\mathbf{u} \cdot (+i\boldsymbol{\sigma})/2} = \mathcal{R}_+$ and the right-adjoint sandwich instead gives $-\theta$. $+\theta$ is recovered by reversing either the exponent or the sandwich. Reflecting the basis also restores Hamilton multiplication, but changes its identification with the physical axes. Replacing a factor by its negative leaves both sandwiches unchanged; the spinor acquires an overall minus sign. Consequently rotations are 2π -periodic on vectors and 4π -periodic on amplitudes.

Rotating a state versus changing the measured observable. With the Hamilton basis, the two consistent descriptions of the same active $+\theta$ rotation are

Choice	Quaternion relation	Rotated observable	Rotated amplitude
$\mathcal{R} = \mathcal{R}_-$	$\mathcal{R} = \mathcal{Q}$	$\mathcal{Z}' = \mathcal{R}\mathcal{Z}\mathcal{R}^H$	$\psi' = \mathcal{R}\psi$
$\mathcal{R} = \mathcal{R}_+$	$\mathcal{R} = \mathcal{Q}^H$	$\mathcal{Z}' = \mathcal{R}^H\mathcal{Z}\mathcal{R}$	$\psi' = \mathcal{R}^H\psi$

Both have physical amplitude factor $e^{-i\theta\mathcal{N}/2}$ and spin generator $\mathcal{N}/2$. Changing only the vector sandwich would reverse its rotation relative to the spin polarization.

For $\mathcal{G} \in \mathbb{C} \oplus \mathbb{C}^3$ with $\det \mathcal{G} \neq 0$ acting as $\psi' = \mathcal{G}\psi$, the definitions (13.3) and (13.6) give

$$\begin{aligned}\rho &= \|\psi\|^2, & \rho' &= \rho \langle \mathcal{G}^H \mathcal{G} \rangle_\psi, \\ \langle \varphi' | \psi' \rangle &= \text{sc}(\varphi^H \mathcal{G}^H \mathcal{G} \psi), \\ \langle \mathcal{Z} \rangle_{\psi'} &= \frac{\langle \mathcal{G}^H \mathcal{Z} \mathcal{G} \rangle_\psi}{\langle \mathcal{G}^H \mathcal{G} \rangle_\psi}.\end{aligned}$$

Here $\varphi' = \mathcal{G}\varphi$, and $\rho > 0$. For a rotation $\mathcal{G}^H \mathcal{G} = 1$:

$$\boxed{\langle \mathcal{Z} \rangle_{\psi'} = \langle \mathcal{G}^H \mathcal{Z} \mathcal{G} \rangle_\psi, \quad \langle \mathcal{G} \mathcal{Z} \mathcal{G}^H \rangle_{\psi'} = \langle \mathcal{Z} \rangle_\psi.} \quad (\text{C.1})$$

The first identity measures fixed $\mathcal{Z}$ in the rotated state; the second rotates the observable with the state. Thus the opposite sandwich orders in (C.1) are required, not competing conventions. For $\mathcal{G} = \mathcal{L}_\pm$, the denominator is $\langle \mathcal{L}_\pm^2 \rangle_\psi$: boosts generally change the local positive density and inner product.

Fields and wave amplitudes under either notation. $\mathcal{G}$ is taken to be the constant physical factor of determinant one, so $\mathcal{X}' = \mathcal{G}\mathcal{X}\mathcal{G}^H$. For notation $\mathcal{T}\mathcal{X}\mathcal{T}^H$, $\mathcal{G} = \mathcal{T}$ is used; for $\mathcal{T}^H\mathcal{X}\mathcal{T}$, $\mathcal{G} = \mathcal{T}^H$ is used. At corresponding events the laws are

$$\begin{aligned}\mathcal{Z}' &= \mathcal{G}\mathcal{Z}\mathcal{G}^H, & \mathcal{Z} &= \mathcal{X}, \mathcal{P}, \mathcal{A}, \mathcal{J}, \\ \mathcal{F}' &= \mathcal{G}\mathcal{F}\mathcal{G}^{-1}, & \partial' &= \mathcal{G}^* \partial \mathcal{G}^{-1}, \\ \psi'_+ &= \mathcal{G}^* \psi_+, & \psi'_- &= \mathcal{G} \psi_-, \\ \psi'_{\text{KG}}(\mathcal{X}') &= \psi_{\text{KG}}(\mathcal{X}).\end{aligned} \quad (\text{C.2})$$

These are the covariance laws of Section 18; $\mathcal{G}^* = (\mathcal{G}^{-1})^H$. The field $\mathcal{F}$ uses similarity, not the four-vector sandwich. Equation (C.2) preserves the Maxwell, Klein–Gordon, and Dirac equations.

For either rotation description, both Dirac entries use $\mathcal{R}_-$. For $\mathcal{G} = \mathcal{L}_\pm$, they use $\mathcal{L}_\pm^{-1}$ and $\mathcal{L}_\pm$, respectively; the factor already contains half the rapidity. With $\mathcal{X}' = \mathcal{T}^H\mathcal{X}\mathcal{T}$, the same laws read $\psi'_+ = \mathcal{T}^{-1}\psi_+$ and $\psi'_- = \mathcal{T}^H\psi_-$. The scalar paired quantity $\text{sc}(\psi_+^H \psi_-)$ is invariant; the sum of the two positive densities generally changes under boosts. The conserved Dirac current transforms as a four-vector; its time component alone is not invariant. For products, the physical factors compose as $\mathcal{G}_2\mathcal{G}_1$. Hence the right-adjoint parameters compose as $\mathcal{T}_2\mathcal{T}_1$, but the left-adjoint parameters compose as $\mathcal{T}_1\mathcal{T}_2$.

Preserved quantities. The sign of the angle or rapidity changes the direction, not these invariance statements. Here $\mathcal{Z}$ is a complex paravector and $\|\mathcal{Z}\|^2 = \text{sc}(\mathcal{Z}^H \mathcal{Z})$; “no” means not generally preserved.

Action	$\det \mathcal{Z}$	$\text{tr } \mathcal{Z}, \text{sc}(\mathcal{Z})$	$\ \mathcal{Z}\ ^2$
Rotation $\mathcal{R}\mathcal{Z}\mathcal{R}^H$	Yes	Yes	Yes
Boost $\mathcal{L}\mathcal{Z}\mathcal{L}^H$	Yes	No	No
Field similarity $\mathcal{G}\mathcal{Z}\mathcal{G}^{-1}$	Yes	Yes	If $\mathcal{G}$ is unitary

For real four-vectors transformed together, both rotations and boosts preserve

$$\text{sc}(\mathcal{X}'^* \mathcal{Y}') = \text{sc}(\mathcal{X}^* \mathcal{Y}).$$

Similarity also preserves $\text{sc}(\mathcal{Z}\mathcal{Y})$ when both factors transform by similarity. The positive amplitude inner product is rotation invariant; boost invariance instead applies to the spacetime metric and the paired Dirac quantity above. Normalized expectations remain equal for a state and observable rotated together, not for an arbitrary fixed observable.

The main convention $\mathcal{R} = \mathcal{Q} = \mathcal{R}_-, \mathcal{X}' = \mathcal{R}\mathcal{X}\mathcal{R}^H, \psi' = \mathcal{R}\psi$ uses the Hamilton quaternion directly. The alternative $\mathcal{R} = \mathcal{Q}^H = \mathcal{R}_+, \mathcal{X}' = \mathcal{R}^H \mathcal{X} \mathcal{R}, \psi' = \mathcal{R}^H \psi$ is equally consistent. Either can use $\mathcal{L}^H \mathcal{X} \mathcal{L} = \mathcal{L} \mathcal{X} \mathcal{L}^H$ with the boost sign fixed by the chosen frame velocity.

References

- (1) W. E. Baylis and G. Sobczyk, “Relativity in Clifford’s geometric algebras of space and spacetime,” *Int. J. Theor. Phys.*, vol. 43, pp. 2061–2079, 2004, doi: [10.1023/B:IJTP.0000049010.53558.b7](https://arxiv.org/pdf/math-ph/0405026). Online. Available: <https://arxiv.org/pdf/math-ph/0405026>.
- (2) J. M. Chappell, A. Iqbal, L. J. Gunn, and D. Abbott, “Functions of multivector variables,” *PLoS ONE*, vol. 10, no. 3, e0116943, 2015, doi: [10.1371/journal.pone.0116943](https://doi.org/10.1371/journal.pone.0116943). Online. Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0116943>.
- (3) W. E. Baylis, “Applications of Clifford algebras in physics,” in *Lectures on Clifford (Geometric) Algebras and Applications*, R. Ablamowicz and G. Sobczyk, Eds. Boston, MA, USA: Birkhäuser, 2004, pp. 91–133, doi: [10.1007/978-0-8176-8190-6_4](https://doi.org/10.1007/978-0-8176-8190-6_4). Online. Published volume: <https://ndl.ethernet.edu.et/bitstream/123456789/32989/1/3.pdf.pdf>.
- (4) N. J. Higham, “Functions of matrices,” University of Manchester, MIMS EPrint 2005.21, Jun. 2006. Online. Available: https://eprints.maths.manchester.ac.uk/310/1/fm_final.pdf.
- (5) W. E. Baylis, “Relativity in introductory physics,” *Can. J. Phys.*, vol. 82, no. 11, pp. 853–873, 2004, doi: [10.1139/p04-058](https://arxiv.org/pdf/physics/0406158). Online. Available: <https://arxiv.org/pdf/physics/0406158>.
- (6) P. Lounesto, “Introduction to Clifford algebras,” in *Lectures on Clifford (Geometric) Algebras and Applications*, R. Ablamowicz and G. Sobczyk, Eds. Boston, MA, USA: Birkhäuser, 2004, pp. 1–29, doi: [10.1007/978-0-8176-8190-6_1](https://doi.org/10.1007/978-0-8176-8190-6_1). Online. Published volume: <https://ndl.ethernet.edu.et/bitstream/123456789/32989/1/3.pdf.pdf>.
- (7) D. Tong, *Electromagnetism*, University of Cambridge, Cambridge, U.K., lecture notes, 2015. Online. Available: <https://davidtong.org/pdfs/teaching/electromagnetism/electro.pdf>.
- (8) D. Hestenes, “Spacetime physics with geometric algebra,” *Am. J. Phys.*, vol. 71, no. 7, pp. 691–714, 2003, doi: [10.1119/1.1571836](https://doi.org/10.1119/1.1571836). Online. Available: <https://davidhestenes.net/geocalc/pdf/SpacetimePhysics.pdf>.
- (9) D. Tong, *Dynamics and Relativity*, University of Cambridge, Cambridge, U.K., lecture notes, 2013. Online. Available: <https://davidtong.org/pdfs/teaching/dynamics-and-relativity/dynrel.pdf>.
- (10) W. E. Baylis, “Surprising symmetries in relativistic charge dynamics,” arXiv:physics/0410197, 2004. Online. Available: <https://arxiv.org/pdf/physics/0410197>.
- (11) W. E. Baylis, R. Cabrera, and D. Keselica, “Quantum/Classical interface: Fermion spin,” arXiv:0710.3144, 2007. Online. Available: <https://arxiv.org/pdf/0710.3144>.
- (12) R. Gurtler and D. Hestenes, “Consistency in the formulation of the Dirac, Pauli, and Schrödinger theories,” *J. Math. Phys.*, vol. 16, no. 3, pp. 573–584, 1975, doi: [10.1063/1.522555](https://doi.org/10.1063/1.522555). Online. Available: <https://davidhestenes.net/geocalc/pdf/Consistency.pdf>.
- (13) J. Sparks, *Quantum Theory*, University of Oxford, Oxford, U.K., lecture notes, 2020. Online. Available: <https://users.ox.ac.uk/~math0391/QT2020.pdf>.

- (14) L. de Broglie, “The wave nature of the electron,” Nobel Lecture, Dec. 12, 1929, in *Nobel Lectures, Physics 1922–1941*. Amsterdam, The Netherlands: Elsevier, 1965, pp. 244–256. Online. Available: <https://www.nobelprize.org/uploads/2016/04/broglie-lecture.pdf>.
- (15) R. A. Millikan, “The electron and the light-quant from the experimental point of view,” Nobel Lecture, May 23, 1924, in *Nobel Lectures, Physics 1922–1941*. Amsterdam, The Netherlands: Elsevier, 1965. Online. Available: <https://www.nobelprize.org/uploads/2018/06/millikan-lecture.pdf>.
- (16) D. Tong, *Quantum Field Theory*, University of Cambridge, Cambridge, U.K., lecture notes, 2006–2007. Online. Available: <https://davidtong.org/pdfs/teaching/quantum-field-theory/qft.pdf>.
- (17) A. Lenz, *Theoretical Physics 3: Quantum Mechanics (PHYS3661)*, IPPP, Durham University, lecture notes, Apr. 8, 2015. Online. Available: https://tp.nt.uni-siegen.de/wp-content/uploads/2024/02/TP3-Notes_Lenz_2015.pdf.
- (18) R. P. Feynman and M. Gell-Mann, “Theory of the Fermi interaction,” *Phys. Rev.*, vol. 109, no. 1, pp. 193–198, 1958, doi: [10.1103/PhysRev.109.193](https://doi.org/10.1103/PhysRev.109.193). Online. Available: <https://harvest.aps.org/v2/journals/articles/10.1103/PhysRev.109.193/fulltext>.
- (19) R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. III. Reading, MA, USA: Addison-Wesley, 1965, ch. 21, “The Schrödinger equation in a classical context: A seminar on superconductivity.” Online. Available: https://www.feynmanlectures.s.caltech.edu/III_21.html.
- (20) R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. III. Reading, MA, USA: Addison-Wesley, 1965, ch. 6, “Spin one-half.” Online. Available: https://www.feynmanlectures.caltech.edu/III_06.html.
- (21) R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. I. Reading, MA, USA: Addison-Wesley, 1963, ch. 17, “Space-time.” Online. Available: https://www.feynmanlectures.caltech.edu/I_17.html.
- (22) J. G. Russo and P. K. Townsend, “Relativistic kinematics and stationary motions,” *J. Phys. A: Math. Theor.*, vol. 42, 445402, 2009, doi: [10.1088/1751-8113/42/44/445402](https://doi.org/10.1088/1751-8113/42/44/445402). Online. Available: <https://arxiv.org/pdf/0902.4243>.
- (23) K. Ko, T. Maekawa, and N. M. Patrikalakis, “Differential geometry of curves,” *Computational Geometry*, Lecture 2, Massachusetts Institute of Technology, 2003. Online. Available: https://ocw.mit.edu/courses/2-158j-computational-geometry-spring-2003/dd2e0ef6278218b74dd1709065814109_lecnotes2.pdf.
- (24) S. M. Carroll, *Lecture Notes on General Relativity*, 1997. Online. Available: <https://arxiv.org/pdf/gr-qc/9712019>.
- (25) D. Bini, A. Geralico, and R. P. Kerr, “The Kerr–Schild ansatz revised,” arXiv:1408.4601, 2014. Online. Available: <https://arxiv.org/pdf/1408.4601>.
- (26) C. Heinicke and F. W. Hehl, “Schwarzschild and Kerr solutions of Einstein’s field equation—an introduction,” arXiv:1503.02172, 2015. Online. Available: <https://arxiv.org/pdf/1503.02172>.